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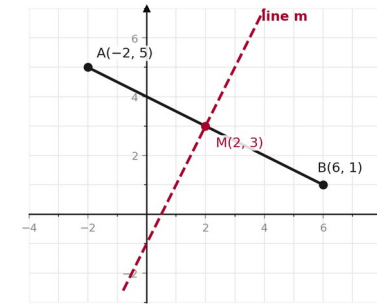
DIGITAL SAT — MATH

Algebra

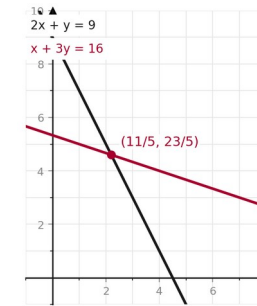
Theory Companion — Explained Rules, Worked Problems & Cheat Sheet

Lines • Systems • Inequalities • Absolute Value • Exponents • Radicals

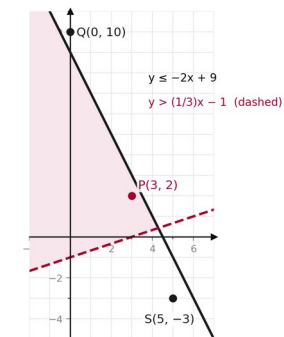
Prepared by the CLC Digital SAT Team



m is the perpendicular bisector of AB

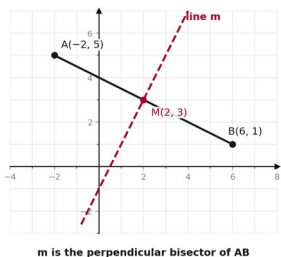


one intersection in Quadrant I



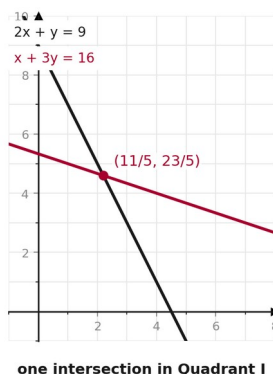
6 Skills in This Deck

Each skill opens with key terms in English and Greek, explained theory with examples, and a Watch Out slide — then 8 Hard / Very Hard worked problems. A cheat sheet closes the deck.



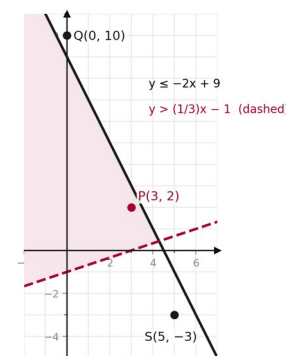
01 Lines

Slope, the four equation forms, and coordinate-geometry formulas.



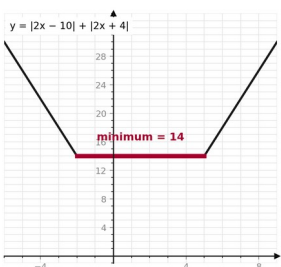
02 Systems of Linear Equations

Solving pairs of equations and counting their solutions.



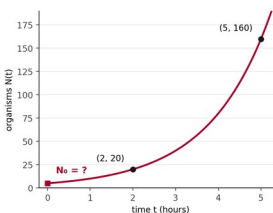
03 Linear Inequalities

One-variable inequalities, compound inequalities, and graphed regions.



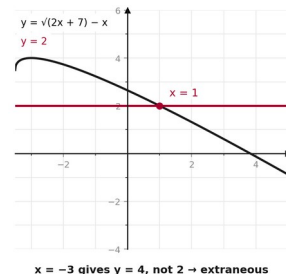
04 Absolute Value

Distance on the number line, two-case equations, and extraneous solutions.



05 Exponent Rules

Simplifying powers and solving exponential equations.



06 Radicals, Rational Exponents & Scientific Notation

Simplifying roots, radical equations, and powers of ten.

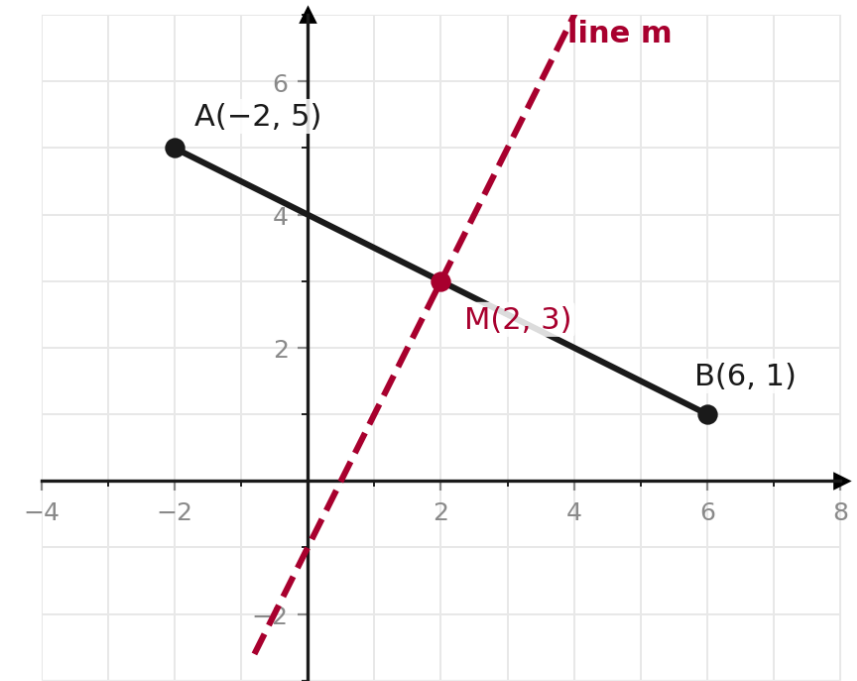
01

Lines

Slope, the four equation forms, and coordinate-geometry formulas.

In this section

- Key terms — English / Ελληνικά
- Explained theory with examples
- 8 worked problems (Hard / Very Hard)



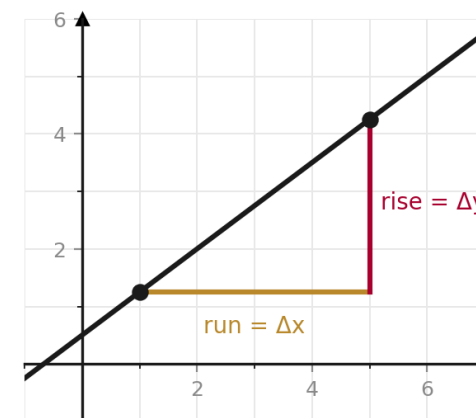
m is the perpendicular bisector of AB

Key Terminology

Term (English)	Ελληνικά
Slope	Κλίση
Y-intercept	Σημείο τομής με τον άξονα y
Slope-intercept form	Μορφή κλίσης-τομής
Parallel lines	Παράλληλες ευθείες
Perpendicular lines	Κάθετες ευθείες
Point-slope form	Μορφή σημείου-κλίσης

A The big picture

- A line = a constant rate of change.
- Every question reduces to slope, an equation form, or a coordinate formula.
- Hard versions hide the slope inside an unsimplified equation.



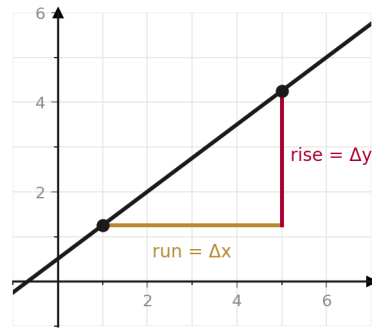
Theory Explained (1/3)

Slope

$$m = (y_2 - y_1) / (x_2 - x_1)$$

WHAT IT MEANS

- Slope = rise ÷ run = change in y per 1 unit of x.
- Positive → rises left to right; negative → falls.
- Horizontal: $m = 0$. Vertical: undefined.



Example

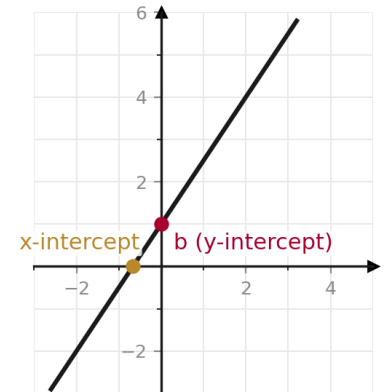
Through (1, 2) and (5, 10): $m = (10 - 2)/(5 - 1) = 8/4 = 2$.

Slope-Intercept Form

$$y = mx + b$$

WHAT IT MEANS

- m = slope (rate); b = y-intercept (value when $x = 0$).
- The fastest form for reading a graph or a word problem.
- In context: b is the starting amount, m the rate.



Example

Taxi: €3 fee + €2 per km → $y = 2x + 3$.

Theory Explained (2/3)

Point-Slope & Standard Form

$$y - y_1 = m(x - x_1) \quad Ax + By = C$$

WHAT IT MEANS

- Point-slope: use when you know one point and the slope.
- Standard form: slope = $-A/B$, y-intercept = C/B .
- Convert to $y = mx + b$ before comparing lines.

Example

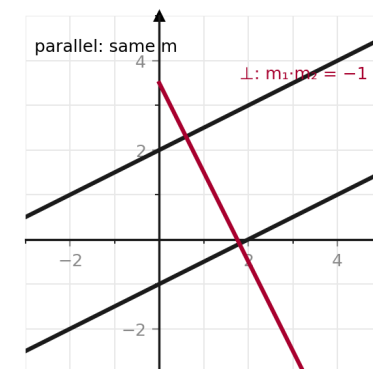
$$3x + 2y = 12 \rightarrow y = -1.5x + 6, \text{ so } m = -1.5, b = 6.$$

Parallel & Perpendicular

$$\text{parallel: } m_1 = m_2 \quad \text{perpendicular: } m_1 \cdot m_2 = -1$$

WHAT IT MEANS

- Parallel lines: same slope, different intercepts.
- Perpendicular: flip the fraction AND change the sign.
- Negative reciprocal of $2/3$ is $-3/2$.



Example

Line $\perp$ to $y = 4x + 1$ has slope $-1/4$.

Theory Explained (3/3)

Intercepts

x-intercept: set $y = 0$ y-intercept: set $x = 0$

WHAT IT MEANS

- Each intercept turns the equation into one-variable algebra.
- Standard form makes both intercepts quick: C/A and C/B .



Example

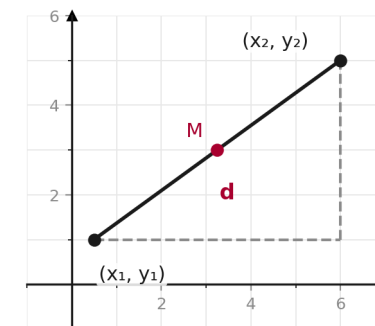
$2x + 5y = 20 \rightarrow$ x-int $(10, 0)$, y-int $(0, 4)$.

Midpoint & Distance

$$M = ((x_1+x_2)/2, (y_1+y_2)/2) \quad d = \sqrt{((x_2-x_1)^2 + (y_2-y_1)^2)}$$

WHAT IT MEANS

- Midpoint: average the x's, average the y's.
- Distance is the Pythagorean theorem on the grid.
- Two points on the same horizontal line:
 $d = |x_2 - x_1|$.



Example

$(1, 2)$ to $(7, 10)$: $M = (4, 6)$; $d = \sqrt{(36 + 64)} = 10$.

Watch Out



Never count grid squares for slope

- Axes are often scaled differently.
- Always use the actual coordinates.



Know the algebra, not only Desmos

- Many questions ask for an exact value a graph can't show.
- Use Desmos to check, not to replace the method.

Problem 1

Very Hard

Segment AB has endpoints A(-2, 5) and B(6, 1). Line m is the perpendicular bisector of segment AB. What is the y-intercept of line m?

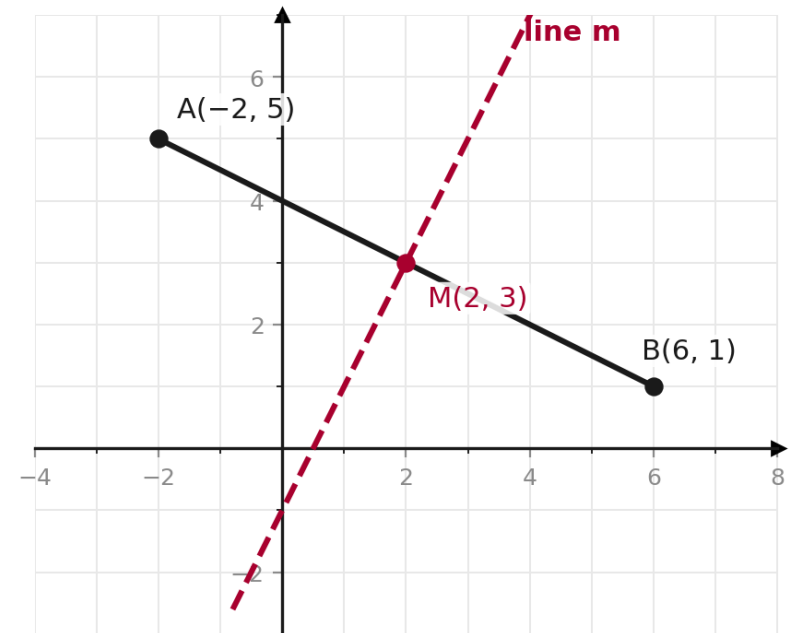
Formula / Rule

Midpoint formula and negative reciprocal slopes (perpendicular bisector).

Solution

1. Find the midpoint of segment AB: $M = ((-2+6)/2, (5+1)/2) = (4/2, 6/2) = (2, 3)$.
2. Find the slope of AB: $m_{AB} = (1 - 5)/(6 - (-2)) = -4/8 = -1/2$.
3. Line m is the perpendicular bisector, so its slope is the negative reciprocal of $-1/2$, which is 2.
4. Line m passes through the midpoint (2, 3) with slope 2. Use point-slope form: $y - 3 = 2(x - 2)$.
5. Simplify: $y - 3 = 2x - 4$, so $y = 2x - 1$.
6. The y-intercept occurs where $x = 0$: $y = 2(0) - 1 = -1$.

✓ **Answer:** y-intercept = -1, so line m crosses the y-axis at (0, -1).



m is the perpendicular bisector of AB



Problem 1 — Strategy & Elimination



Tip

For a perpendicular bisector problem, always compute the midpoint AND the perpendicular slope before writing the equation — both pieces are required.



Shortcut

Since only the y-intercept is needed, you can skip fully simplifying to standard form — just solve the point-slope equation for y and read off the constant term directly.



Trap

A common error is building the bisector line through one of the original endpoints (A or B) instead of the midpoint, which produces a different, incorrect y-intercept.



Desmos Tip

Plot A(-2,5) and B(6,1), graph the segment, then graph $y = 2x - 1$ and confirm it crosses AB at a right angle exactly at the midpoint (2,3).



Process of Elimination

Correct: **y-intercept = -1**, so line m crosses the y-axis at (0, -1).

Tempting wrong answer	Exactly what went wrong
y-intercept = 9	Used endpoint A(-2, 5) instead of the midpoint (2, 3) as the point for the bisector line, keeping the correct perpendicular slope of 2 but building the equation through the wrong point.
y-intercept = 4	Used the slope of segment AB itself (-1/2) instead of its negative reciprocal (2) when writing the bisector's equation through the midpoint.
y-intercept = 7	Made a sign error finding the slope of AB, computing $(1-5)/(6+2)$ as $4/8 = 1/2$ instead of $-4/8 = -1/2$, which flipped the perpendicular slope to -2 instead of 2.

Problem 2

Hard

Line p satisfies the equation $3(2y - x) = 4(y + 6)$. Line q is parallel to line p and passes through the point $(4, -1)$. Write the equation of line q in standard form $Ax + By = C$, where A, B, and C are integers and A is positive.

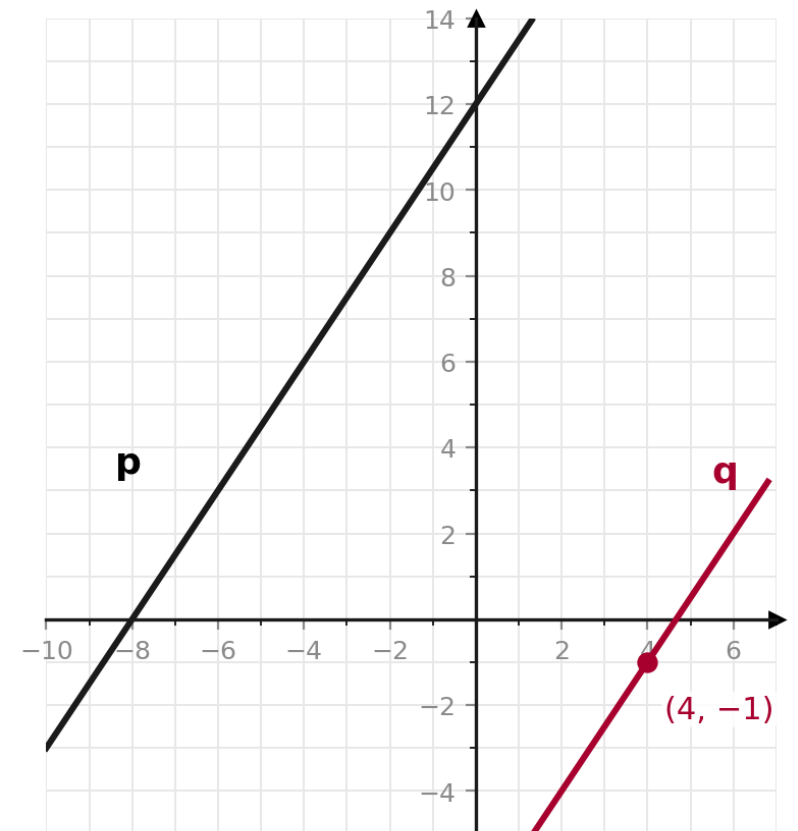
Formula / Rule

Simplify to slope-intercept form; parallel lines share slope; convert to standard form.

Solution

1. Distribute both sides of line p's equation: $6y - 3x = 4y + 24$.
2. Collect y-terms on one side: $6y - 4y = 3x + 24$, so $2y = 3x + 24$.
3. Solve for y: $y = (3/2)x + 12$, so line p has slope $3/2$.
4. Line q is parallel to line p, so line q also has slope $3/2$.
5. Use point-slope form with $(4, -1)$: $y - (-1) = (3/2)(x - 4)$, so $y + 1 = (3/2)(x - 4)$.
6. Multiply both sides by 2 to clear the fraction: $2y + 2 = 3(x - 4) = 3x - 12$.
7. Rearrange into standard form: $3x - 2y = 14$.

✓ **Answer:** $3x - 2y = 14$



$p \parallel q \rightarrow$ equal slopes



Problem 2 — Strategy & Elimination



Tip

When an equation is wrapped in parentheses or mixes x and y on both sides, distribute and collect like terms first — don't try to read off the slope before it's isolated.



Shortcut

Once you know line q needs the same slope as line p , you can skip fully solving line p for y — just read the slope directly as the ratio of coefficients from $2y = 3x + 24$ ($3/2$) without finishing the simplification.



Trap

A rushed student distributes only part of the equation or mishandles the coefficient when isolating y , landing on slope 3 instead of the correct slope $3/2$, which produces the wrong final equation for line q .



Desmos Tip

Graph $3(2y - x) = 4(y + 6)$ directly (Desmos accepts implicit equations) alongside $3x - 2y = 14$ and confirm the two lines never intersect and that the second passes through $(4, -1)$.



Process of Elimination

Correct: $3x - 2y = 14$

Tempting wrong answer	Exactly what went wrong
$3x - y = 13$	Mishandled isolating y from $2y = 3x + 24$, dividing only the x -term by 2 and not the constant, landing on slope 3 instead of $3/2$, then building line q 's equation with that wrong slope.
$3x - 2y = 10$	Used the correct slope $3/2$ but dropped the negative sign on -1 when substituting the point, writing $y - 1 = (3/2)(x - 4)$ instead of $y - (-1) = (3/2)(x - 4)$.
$3x - 2y = 6$	Forgot to multiply the constant by 3 when distributing on the right side, writing $3(x - 4)$ as $3x - 4$ instead of $3x - 12$, which shifted the final constant term.

Problem 3

Very Hard

Point P has coordinates $(k, 3)$, and point Q has coordinates $(-2, -3)$. The distance between P and Q is 10 units. What is the sum of all possible values of k ?

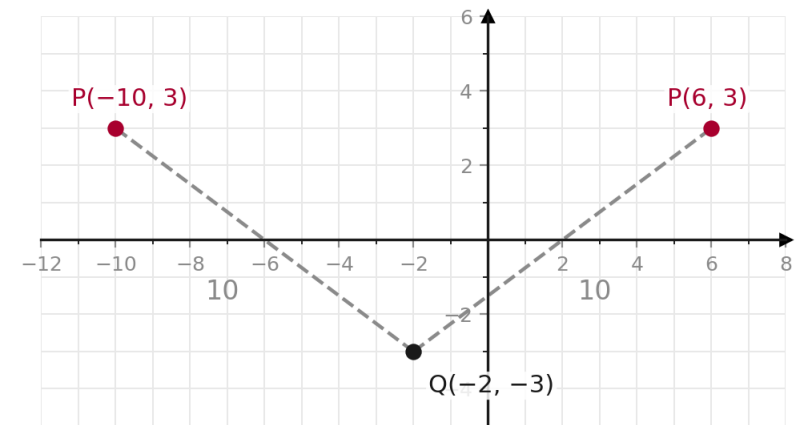
Formula / Rule

Distance formula, solved as a quadratic for an unknown coordinate.

Solution

1. Apply the distance formula: $d = \sqrt{((k - (-2)))^2 + (3 - (-3))^2}$.
2. Substitute $d = 10$: $10 = \sqrt{((k + 2))^2 + 6^2}$.
3. Square both sides: $100 = (k + 2)^2 + 36$.
4. Isolate the squared term: $(k + 2)^2 = 64$.
5. Take the square root of both sides, keeping both signs: $k + 2 = 8$ or $k + 2 = -8$.
6. Solve each case: $k = 6$ or $k = -10$.
7. Add the two possible values: $6 + (-10) = -4$.

✓ **Answer:** -4





Problem 3 — Strategy & Elimination



Tip

Whenever a distance-formula problem gives you the distance and asks for a coordinate, expect a squared equation with two valid solutions — solve it like any other quadratic and keep both roots unless the context rules one out.



Shortcut

Because the problem only asks for the SUM of the possible k-values, you don't need to solve both equations individually: the two roots of $(k+2)^2 = 64$ are symmetric around $k = -2$, so their sum is always $2(-2) = -4$.



Trap

Many students take the square root of $(k+2)^2 = 64$ and keep only the positive root ($k = 6$), missing $k = -10$ entirely and reporting an incomplete answer.



Desmos Tip

Graph the circle $(x + 2)^2 + (y + 3)^2 = 100$ centered at Q with radius 10, then find where it crosses the horizontal line $y = 3$ — the x-coordinates of the two intersection points are the two values of k.



Process of Elimination

Correct: -4

Tempting wrong answer	Exactly what went wrong
6	Took the square root of $(k+2)^2 = 64$ and kept only the positive root ($k+2 = 8$, so $k = 6$), discarding the negative branch entirely and reporting that single value instead of the sum of both roots.
4	Misread the x-coordinate of Q as +2 instead of -2, so assumed the two k-values were symmetric about $k = 2$ and reported their sum as $2(2) = 4$ instead of $2(-2) = -4$.
-60	Correctly found both roots $k = 6$ and $k = -10$, but multiplied them together instead of adding them, computing the product (-60) instead of the requested sum (-4).

Problem 4

Hard

A bike-rental kiosk in Rotterdam charges a fixed unlocking fee plus a constant rate per minute. Nikos rents a bike for 15 minutes and pays 4.75 euros total. His friend Elena rents a bike for 40 minutes and pays 9.00 euros total. Assuming cost is a linear function of rental time, how much would a 70-minute rental cost, in euros?

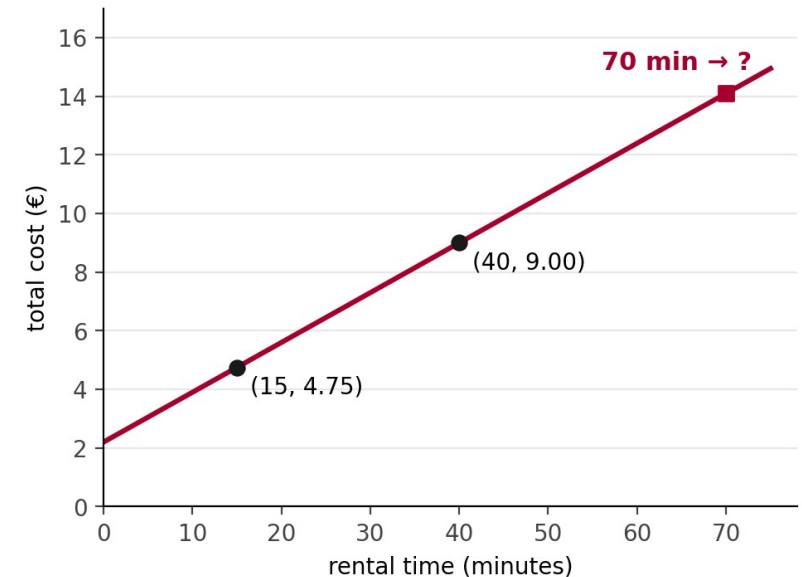
Formula / Rule

Build a linear model $C = mt + b$ from two data points, then evaluate it.

Solution

1. Translate the two rentals into coordinate points (time, cost): (15, 4.75) and (40, 9.00).
2. Find the rate (slope) using these points: $m = (9.00 - 4.75)/(40 - 15) = 4.25/25 = 0.17$ euros per minute.
3. Use point-slope form with (15, 4.75) to find the fixed fee (y-intercept): $4.75 = 0.17(15) + b$.
4. Compute $0.17(15) = 2.55$, so $4.75 = 2.55 + b$, giving $b = 2.20$ euros.
5. Write the full cost model: $C = 0.17t + 2.20$.
6. Substitute $t = 70$: $C = 0.17(70) + 2.20 = 11.90 + 2.20$.
7. Add: $C = 14.10$ euros.

✓ **Answer: 14.10 euros**





Problem 4 — Strategy & Elimination



Tip

In a linear word problem, immediately convert each described scenario into an (input, output) ordered pair — once you have two points, it's a standard slope-and-intercept problem.



Shortcut

No shortcut beats setting up the two points and solving for m and b directly — this is already the fastest reliable path for a two-point linear word problem.



Trap

A student who only computes the per-minute rate and multiplies it by 70 (getting $0.17 \times 70 = 11.90$) forgets the fixed unlocking fee entirely and arrives at the wrong total.



Desmos Tip

Enter the two points (15, 4.75) and (40, 9.00) in a Desmos table, graph $y = 0.17x + 2.20$ through them, and read the y-value at $x = 70$ directly off the graph.



Process of Elimination

Correct: **14.10 euros**

Tempting wrong answer	Exactly what went wrong
11.90 euros	Computed only the per-minute rate (0.17) and multiplied it by 70, forgetting to add the fixed unlocking fee of 2.20 euros.
15.75 euros	Rounded the per-minute rate from 0.17 to 0.2 too early, then carried that rounding error through the intercept ($b = 1.75$) and the final calculation instead of keeping the exact rate.
11.55 euros	Misread the question as asking for the cost of 70 additional minutes beyond Nikos's original 15-minute rental, substituting $t = 55$ instead of $t = 70$ into the cost model.

Problem 5

Hard

Triangle ABC has vertices A(-4, -3), B(6, -3), and C(2, 9). Side AB is horizontal, and the slope of side BC is -3. Find the perimeter of triangle ABC, in exact simplified radical form.

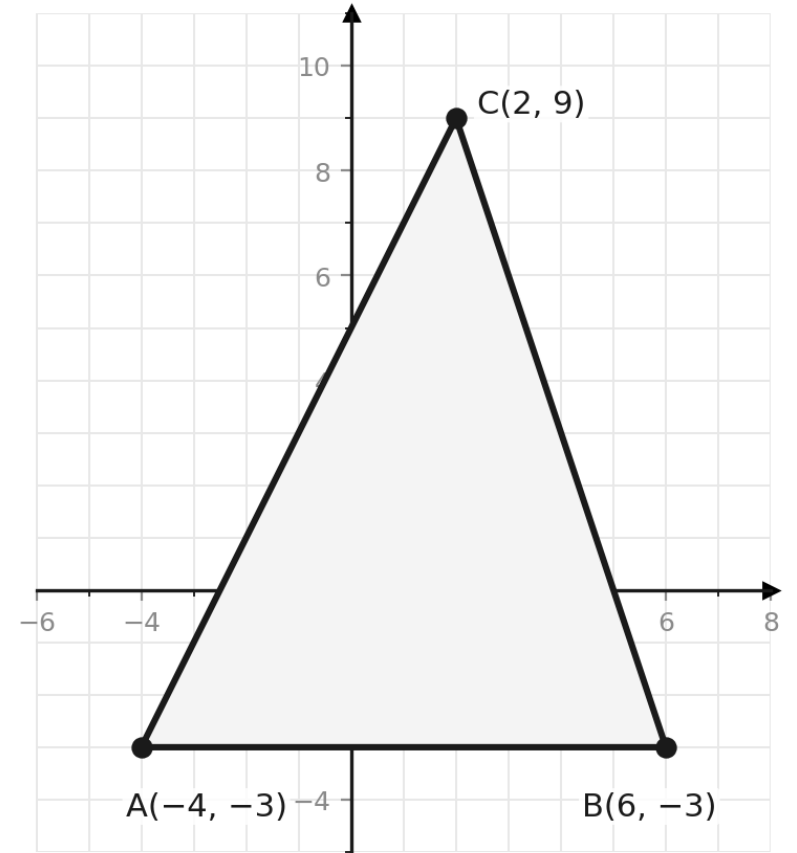
Formula / Rule

Distance formula applied to each side of a triangle to find perimeter.

Solution

1. Since A and B share the same y-coordinate (-3), side AB is horizontal, so its length is simply the difference in x-coordinates: $AB = |6 - (-4)| = 10$.
2. The slope of BC is given but not needed to find lengths — apply the distance formula directly to B(6,-3) and C(2,9): $BC = \sqrt{(2-6)^2 + (9-(-3))^2}$.
3. Compute inside the square root: $(2-6)^2 = 16$ and $(9-(-3))^2 = 144$, so $BC = \sqrt{16 + 144} = \sqrt{160}$.
4. Simplify $\sqrt{160} = \sqrt{16 \times 10} = 4\sqrt{10}$.
5. Apply the distance formula to A(-4,-3) and C(2,9): $AC = \sqrt{(2-(-4))^2 + (9-(-3))^2}$.
6. Compute inside the square root: $(2-(-4))^2 = 36$ and $(9-(-3))^2 = 144$, so $AC = \sqrt{36 + 144} = \sqrt{180}$.
7. Simplify $\sqrt{180} = \sqrt{36 \times 5} = 6\sqrt{5}$.
8. Add all three sides for the perimeter: $\text{Perimeter} = 10 + 4\sqrt{10} + 6\sqrt{5}$.

✓ **Answer:** $10 + 4\sqrt{10} + 6\sqrt{5}$ units (approximately 36.0 units)





Problem 5 — Strategy & Elimination



Tip

For a perimeter or side-length problem, apply the distance formula to each pair of vertices directly — don't try to use slope information, which only tells you direction, not length.



Shortcut

Recognizing that AB is horizontal (equal y-coordinates) lets you find that side by simple subtraction of x-values instead of running the full distance formula, saving a step on one of the three sides.



Trap

The slope of BC (-3) is a tempting piece of data, and a student might try to fold it into a length calculation (treating it as a ratio of legs), which doesn't give the actual side length and wastes time or produces a wrong value.



Desmos Tip

Plot the three vertices in Desmos, draw the triangle with the polygon feature, and use the distance measurement between each pair of points to confirm $BC = \sqrt{160}$ and $AC = \sqrt{180}$ numerically.



Process of Elimination

Correct: $10 + 4\sqrt{10} + 6\sqrt{5}$ units (approximately 36.0 units)

Tempting wrong answer	Exactly what went wrong
$2 + 4\sqrt{10} + 6\sqrt{5}$ units (approximately 28.1 units)	Misread vertex A's x-coordinate as 4 instead of -4, computing $AB = 6 - 4 = 2$ instead of the correct $AB = 6 - (-4) = 10$.
$10 + 4\sqrt{10} + 3\sqrt{2}$ units (approximately 27.9 units)	When computing AC, added the coordinate differences directly instead of squaring them first: $\sqrt{(6 + 12)} = \sqrt{18} = 3\sqrt{2}$, instead of correctly squaring to get $\sqrt{(36 + 144)} = \sqrt{180} = 6\sqrt{5}$.
$10 + 4\sqrt{10}$ units (approximately 22.6 units)	Computed AB and BC correctly but forgot to include side AC in the perimeter sum, stopping after only two of the three sides.

Problem 6

Very Hard

Line r is given by $kx + 3y = 12$, and line s is given by $2x - 5y = 10$. If line r is perpendicular to line s , what is the x -intercept of line r ?

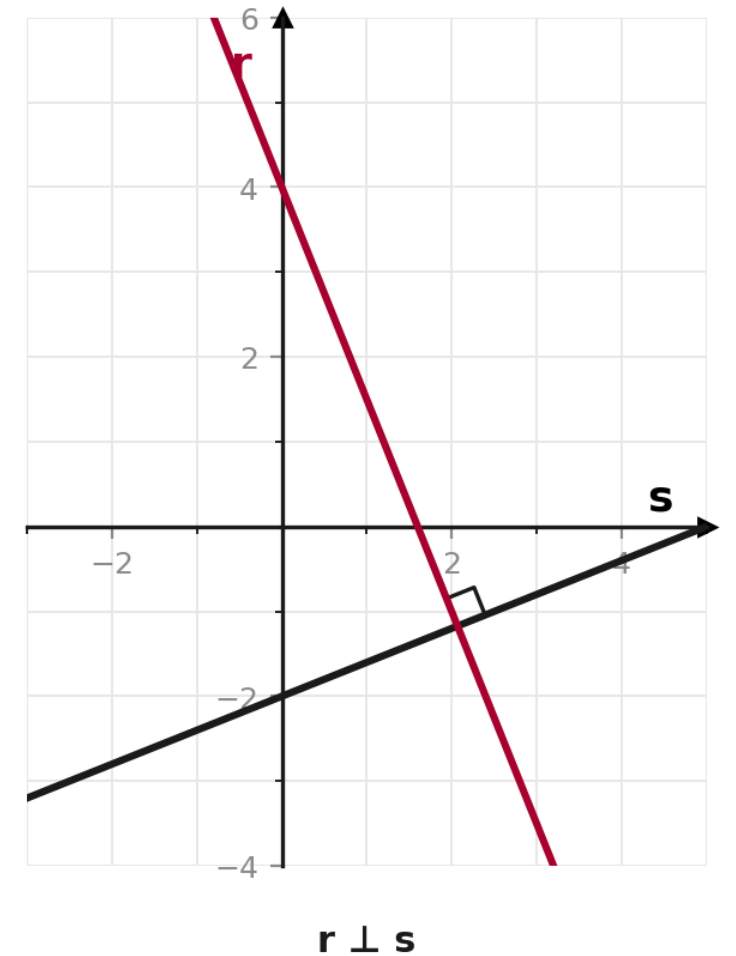
Formula / Rule

Standard-form slope $(-A/B)$, perpendicular negative reciprocal, then x -intercept.

Solution

1. Find the slope of line s from standard form $Ax + By = C$, where slope $= -A/B$: slope of $s = -2/(-5) = 2/5$.
2. Since line r is perpendicular to line s , its slope must be the negative reciprocal of $2/5$, which is $-5/2$.
3. Find the slope of line r in terms of k from its standard form $kx + 3y = 12$: slope of $r = -k/3$.
4. Set the two expressions for line r 's slope equal: $-k/3 = -5/2$.
5. Solve for k : $k/3 = 5/2$, so $k = 15/2$.
6. Substitute $k = 15/2$ back into line r 's equation: $(15/2)x + 3y = 12$.
7. Multiply every term by 2 to clear the fraction: $15x + 6y = 24$, which simplifies (divide by 3) to $5x + 2y = 8$.
8. Find the x -intercept by setting $y = 0$: $5x = 8$, so $x = 8/5$.

✓ **Answer:** x -intercept = $(8/5, 0)$





Problem 6 — Strategy & Elimination



Tip

When a line's equation contains an unknown coefficient and a perpendicularity or parallelism condition, first write both slopes in terms of A and B (slope = $-A/B$ for standard form), then set up one equation to solve for the unknown before touching the rest of the problem.



Shortcut

There isn't a faster route than solving for k first — the x -intercept genuinely depends on k , so both steps (solve for k , then find the intercept) are necessary in this order.



Trap

A student who confuses the perpendicular condition with the parallel condition sets $-k/3 = 2/5$ (matching slopes instead of using negative reciprocals), which yields the wrong value of k and a wrong final intercept.



Desmos Tip

After solving for k , graph $5x + 2y = 8$ and $2x - 5y = 10$ together in Desmos and confirm visually that the lines cross at a right angle and that the first line crosses the x -axis at $8/5 = 1.6$.



Process of Elimination

Correct: **x -intercept** = $(8/5, 0)$

Tempting wrong answer	Exactly what went wrong
x-intercept = $(-10, 0)$	Confused the perpendicular condition with the parallel condition, setting $-k/3 = 2/5$ (matching slopes) instead of $-k/3 = -5/2$ (negative reciprocal), which gave $k = -6/5$ and a completely different line.
x-intercept = $(72/5, 0)$	Correctly set up $k/3 = 5/2$ but solved it by dividing $5/2$ by 3 instead of multiplying by 3, getting $k = 5/6$ instead of the correct $k = 15/2$.
x-intercept = $(0, 4)$	Found the correct simplified line $5x + 2y = 8$ but set $x = 0$ instead of $y = 0$, computing the y -intercept $(0, 4)$ and mislabeling it as the x -intercept.

Problem 7

Very Hard

For what value of k does the system of equations $4x - 6y = 10$ and $kx + 9y = -20$ have no solution?

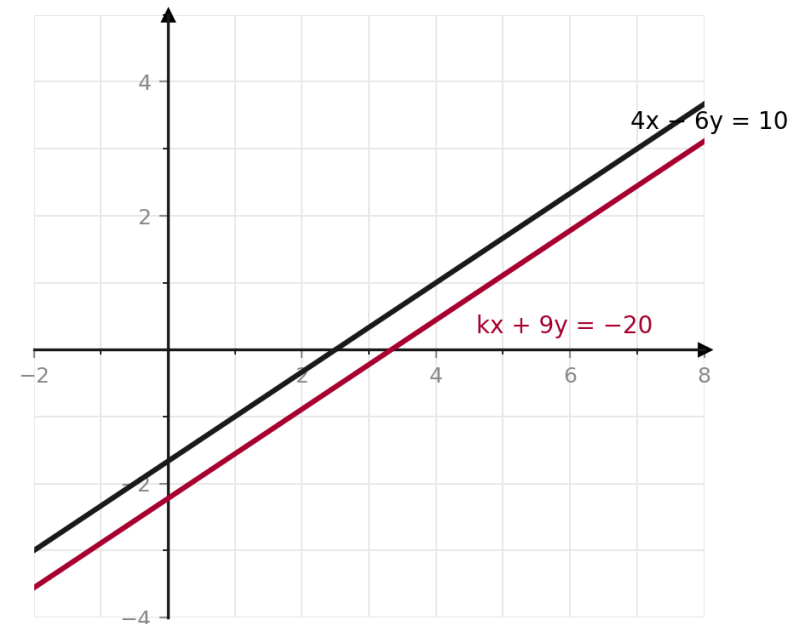
Formula / Rule

Standard-form slope ($-A/B$); parallel lines share slope but must have different constants for a true 'no solution' case.

Solution

1. Find the slope of the first line from standard form: $4x - 6y = 10$ has slope $= -A/B = -4/(-6) = 2/3$.
2. For the system to have no solution, the two lines must be parallel (equal slopes) but not identical. Find the slope of the second line in terms of k : $kx + 9y = -20$ has slope $= -k/9$.
3. Set the slopes equal: $-k/9 = 2/3$.
4. Solve for k : multiply both sides by 9: $-k = 6$, so $k = -6$.
5. Check that the lines aren't identical when $k = -6$ (which would give infinitely many solutions instead of no solution): substitute $k = -6$, so the second line becomes $-6x + 9y = -20$.
6. Scale the first line ($4x - 6y = 10$) by -1.5 to match the second line's x -coefficient: $-6x + 9y = -15$.
7. Since -15 does not equal -20 , the two lines are parallel but distinct, confirming the system truly has no solution rather than infinitely many.
8. Therefore $k = -6$.

✓ **Answer:** $k = -6$



no solution → parallel lines (equal slopes)



Problem 7 — Strategy & Elimination



Tip

Whenever a system-of-equations problem asks for 'no solution', translate it as 'same slope, different intercept' — find both lines' slopes from standard form using $-A/B$ and set them equal.



Shortcut

Skip computing each slope separately — cross-multiply the coefficients directly (parallel condition $A_1 \times B_2 = A_2 \times B_1$): $4(9) = k(-6)$, so $36 = -6k$, giving $k = -6$ in one line.



Trap

The most common error is misreading the standard-form slope formula $-A/B$ when B is already negative (here $B = -6$): students often compute $-4/6 = -2/3$ instead of correctly canceling the double negative to get $2/3$, which flips the sign of the final answer.



Desmos Tip

Graph $4x - 6y = 10$ and $kx + 9y = -20$ for a few trial k -values near -6 and watch the second line rotate until it looks parallel to the first, then confirm $k = -6$ makes the lines never touch.



Process of Elimination

Correct: $k = -6$

Tempting wrong answer	Exactly what went wrong
$k = 6$	Miscomputed the slope of the first line from standard form as $-4/6 = -2/3$ (mishandling the double negative from $B = -6$), instead of the correct $2/3$, which flipped the sign of the final answer.
$k = 13.5$	Used the perpendicular condition (product of slopes = -1) instead of the parallel/no-solution condition (equal slopes), solving $(2/3)(-k/9) = -1$ instead of $-k/9 = 2/3$.
$k = -9$	Made an arithmetic slip computing $9 \times (2/3)$, getting 9 instead of 6 , so $-k = 9$ was solved as $k = -9$ instead of the correct $k = -6$.

Problem 8

Hard

Line w passes through the points $(2, 5)$ and $(6, -3)$. Line w crosses the x -axis at point R and the y -axis at point S . What is the area of triangle ORS , where O is the origin?

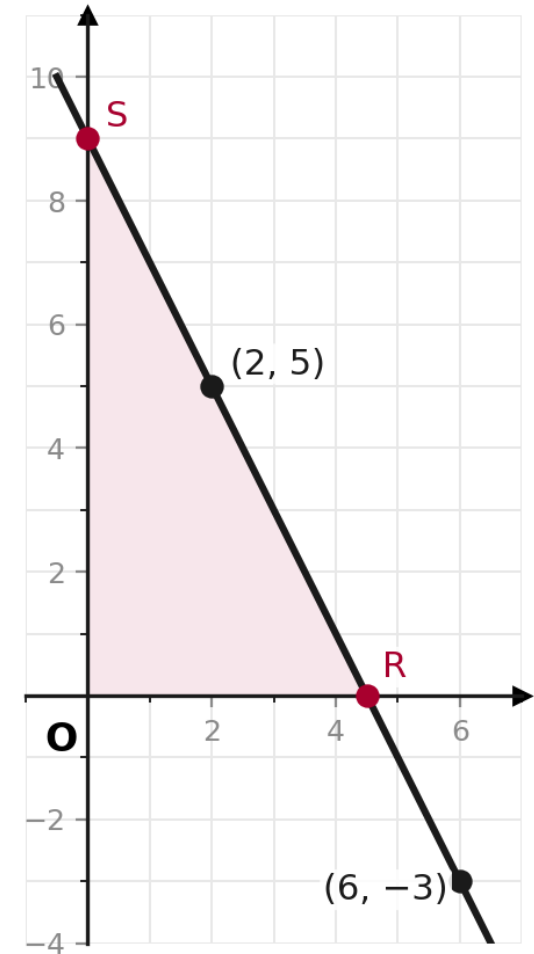
Formula / Rule

Slope formula, point-slope form, x - and y -intercepts, triangle area = $(1/2)(\text{base})(\text{height})$.

Solution

1. Find the slope of line w using the two given points, keeping the same point order in numerator and denominator: $m = (-3 - 5)/(6 - 2) = -8/4 = -2$.
2. Use point-slope form with $(2, 5)$: $y - 5 = -2(x - 2)$.
3. Simplify: $y - 5 = -2x + 4$, so $y = -2x + 9$.
4. Find the y -intercept (point S) by setting $x = 0$: $y = -2(0) + 9 = 9$, so $S = (0, 9)$.
5. Find the x -intercept (point R) by setting $y = 0$: $0 = -2x + 9$, so $2x = 9$, $x = 9/2 = 4.5$, so $R = (4.5, 0)$.
6. Triangle ORS is a right triangle with legs OS (along the y -axis, length 9) and OR (along the x -axis, length 4.5), since the axes meet at the origin at a right angle.
7. Area = $(1/2)(\text{base})(\text{height}) = (1/2)(4.5)(9) = (1/2)(40.5) = 20.25$.

✓ **Answer:** Area = 20.25 square units ($81/4$).





Problem 8 — Strategy & Elimination



Tip

When a triangle is formed by a line and both coordinate axes, find the line's x- and y-intercepts first — those two intercepts are automatically the legs of a right triangle with the origin, so the area formula applies directly.



Shortcut

You can't skip any of the three formulas here, but you can save a step by reading the y-intercept b directly from $y = mx + b$ using either point ($5 = -2(2) + b$, so $b = 9$) instead of writing out full point-slope form first.



Trap

The most common slip is losing track of point order when computing slope from two points — pairing the y-values in one order but the x-values in the reverse order silently flips the sign of the slope, which then shifts both intercepts and produces a completely different (and much smaller) triangle.



Desmos Tip

Graph the line through $(2,5)$ and $(6,-3)$ in Desmos — it will automatically display where the line crosses each axis, so you can read R and S directly and check the area with Desmos's polygon/area tool on triangle O, R, S .



Process of Elimination

Correct: **Area = 20.25 square units (81/4).**

Tempting wrong answer	Exactly what went wrong
Area = 0.25 (1/4) square units	Mismatched the point order in the slope formula, computing $m = (-3-5)/(2-6) = 2$ instead of pairing coordinates in the same order ($m = (-3-5)/(6-2) = -2$), which flipped the sign of the slope and shrank both intercepts.
Area = 40.5 square units	Found both intercepts correctly ($R = (4.5, 0)$, $S = (0, 9)$) but forgot to multiply by $1/2$ in the triangle area formula, computing base $\times$ height instead of $(1/2)(\text{base})(\text{height})$.
Area = 22.5 square units	Correctly found the line $y = -2x + 9$, but rounded the x-intercept $9/2 = 4.5$ up to 5 instead of keeping the exact value, then computed $(1/2)(5)(9) = 22.5$ instead of $(1/2)(4.5)(9) = 20.25$.

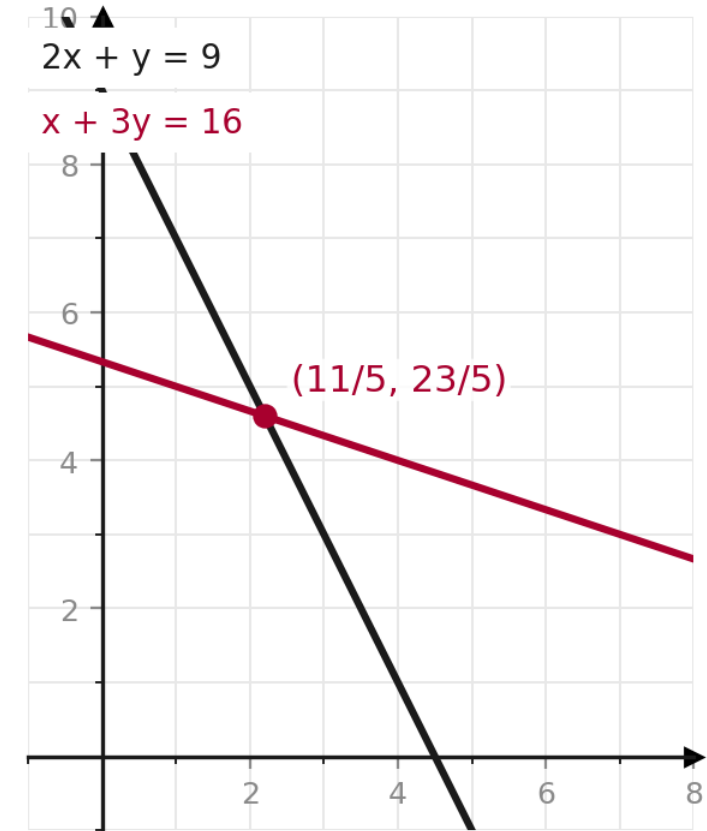
02

Systems of Linear Equations

Solving pairs of equations and counting their solutions.

In this section

- Key terms — English / Ελληνικά
- Explained theory with examples
- 8 worked problems (Hard / Very Hard)



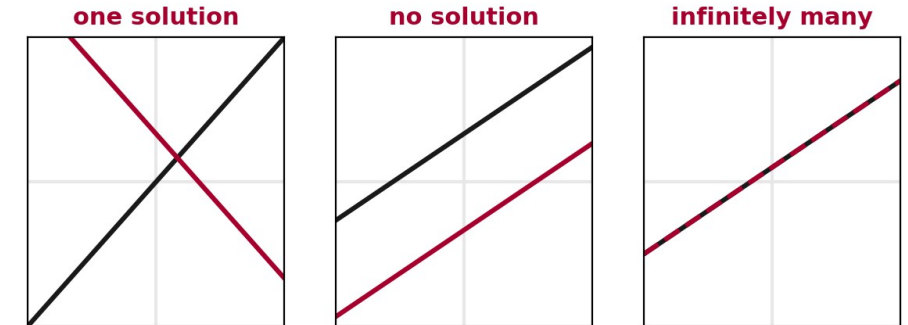
one intersection in Quadrant I

Key Terminology

Term (English)	Ελληνικά
System of equations	Σύστημα εξισώσεων
Solution (of a system)	Λύση συστήματος
Substitution method	Μέθοδος αντικατάστασης
Elimination method	Μέθοδος απαλοιφής
No solution / infinite solutions	Καμία λύση / άπειρες λύσεις

A ㄱ The big picture

- Two equations, two unknowns — find where the lines meet.
- Hard questions ask how MANY solutions exist.
- Often you need a combination (like $3x + 4y$), not x and y .



Theory Explained (1/3)

Substitution

solve one equation for a variable → plug in

WHAT IT MEANS

- Best when a variable already has coefficient 1.
- Leaves one equation with one unknown.
- Distribute carefully after substituting.



Example

$$y = 9 - 2x \text{ into } x + 3y = 16 \rightarrow x + 27 - 6x = 16 \rightarrow x = 11/5.$$

Elimination

scale → add or subtract → one variable cancels

WHAT IT MEANS

- Multiply one or both equations so a coefficient matches.
- Add if signs are opposite, subtract if they match.
- Often faster than substitution on the SAT.



Example

$$(2x + y = 9) + (x + 3y = 16) \rightarrow 3x + 4y = 25 \text{ directly.}$$

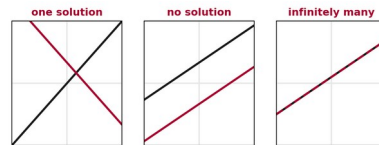
Theory Explained (2/3)

Number of Solutions

one · none · infinitely many

WHAT IT MEANS

- One: different slopes → lines cross once.
- None: same slope, different intercept → parallel.
- Infinite: same line (every coefficient in the same ratio).



Example

$2x + 3y = 5$ and $4x + 6y = 12 \rightarrow$ parallel $\rightarrow$ no solution.

The Ratio Test

$a_1/a_2 = b_1/b_2 \neq c_1/c_2 \rightarrow$ none; $a_1/a_2 = b_1/b_2 = c_1/c_2 \rightarrow$ infinite

WHAT IT MEANS

- Write both equations in $Ax + By = C$ form first.
- Compare x-coefficients, y-coefficients, constants.
- Solves 'for what value of k' questions in one line.



Example

$6x - 9y = 3k$, $8x - 12y = 20$: $6/8 = 3k/20 \rightarrow k = 5$.

Theory Explained (3/3)

Word Problems

one equation per relationship

WHAT IT MEANS

- Name variables clearly (s = standard price).
- Translate 'more than', 'twice' carefully: $e = 2s + 15$.
- Answer exactly what is asked — total, difference, or one value.



Example

3 standard + 2 express = €205 $\rightarrow 3s + 2e = 205$.

Watch Out



Answer the question asked

- $x + y$, $x - y$, or $3x + 4y$ are common targets.
- Look for a shortcut combination before solving fully.



Check the form before the ratio test

- Move every term into $Ax + By = C$ first.
- A sign slip flips the answer.

Problem 1

Very Hard

Consider the system $2x = 11 - 5y$ and $ky = 30 - 6x$, where k is a constant. For what value of k does this system have no solution?

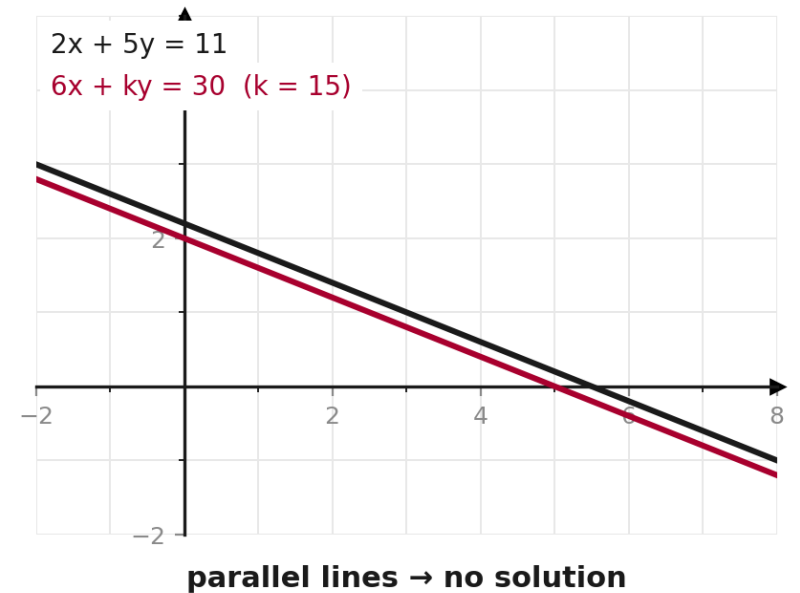
Formula / Rule

No-solution condition for a linear system: $A_1/A_2 = B_1/B_2$ does not equal C_1/C_2 in standard form $Ax + By = C$.

Solution

1. Rewrite the first equation in standard form: $2x + 5y = 11$.
2. Rewrite the second equation in standard form: $6x + ky = 30$.
3. For no solution, the coefficient ratios must match: $2/6 = 5/k$.
4. Cross-multiply: $2k = 30$, so $k = 15$.
5. Check that the constant ratio is genuinely different (required for no solution, not infinite solutions): $C_1/C_2 = 11/30$, while $A_1/A_2 = 2/6 = 1/3 = 10/30$.
6. Since $11/30$ does not equal $10/30$, the lines are parallel but distinct at $k = 15$, confirming no solution.

✓ **Answer:** $k = 15$





Problem 1 — Strategy & Elimination



Tip

Before comparing ratios, rearrange every equation into $Ax + By = C$ form — don't try to compare coefficients while variables and constants are mixed on both sides.



Trap

A student who finds $k = 15$ from the coefficient ratio alone often stops and assumes matching slopes automatically means infinitely many solutions; they must also check that the constant ratio ($11/30$) differs from the coefficient ratio ($1/3$) to confirm this is actually a no-solution case.



Shortcut

Once both equations are in standard form, skip solving the system entirely — just cross-multiply the x- and y-coefficient ratio ($2/6 = 5/k$) to get $k = 15$ in one line.



Desmos Tip

Graph $2x + 5y = 11$ and $6x + 15y = 30$ (solved for y) in Desmos and confirm the two lines are visibly parallel and never touch.



Process of Elimination

Correct: **$k = 15$**

Tempting wrong answer	Exactly what went wrong
$k = -15$	Rearranged $ky = 30 - 6x$ incorrectly as $6x - ky = 30$ (sign error on the y-term) instead of $6x + ky = 30$, which flips the sign in the ratio and produces $k = -15$ instead of $k = 15$.
$k = 12/5$	Mispaired the coefficients when writing the ratio, using $5/6 = 2/k$ (y-coefficient of equation 1 over x-coefficient of equation 2) instead of the correct $2/6 = 5/k$, mixing up which coefficients correspond to which equation.
$k = 5/3$	Cross-multiplied the proportion $2/6 = 5/k$ backwards, computing $2 \times 5 = 6 \times k$ instead of $2 \times k = 6 \times 5$, giving $k = 5/3$ instead of 15.

Problem 2

Hard

In the system below, the two lines intersect at a point located in Quadrant I of the xy -plane. $2x + y = 9$ and $x + 3y = 16$. What is the value of $3x + 4y$?

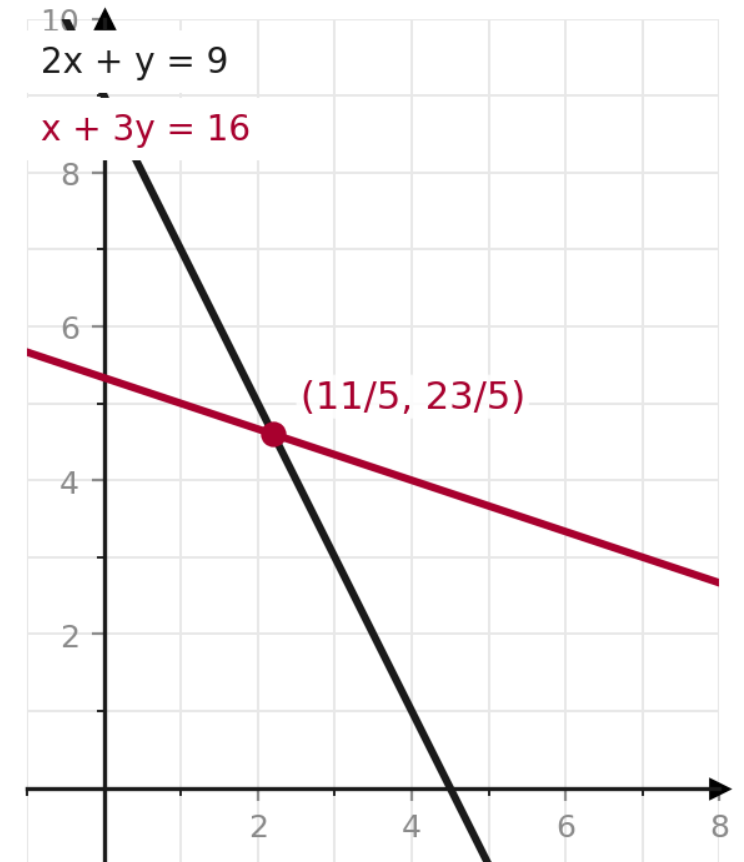
Formula / Rule

Linear combination shortcut: add or scale the two equations directly to build the target expression without solving for x and y .

Solution

1. Notice the target expression $3x + 4y$ can be built by adding the two equations as given: $(2x + y) + (x + 3y) = 3x + 4y$.
2. Add the left sides: $2x + y + x + 3y = 3x + 4y$.
3. Add the right sides: $9 + 16 = 25$.
4. Therefore $3x + 4y = 25$ directly, with no need to solve for x and y individually.
5. As a check, solve the system fully: from equation 1, $y = 9 - 2x$; substitute into equation 2: $x + 3(9 - 2x) = 16$, so $x + 27 - 6x = 16$, giving $-5x = -11$ and $x = 11/5$.
6. Then $y = 9 - 2(11/5) = 9 - 22/5 = 23/5$, and both values are positive, matching the Quadrant I detail (which is not needed to answer the question).
7. Confirm: $3(11/5) + 4(23/5) = 33/5 + 92/5 = 125/5 = 25$, matching the shortcut answer.

✓ **Answer:** $3x + 4y = 25$



one intersection in Quadrant I



Problem 2 — Strategy & Elimination



Tip

Before solving a system fully, check whether the target expression is already a simple sum, difference, or scaled combination of the two given equations — if it is, add or subtract directly instead of finding x and y .



Trap

A student who sees a two-variable system and a geometric detail like 'Quadrant I' often assumes they must fully solve for x and y (landing on messy fractions like $x = 11/5$) before computing $3x + 4y$, wasting time and risking an arithmetic slip — the quadrant location is never actually used.



Shortcut

Just add the two equations as given — their sum already equals $3x + 4y$, so the answer appears in one line without ever isolating x or y .



Desmos Tip

Graph both lines in Desmos, find the intersection point $(2.2, 4.6)$, then compute $3(2.2) + 4(4.6) = 25$ directly from the coordinates as a check.



Process of Elimination

Correct: $3x + 4y = 25$

Tempting wrong answer	Exactly what went wrong
-7	Subtracted the two equations instead of adding them, computing $(2x + y) - (x + 3y) = 9 - 16 = -7$, then mistakenly reported -7 as the value of $3x + 4y$ instead of recognizing only addition produces that target expression.
-19	While solving the system fully as a check, distributed $3(9 - 2x)$ incorrectly as $27 - 2x$ instead of $27 - 6x$, producing wrong values $x = 11$ and $y = -13$, and an incorrect final combination of $3(11) + 4(-13) = -19$.
26	Misled by the Quadrant I detail into assuming the coordinates must be 'nice' integers, rounded $x = 11/5$ to 2 and used $y = 9 - 2(2) = 5$, then computed $3(2) + 4(5) = 26$ instead of using the exact fractional solution.

Problem 3

Very Hard

Solve the system for the ordered pair (x, y) : $x/2 + 5y/6 = 9/2$ and $4x/3 - y = 7/3$.

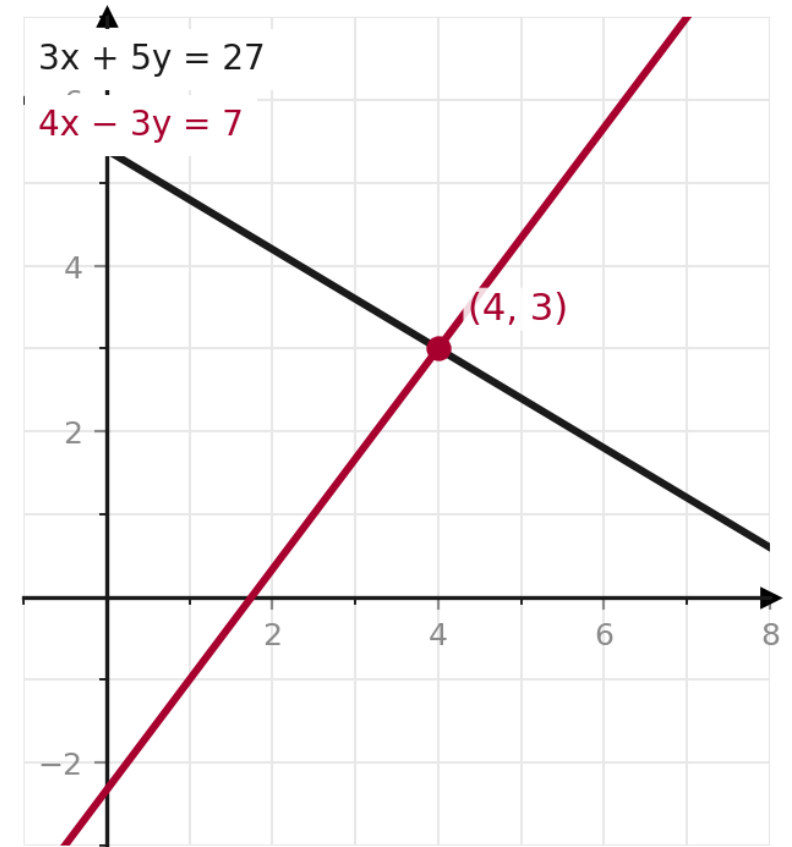
Formula / Rule

Clear fractions by multiplying each equation by its LCD, then eliminate a variable by multiplying both resulting equations by different constants.

Solution

1. Clear the first equation by multiplying every term by 6 (the LCD of 2 and 6): $3x + 5y = 27$.
2. Clear the second equation by multiplying every term by 3 (the LCD of 3 and 1): $4x - 3y = 7$.
3. To eliminate y , find a common multiple of 5 and 3 (which is 15): multiply the first cleared equation by 3 to get $9x + 15y = 81$.
4. Multiply the second cleared equation by 5 to get $20x - 15y = 35$.
5. Add the two new equations: $9x + 15y + 20x - 15y = 81 + 35$, so $29x = 116$.
6. Solve for x : $x = 116/29 = 4$.
7. Substitute $x = 4$ into $4x - 3y = 7$: $16 - 3y = 7$, so $-3y = -9$ and $y = 3$.

✓ **Answer:** $(x, y) = (4, 3)$





Problem 3 — Strategy & Elimination



Tip

Clear every fraction by multiplying each equation by its own LCD before attempting elimination — never try to eliminate variables while denominators are still in play.



Trap

A student often tries to eliminate y by scaling only one cleared equation (since 5 and -3 share no simple single multiplier), instead of multiplying BOTH cleared equations (by 3 and by 5) to reach matching y -coefficients of 15, which leads to leftover fractions and arithmetic errors.



Shortcut

No shortcut beats clearing fractions first — trying to eliminate a variable while fractions remain multiplies the chance of arithmetic slips, so multiplying by the LCD immediately is itself the fastest path.



Desmos Tip

Graph $x/2 + 5y/6 = 9/2$ and $4x/3 - y = 7/3$ directly in Desmos and use the intersection point to confirm the solution $(4, 3)$.



Process of Elimination

Correct: $(x, y) = (4, 3)$

Tempting wrong answer	Exactly what went wrong
$(x, y) = (4, -23/3)$	After correctly finding $x = 4$ and substituting into $4x - 3y = 7$ to get $16 - 3y = 7$, subtracted in the wrong direction (computed $-3y = 7 + 16$ instead of $-3y = 7 - 16$), giving $y = -23/3$ instead of 3.
$(x, y) = (3, 4)$	Correctly computed the two numeric values 4 and 3 through elimination but swapped which value is x and which is y when writing the final ordered pair.
$(x, y) = (62/23, 87/23)$	Cleared $4x/3 - y = 7/3$ by multiplying only the x -term and the constant by 3, forgetting to also multiply the y -term by 3, producing the wrong equation $4x - y = 7$ instead of $4x - 3y = 7$, which throws off the entire solution.

Problem 4

Hard

The system $5x - 2y = 16$ and $bx + 3y = -1$ has a solution in which $y = -3$. What is the value of $b - x$?

Formula / Rule

Substitute the known value into the equation without an unknown constant first, then use the result to solve for the unknown constant in the other equation.

First move

When one equation contains an unknown constant and the other doesn't, always use the equation without the unknown constant first to pin down the known variable before touching the equation that has two unknowns.

Solution

1. Substitute $y = -3$ into the first equation, which has no unknown constant: $5x - 2(-3) = 16$.
2. Simplify: $5x + 6 = 16$, so $5x = 10$ and $x = 2$.
3. Substitute $x = 2$ and $y = -3$ into the second equation: $b(2) + 3(-3) = -1$.
4. Simplify: $2b - 9 = -1$, so $2b = 8$ and $b = 4$.
5. Compute the requested expression: $b - x = 4 - 2 = 2$.

 **Answer:** $b - x = 2$



Problem 4 — Strategy & Elimination



Tip

When one equation contains an unknown constant and the other doesn't, always use the equation without the unknown constant first to pin down the known variable before touching the equation that has two unknowns.



Trap

A student who plugs $y = -3$ into the second equation first gets stuck with two unknowns (b and x) in one equation and often guesses or mishandles the algebra; working equation 1 first avoids this entirely.



Shortcut

There's no way around the two substitutions here, but always resolving the 'clean' equation (the one without the unknown constant) first keeps you from ever juggling two unknowns at once.



Desmos Tip

After finding $x = 2$, graph $bx + 3y = -1$ as $y = (-1 - bx)/3$ and adjust b until the line passes through $(2, -3)$ to visually confirm $b = 4$.



Process of Elimination

Correct: $b - x = 2$

Tempting wrong answer	Exactly what went wrong
-7	Correctly found $x = 2$ but substituted $y = -3$ into the second equation with a sign flip, writing $2b + 3(3) = -1$ instead of $2b + 3(-3) = -1$, getting $b = -5$ instead of $b = 4$, then computed $-5 - 2 = -7$.
-2	Correctly found $b = 4$ and $x = 2$ but computed $x - b$ instead of $b - x$, reversing the order the question asks for.
6	Correctly found $b = 4$ and $x = 2$ but added the two values instead of subtracting, computing $b + x = 6$ instead of $b - x = 2$.

Problem 5

Hard

Cycladic Ferries sells standard and express tickets for its Piraeus-Naxos route. The express ticket costs 15 euros more than twice the price of a standard ticket. Last week, a travel agency bought 3 standard tickets and 2 express tickets for a total of 205 euros. What is the combined price, in euros, of one standard ticket and one express ticket?

Formula / Rule

Translate verbal relationships into algebraic equations, then substitute to reduce the system to one variable.

First move

Translate the comparison sentence into an equation first ($e = 2s + 15$), then substitute it directly into the purchase equation instead of building two equations and eliminating — it turns the system into a single-variable equation immediately.

Solution

1. Let s = the price of a standard ticket and e = the price of an express ticket.
2. Translate 'the express ticket costs 15 euros more than twice the standard price' into $e = 2s + 15$.
3. Translate the purchase into an equation: $3s + 2e = 205$.
4. Substitute $e = 2s + 15$ into the second equation: $3s + 2(2s + 15) = 205$.
5. Expand: $3s + 4s + 30 = 205$, so $7s + 30 = 205$.
6. Solve for s : $7s = 175$, so $s = 25$.
7. Find e : $e = 2(25) + 15 = 65$.
8. Compute the combined price of one of each: $s + e = 25 + 65 = 90$ euros.

 **Answer:** 90 euros (standard = 25 euros, express = 65 euros)



Problem 5 — Strategy & Elimination



Tip

Translate the comparison sentence into an equation first ($e = 2s + 15$), then substitute it directly into the purchase equation instead of building two equations and eliminating — it turns the system into a single-variable equation immediately.



Shortcut

Since one equation already isolates e in terms of s , substitution is faster than elimination here — there's no need to multiply or align coefficients at all.



Trap

A student may translate 'costs 15 euros more than twice the standard price' backwards as $s = 2e + 15$ instead of $e = 2s + 15$, swapping which ticket is expressed in terms of the other and producing incorrect prices for both tickets.



Desmos Tip

Graph $e = 2s + 15$ and $3s + 2e = 205$ (solved for e) in Desmos; the intersection point (25, 65) gives both ticket prices directly, and their sum confirms 90.



Process of Elimination

Correct: **90 euros** (standard = 25 euros, express = 65 euros)

Tempting wrong answer	Exactly what went wrong
75 euros	Translated the comparison sentence backwards as $s = 2e + 15$ instead of $e = 2s + 15$, swapping which ticket price is expressed in terms of the other, giving standard = 55 and express = 20 (combined 75) instead of the correct 90.
675/7 euros (about 96.43)	Substituted $e = 2s + 15$ into $3s + 2e = 205$ but distributed incorrectly as $3s + 4s + 15 = 205$ (forgot to double the 15), instead of the correct $3s + 4s + 30 = 205$, throwing off both prices.
65 euros	Correctly solved for both prices ($s = 25$, $e = 65$) but reported only the express ticket price instead of computing the combined total $s + e = 90$.

Problem 6

Very Hard

The system $6x - 9y = 3k$ and $8x - 12y = 20$ has infinitely many solutions, where k is a constant. What is the value of $k^2 - 5$?

Formula / Rule

Infinite-solutions ratio shortcut: $A1/A2 = B1/B2 = C1/C2$ for the two equations to represent the same line.



First move

Whenever a system is described as having infinitely many solutions, immediately set up $A1/A2 = B1/B2 = C1/C2$ instead of trying to 'solve' the system — there is no unique point to find, only a constant to pin down.

Solution

1. Compare the x-coefficients: $A1/A2 = 6/8 = 3/4$.
2. Compare the y-coefficients: $B1/B2 = -9/-12 = 3/4$, which matches, so the two lines already share a slope.
3. For infinitely many solutions, the constant ratio must also equal $3/4$: $3k/20 = 3/4$.
4. Cross-multiply: $3k$ times $4 = 3$ times 20 , so $12k = 60$.
5. Solve for k : $k = 5$.
6. Substitute into the requested expression: $k^2 - 5 = 5^2 - 5 = 25 - 5 = 20$.



Answer: 20



Problem 6 — Strategy & Elimination



Tip

Whenever a system is described as having infinitely many solutions, immediately set up $A1/A2 = B1/B2 = C1/C2$ instead of trying to 'solve' the system — there is no unique point to find, only a constant to pin down.



Trap

A student who overlooks the coefficient 3 in front of k and solves $6/8 = k/20$ instead of $6/8 = 3k/20$ gets $k = 15$ (not 5), which leads to a final answer of 220 instead of the correct 20.



Shortcut

You never need to find x or y at all — the entire problem collapses to one proportion ($3k/20 = 3/4$) once the coefficient ratios are confirmed to already match.



Desmos Tip

Substitute $k = 5$ back into $6x - 9y = 15$ and graph it alongside $8x - 12y = 20$ (solved for y) to confirm the two lines overlap exactly, verifying infinitely many solutions.



Process of Elimination

Correct: 20

Tempting wrong answer	Exactly what went wrong
220	Set up the constant ratio as $6/8 = k/20$ instead of $6/8 = 3k/20$, forgetting the coefficient 3 in front of k , getting $k = 15$ instead of 5, then computed $15^2 - 5 = 220$.
0	Correctly found $k = 5$ but misread the requested expression $k^2 - 5$ as $(k - 5)^2$, computing $(5 - 5)^2 = 0$ instead of $5^2 - 5 = 20$.
5995/81 (about 74.0)	Cross-multiplied the proportion $3k/20 = 3/4$ backward, computing $3k \times 3 = 20 \times 4$ instead of $3k \times 4 = 20 \times 3$, giving $k = 80/9$ instead of 5, then computed $(80/9)^2 - 5$.

Problem 7

Very Hard

Let m equal the value of y in the solution to the system $x + y = 15$ and $x - y = 3$. Using this value of m , for what value of n does the system $3x + 4y = 7$ and $mx + ny = -2$ have no solution?

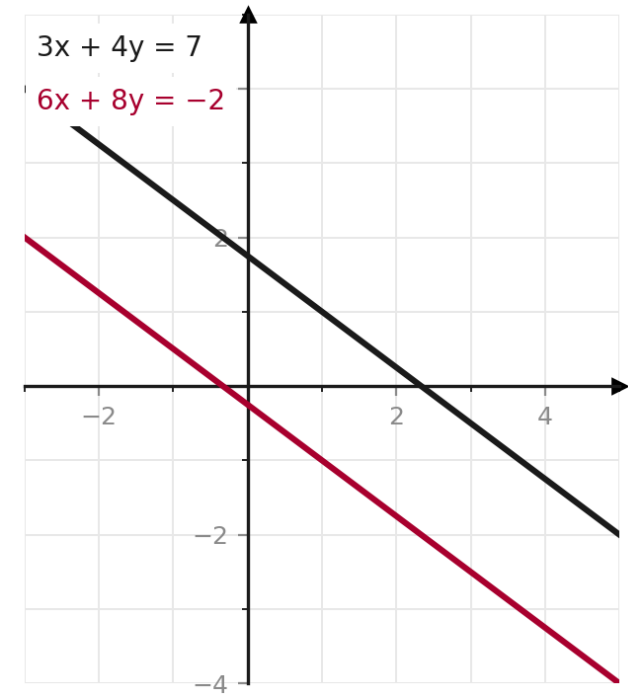
Formula / Rule

Solve a preliminary system to fix an unknown coefficient, then apply the no-solution ratio condition $A_1/A_2 = B_1/B_2$, with C_1/C_2 different, to the resulting system.

Solution

1. Solve the first system: add the two equations, $(x + y) + (x - y) = 15 + 3$, so $2x = 18$ and $x = 9$.
2. Subtract the two equations instead to get y directly: $(x + y) - (x - y) = 15 - 3$, so $2y = 12$ and $y = 6$.
3. So $m = 6$, the value of y from the first system.
4. Substitute $m = 6$ into the second system, which becomes $3x + 4y = 7$ and $6x + ny = -2$.
5. For this system to have no solution, the coefficient ratios must be equal: $3/6 = 4/n$.
6. Simplify the left side: $3/6 = 1/2$, so $1/2 = 4/n$.
7. Cross-multiply: $n = 8$.
8. Confirm this is a true no-solution case (not infinitely many): the constant ratio is $7/(-2) = -3.5$, which does not equal $1/2$, so the lines are parallel but distinct.

✓ **Answer:** $n = 8$



same slope, different intercept → no solution



Problem 7 — Strategy & Elimination



Tip

When a coefficient in a ratio-condition problem is defined as 'the value of' a specific variable from a separate system, fully solve that system first and clearly identify which variable (x or y) supplies the coefficient before setting up any ratio.



Trap

The two-stage setup tempts students to rush the first system and grab the wrong variable (x instead of y) for m, or to mismatch which coefficient belongs in the numerator versus denominator when writing the ratio $3/6 = 4/n$, since the problem never explicitly labels which equation is 'first' and which is 'second'.



Shortcut

The first system's equations sum directly to $2x = 18$ and subtract directly to $2y = 12$, so you can read off both x and y in one line each without full elimination setup — useful for confirming $m = 6$ quickly.



Desmos Tip

Graph $x + y = 15$ and $x - y = 3$ to confirm the intersection (9, 6), then graph $3x + 4y = 7$ and $6x + 8y = -2$ (solved for y) to visually confirm the lines are parallel and never intersect.



Process of Elimination

Correct: **n = 8**

Tempting wrong answer	Exactly what went wrong
n = 12	Used the x-value (9) from the first system as m instead of the y-value (6), then solved the ratio $3/9 = 4/n$.
n = 2	Correctly found $m = 6$ but inverted the coefficient ratio, comparing the second equation's x-coefficient to the first's (6/3) instead of the first's to the second's (3/6), while keeping the y-coefficients in the original order.
n = -8	Correctly found $m = 6$ but made a sign error setting up the ratio equation, writing $3/6 = -4/n$ instead of $3/6 = 4/n$, which flips the sign of the resulting n.

Problem 8

Hard

Solve the system $3x + 4y = 7$ and $2x - 3y = 16$. What is the value of $x - y$?

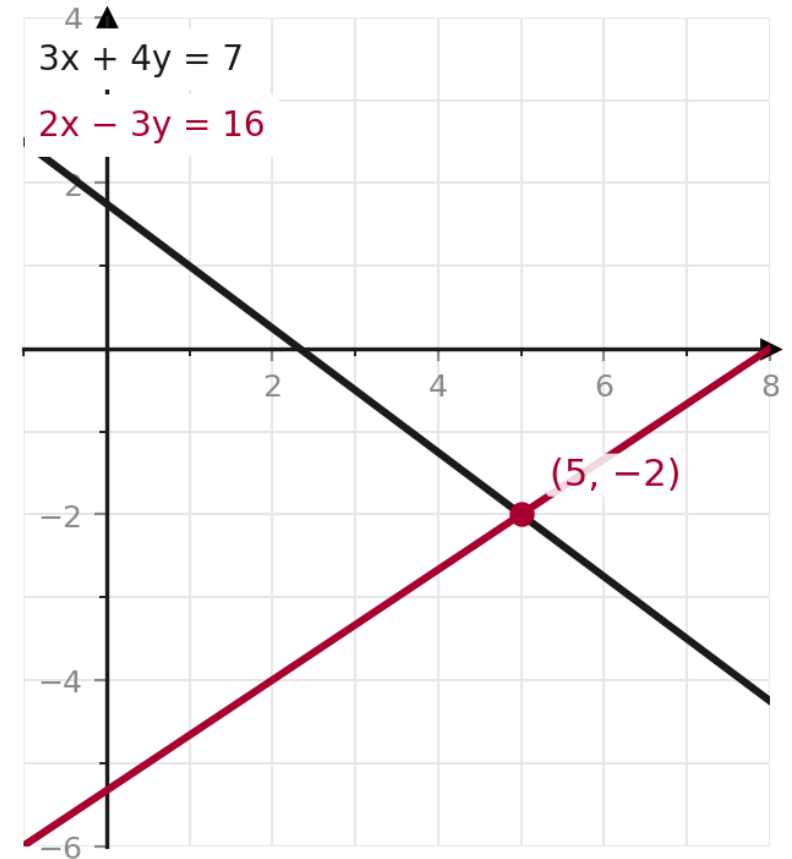
Formula / Rule

Elimination by scaling both equations to a common coefficient, then compute the specific combination the question actually asks for.

Solution

1. To eliminate y , find the LCM of 4 and 3, which is 12: multiply the first equation by 3 to get $9x + 12y = 21$.
2. Multiply the second equation by 4 to get $8x - 12y = 64$.
3. Add the two new equations: $9x + 12y + 8x - 12y = 21 + 64$, so $17x = 85$.
4. Solve for x : $x = 85/17 = 5$.
5. Substitute $x = 5$ into $2x - 3y = 16$: $10 - 3y = 16$, so $-3y = 6$ and $y = -2$.
6. Compute the requested expression: $x - y = 5 - (-2) = 5 + 2 = 7$.

✓ **Answer:** $x - y = 7$





Problem 8 — Strategy & Elimination



Tip

Once you solve for x and y , reread exactly what expression the question wants ($x - y$, not x alone, not y alone, not $y - x$) before writing your final answer.



Trap

This problem is built so that $x = 5$ and $y = -2$ are both 'clean' numbers on their own, tempting a student to report just $x = 5$ (answering 'what is x ' instead of 'what is $x - y$ ') or to subtract in the wrong order — either looks like a perfectly reasonable final answer if you stop reading carefully.



Shortcut

No shortcut beats solving fully here — since the question asks for a specific combination ($x - y$) rather than a simple sum or difference of the two original equations as given, you must find x and y individually before combining them.



Desmos Tip

Graph $3x + 4y = 7$ and $2x - 3y = 16$ in Desmos, read the intersection point $(5, -2)$, then compute $5 - (-2) = 7$ directly from the coordinates.



Process of Elimination

Correct: $x - y = 7$

Tempting wrong answer	Exactly what went wrong
5	Solved correctly for $x = 5$ but reported it directly, forgetting the question asks for $x - y$, not x alone.
-2	Solved correctly for $y = -2$ but reported it directly, mistakenly treating y as the requested value instead of computing $x - y$.
-7	Correctly found $x = 5$ and $y = -2$ but computed $y - x$ instead of $x - y$, subtracting in the reversed order and getting -7 instead of 7.

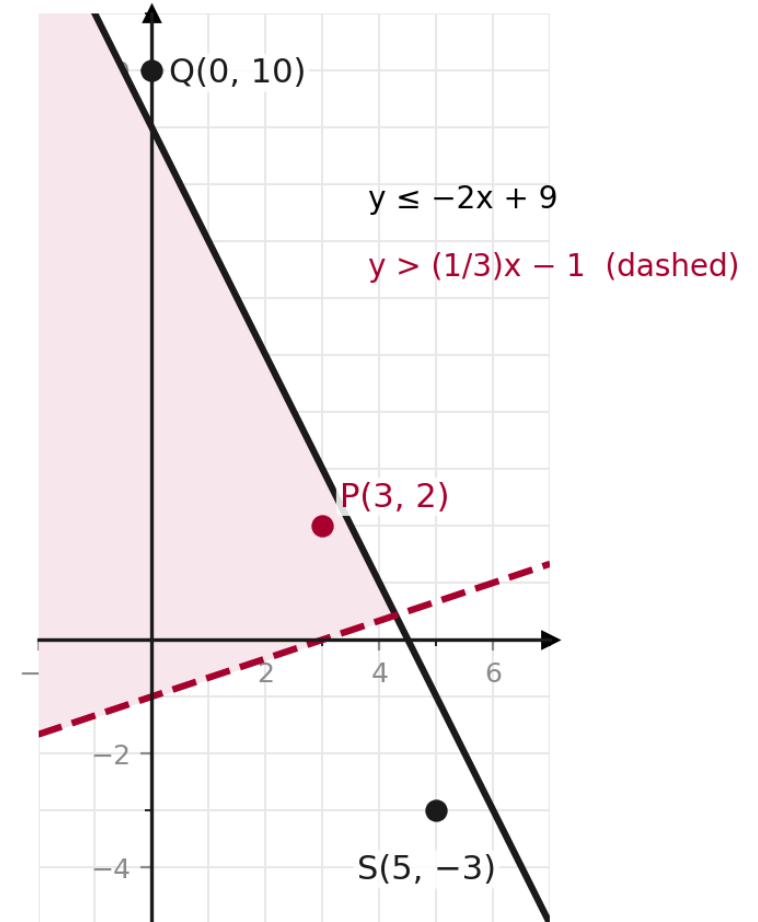
03

Linear Inequalities

One-variable inequalities, compound inequalities, and graphed regions.

In this section

- Key terms — English / Ελληνικά
- Explained theory with examples
- 8 worked problems (Hard / Very Hard)



Key Terminology

Term (English)	Ελληνικά
Inequality	Ανίσωση
Solution set	Σύνολο λύσεων
Compound inequality	Σύνθετη ανίσωση
Flip the inequality sign	Αναστροφή συμβόλου ανίσωσης
Boundary line / shaded region	Οριακή ευθεία / σκιασμένη περιοχή

A The big picture

- Solve like an equation — with one extra rule.
- The answer is a set of values, not a single value.
- Word problems often need rounding in the right direction.

$1 \leq x < 5$: ● included, ○ excluded



Theory Explained (1/3)

The Flip Rule

$\times$ or $\div$ by a negative $\rightarrow$ reverse the sign

WHAT IT MEANS

- Adding/subtracting never flips the sign.
- Multiplying/dividing by a negative always does.
- Swapping sides also reverses the symbol.



Example

$$-3x > 12 \rightarrow x < -4.$$

Compound Inequalities

$a < \text{expression} < b \rightarrow$ do each step to all three parts

WHAT IT MEANS

- Clear fractions by multiplying ALL parts.
- If you divide by a negative, flip both signs and reorder.
- Write the final answer smallest $\rightarrow$ largest.

$1 \leq x < 5$: ● included, ○ excluded



Example

$$-3 < (2 - 5x)/3 \leq 4 \rightarrow -2 \leq x < 11/5.$$

Theory Explained (2/3)

Open vs Closed

$<, >$ → open circle $\leq, \geq$ → closed circle

WHAT IT MEANS

- Closed: the endpoint IS a solution.
- Open: the endpoint is NOT a solution.
- Counting integers: check each endpoint separately.



Example

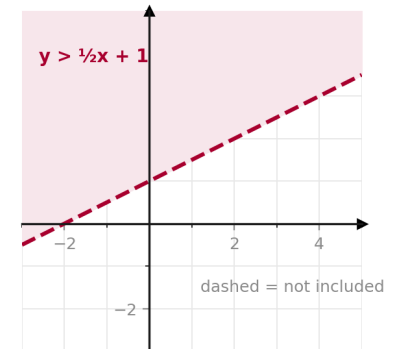
$2x - 5 < 9 \rightarrow x < 7$, so 7 itself is excluded.

Graphing Regions

$y > mx + b$ → shade above; $y < mx + b$ → shade below

WHAT IT MEANS

- Solve for y first, then read the symbol.
- Dashed line for $<$ or $>$, solid for $\leq$ or $\geq$.
- A system's solution = where the shadings overlap.



Example

Test a point: plug it into EVERY inequality.

Theory Explained (3/3)

Word Problems

'at least' $\rightarrow \geq$ 'at most' $\rightarrow \leq$ 'more than' $\rightarrow >$

WHAT IT MEANS

- Build the inequality from the sentence, then solve.
- Minimum count $\rightarrow$ round UP; maximum count $\rightarrow$ round DOWN.
- Ignore extra numbers that don't affect the condition.



Example

$13n \geq 840 \rightarrow n \geq 64.6 \rightarrow$ at least 65.

Watch Out



Negative divisor = flip

- The single most common inequality error.
- Check with a test value from your answer.



Round in the direction of the condition

- 'At least' questions round up.
- 'At most' questions round down.

Problem 1

Hard

The inequality $3(x - 4) - 5x \geq k - 15$ has solution set $x \leq 7$. What is the value of k ?

Formula / Rule

Distribute and combine like terms, then divide by a negative number (which flips the inequality) to isolate x ; reverse-engineer an unknown constant from a given solution set.

Solution

1. Expand the left side: $3(x - 4) - 5x = 3x - 12 - 5x = -2x - 12$, so the inequality becomes $-2x - 12 \geq k - 15$.
2. Add 15 to both sides: $-2x - 12 + 15 \geq k$, which simplifies to $-2x + 3 \geq k$, or equivalently $-2x \geq k - 3$.
3. Divide both sides by -2 to isolate x . Dividing by a negative number reverses the inequality sign: $x \leq (k - 3)/(-2)$, which simplifies to $x \leq (3 - k)/2$.
4. The problem states the solution set is $x \leq 7$, so set the boundary expressions equal: $(3 - k)/2 = 7$.
5. Multiply both sides by 2: $3 - k = 14$.
6. Solve for k : $-k = 14 - 3 = 11$, so $k = -11$.
7. Check: substitute $k = -11$ into the original inequality: $3(x - 4) - 5x \geq -11 - 15 = -26$. The left side simplifies to $-2x - 12$, so $-2x - 12 \geq -26$, giving $-2x \geq -14$, and dividing by -2 (flip) gives $x \leq 7$. Confirmed.

✓ **Answer: $k = -11$**

solution set: $x \leq 7$





Problem 1 — Strategy & Elimination



Tip

When a solution set is given and you need to find a hidden constant, solve the inequality symbolically in terms of that constant first, then match the resulting boundary expression to the given boundary value.



Shortcut

Skip solving the general inequality entirely: plug the boundary value $x = 7$ directly into the original expression as an equality, $-2(7) - 12 = k - 15$, which gives $k = -11$ in one line.



Trap

A student who forgets to flip the inequality sign when dividing by -2 will set up the equation $(k-3)/2 = 7$ instead of $(3-k)/2 = 7$, leading to the wrong value $k = 11$ instead of the correct $k = -11$.



Desmos Tip

Graph $y = -2x - 12$ and $y = -26$ (using $k = -11$) and confirm the graphs intersect at $x = 7$, matching the given boundary.



Process of Elimination

Correct: **$k = -11$**

Tempting wrong answer	Exactly what went wrong
$k = 11$	Divided $-2x \geq k - 3$ by -2 without reversing the inequality, setting up $(k-3)/2 = 7$ instead of $(3-k)/2 = 7$, giving $k = 11$ instead of -11 .
$k = -3$	Distributed $3(x-4)$ incorrectly as $3x - 4$ (forgetting to multiply the -4 by 3 as well), which shifts the constant term and yields $k = -3$ instead of -11 .
$k = 13$	Mis-copied the -12 term as $+12$ when isolating k , writing $-2x + 12 \geq k - 15$ instead of $-2x - 12 \geq k - 15$, which produces $k = 13$ instead of -11 .

Problem 2

Very Hard

Solve the compound inequality for x , and express the solution set using inequality notation: $-3 < (2 - 5x)/3 \leq 4$.

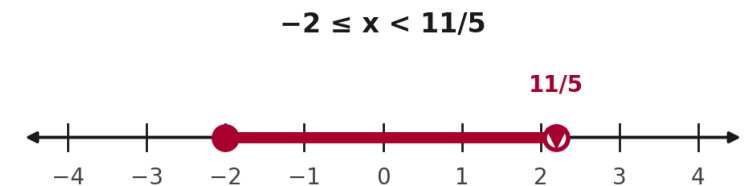
Formula / Rule

Compound (three-part) inequality: perform the same operation on all three parts, reversing both inequality signs and swapping the endpoint order when multiplying or dividing by a negative number.

Solution

1. Multiply all three parts by 3 to clear the denominator: $-3 \times 3 < 2 - 5x \leq 4 \times 3$, giving $-9 < 2 - 5x \leq 12$.
2. Subtract 2 from all three parts: $-9 - 2 < -5x \leq 12 - 2$, giving $-11 < -5x \leq 10$.
3. Divide all three parts by -5 . Since -5 is negative, reverse both inequality signs and flip the order of the parts: $-11/-5 > x \geq 10/-5$.
4. Simplify each side: $-11/-5 = 11/5$ and $10/-5 = -2$, so the inequality reads $11/5 > x \geq -2$.
5. Rewrite in standard increasing order (smallest to largest, left to right): $-2 \leq x < 11/5$.
6. Check an endpoint: at $x = -2$, $(2 - 5(-2))/3 = (2 + 10)/3 = 12/3 = 4$, which satisfies ≤ 4 with equality, confirming $x = -2$ is correctly included.
7. Check the other endpoint: at $x = 11/5$, $(2 - 5(11/5))/3 = (2 - 11)/3 = -9/3 = -3$, which is NOT strictly greater than -3 (it equals -3), confirming $x = 11/5$ is correctly excluded since the original inequality was strict ($>$) on that side.

Answer: $-2 \leq x < 11/5$





Problem 2 — Strategy & Elimination



Tip

Clear the fraction first by multiplying every part by the denominator, then isolate x exactly as with a normal compound inequality, remembering to flip both inequality signs and reverse the order of the endpoints whenever you multiply or divide by a negative number.



Shortcut

No shortcut beats clearing the fraction first — trying to divide by 3 termwise before combining like terms just recreates the same fraction later, so multiplying through immediately is already the fastest path.



Trap

A student who divides by -5 but forgets to also swap which endpoint is 'less than' and which is 'greater than' will report the answer as $11/5 \leq x < -2$, which is an empty (impossible) set since $11/5$ is greater than -2 .



Desmos Tip

Graph $y = (2 - 5x)/3$ along with the horizontal lines $y = -3$ and $y = 4$, then read off the x -interval where the slanted line stays between the two horizontal lines (open circle at $x = 11/5$, closed circle at $x = -2$).



Process of Elimination

Correct: $-2 \leq x < 11/5$

Tempting wrong answer	Exactly what went wrong
$11/5 \leq x < -2$	Flipped the inequality symbols correctly when dividing by -5 but forgot to also swap the left-right order of the two endpoints, producing an impossible set where the 'lower' bound is larger than the 'upper' bound.
$-2/5 \leq x < 1$	Skipped clearing the fraction (never multiplied all three parts by 3) and instead subtracted 2 and divided by -5 directly on the un-cleared expression $(2 - 5x)/3$, shifting every boundary value.
$-2 < x \leq 11/5$	Correctly flipped the inequality direction and endpoint order but attached the strict ' $<$ ' symbol to the wrong endpoint, swapping which side is open and which is closed.

Problem 3

Hard

A region R in the xy-plane is defined by the system of inequalities: $y \leq -2x + 9$ and $y > (1/3)x - 1$. Determine which of the points P(3, 2), Q(0, 10), or S(5, -3) lies in region R.

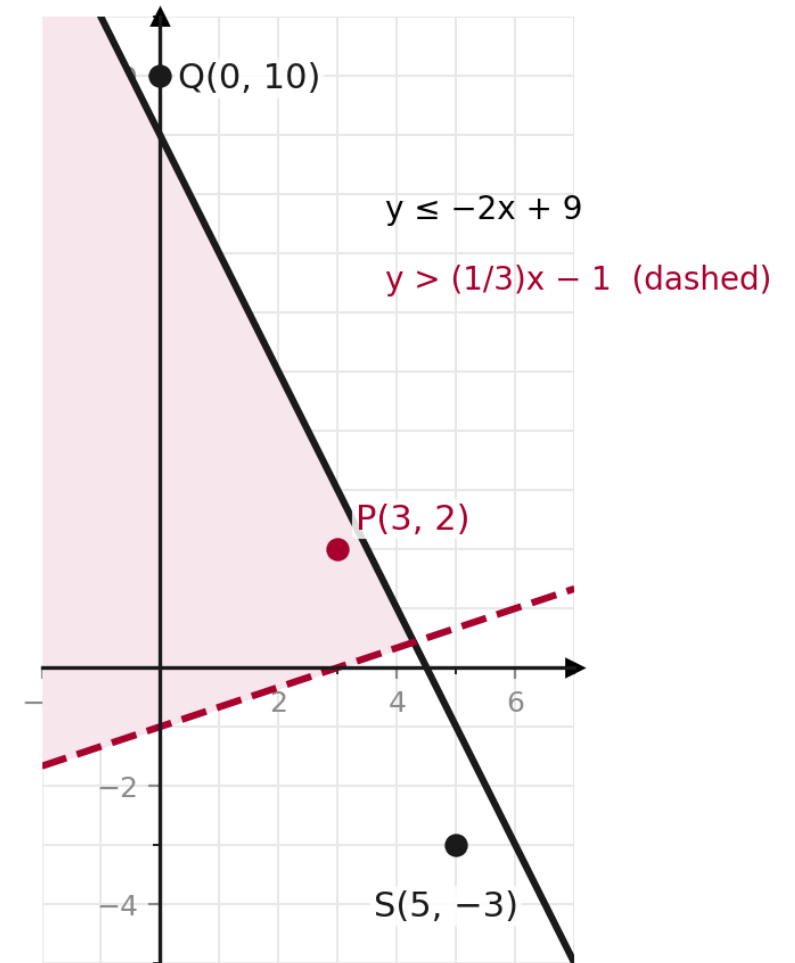
Formula / Rule

A point satisfies a system of inequalities only if it satisfies every inequality in the system simultaneously — substitute the point's coordinates directly into each inequality.

Solution

1. Test point P(3, 2) in the first inequality: $2 \leq -2(3) + 9 = -6 + 9 = 3$. Since $2 \leq 3$ is true, P satisfies the first inequality.
2. Test point P(3, 2) in the second inequality: $2 > (1/3)(3) - 1 = 1 - 1 = 0$. Since $2 > 0$ is true, P satisfies the second inequality too, so P is in region R.
3. Test point Q(0, 10) in the first inequality: $10 \leq -2(0) + 9 = 9$. Since $10 \leq 9$ is false, Q fails the first inequality and is NOT in region R (no need to check the second inequality).
4. Test point S(5, -3) in the first inequality: $-3 \leq -2(5) + 9 = -10 + 9 = -1$. Since $-3 \leq -1$ is true, S passes the first inequality.
5. Test point S(5, -3) in the second inequality: $-3 > (1/3)(5) - 1 = 5/3 - 1 = 2/3$. Since $-3 > 2/3$ is false, S fails the second inequality, so S is NOT in region R.
6. Only point P(3, 2) satisfies both inequalities simultaneously, so P is the point that lies in region R.

✓ **Answer:** P(3, 2)





Problem 3 — Strategy & Elimination



Tip

Never rearrange or graph the inequalities to check a specific point — just substitute its x - and y -coordinates directly into each inequality as given and evaluate; it's faster and avoids sign errors.



Trap

A student who only checks the first inequality for each point (since both P and S pass it) might stop early and mistakenly accept S as a valid answer without testing the strict second inequality, missing that S fails $y > (1/3)x - 1$.



Shortcut

Since Q obviously fails the first inequality's y -value at $x = 0$ (10 versus the boundary value 9), testing Q first eliminates it fastest, leaving only two candidates left to fully check.



Desmos Tip

Graph both boundary lines $y = -2x + 9$ (solid) and $y = (1/3)x - 1$ (dashed) with the correct shading, then plot the three points to see visually that only P falls inside the overlapping shaded region.



Process of Elimination

Correct: **P(3, 2)**

Tempting wrong answer	Exactly what went wrong
S(5, -3)	Checked only the first inequality $y \leq -2x + 9$ for each point (which both P and S satisfy) and stopped there without testing the strict second inequality, incorrectly accepting S even though it fails $y > (1/3)x - 1$.
Q(0, 10)	Checked only the second inequality $y > (1/3)x - 1$ for each point — Q passes it easily ($10 > -1$) — and picked Q without ever testing it against the first inequality, where it actually fails ($10 \leq 9$ is false).
No point lies in R	Misapplied the first inequality's substitution for P, computing $-2(3 + 9)$ instead of $-2(3) + 9$ (distributing across the sum instead of just the x -term), making P appear to fail the first inequality, so no point seemed to satisfy both conditions.

Problem 4

Hard

A student committee in Rotterdam is chartering a bus for a class trip. The bus rental costs a flat 340 euros plus 12 euros per student who rides it. Each student pays a 25 euro activity fee to the committee. The committee also printed 80 information flyers for parents. The committee wants the total fees collected to be at least 500 euros more than the total bus rental cost. What is the minimum number of students needed for this to happen?

Formula / Rule

Translate 'at least' as $\geq$, set up a linear inequality in the unknown count, then round the solution to the correct integer in the direction that keeps the inequality true.



First move

In minimum-value word problems with 'at least,' always solve the inequality first, then round the decimal boundary UP to the next whole number, since rounding down would fall short of the requirement.

Solution

1. Let n represent the number of students riding the bus.
2. Total fees collected by the committee: $25n$.
3. Total bus rental cost: $340 + 12n$.
4. Translate 'at least 500 euros more than the cost' into an inequality: $25n \geq (340 + 12n) + 500$.
5. Simplify the right side: $25n \geq 840 + 12n$.
6. Subtract $12n$ from both sides: $25n - 12n \geq 840$, so $13n \geq 840$.
7. Divide both sides by 13 (positive, no flip needed): $n \geq 840/13 = 64.615\dots$
8. Since n must be a whole number of students, round UP to the next integer: $n = 65$ (rounding down to 64 would give $13(64) = 832$, which is less than 840 and fails the requirement).
9. Note that the flyer count (80) is never used in the calculation — it is irrelevant to the fee-versus-cost relationship.



Answer: 65 students



Problem 4 — Strategy & Elimination



Tip

In minimum-value word problems with 'at least,' always solve the inequality first, then round the decimal boundary UP to the next whole number, since rounding down would fall short of the requirement.



Trap

A student might round 64.615... down to 64 out of habit, or try to work the irrelevant flyer count of 80 into the equation — explicitly verifying that $n = 64$ gives $832 < 840$ (a failure) shows why 65 is required.



Shortcut

Instead of expanding $(340 + 12n) + 500$ fully, immediately combine the constants $340 + 500 = 840$ mentally before writing the inequality, saving a simplification step.



Desmos Tip

Graph $y = 25x$ and $y = 840 + 12x$ and find where the first crosses above the second; any integer x at or to the right of that intersection point (rounded up) is a valid answer.



Process of Elimination

Correct: 65 students

Tempting wrong answer	Exactly what went wrong
64 students	Computed $n \geq 840/13 = 64.615...$ but rounded down to the nearest whole number out of habit instead of up, submitting 64, which actually gives $13(64) = 832 < 840$ and fails the requirement.
27 students	Dropped the '500 euros more than' condition entirely, setting up $25n \geq 340 + 12n$ instead of $25n \geq 340 + 12n + 500$, which gives $n = 27$, far too low to satisfy the actual requirement.
23 students	Combined the $12n$ term incorrectly by adding instead of subtracting when isolating n , turning $25n - 12n \geq 840$ into $25n + 12n \geq 840$, which gives $37n \geq 840$ and $n = 23$ instead of 65.

Problem 5

Very Hard

For what values of k is the slope of the line through the points $(2, k)$ and $(7, -3)$ less than -2 ?

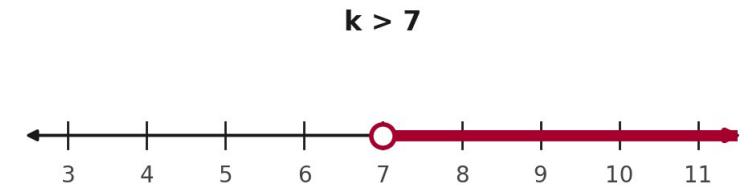
Formula / Rule

Slope formula $m = (y_2 - y_1)/(x_2 - x_1)$, combined with solving a linear inequality where dividing by a negative number reverses the inequality.

Solution

1. Write the slope formula using the two given points: $m = (-3 - k)/(7 - 2) = (-3 - k)/5$.
2. Set up the inequality using the given condition that the slope is less than -2 : $(-3 - k)/5 < -2$.
3. Multiply both sides by 5 (positive, no flip needed): $-3 - k < -10$.
4. Add 3 to both sides: $-k < -10 + 3 = -7$.
5. Divide both sides by -1 to isolate k . Since we're dividing by a negative number, reverse the inequality sign: $k > 7$.
6. Check with a test value: let $k = 8$. Slope $= (-3 - 8)/5 = -11/5 = -2.2$, and $-2.2 < -2$ is true, confirming $k > 7$ is correct.
7. Check the boundary: at $k = 7$, slope $= (-3 - 7)/5 = -10/5 = -2$, which is NOT less than -2 (it's equal), confirming $k = 7$ itself must be excluded, matching the strict inequality $k > 7$.

✓ **Answer:** $k > 7$





Problem 5 — Strategy & Elimination



Tip

When slope is described using an inequality, first write the slope formula normally as an expression in terms of the unknown, then solve that expression as an ordinary inequality — treat the slope formula as just another algebraic expression.



Shortcut

Multiply the inequality $(-3 - k)/5 < -2$ by 5 and then by -1 in one combined move, immediately reversing to $3 + k > 10$, then solve $k > 7$ directly without a separate 'add 3' step.



Trap

A student who mishandles the final divide-by -1 step (forgetting to flip the sign) will conclude $k < 7$ instead of the correct $k > 7$ — exactly backwards.



Desmos Tip

Graph $y = (-3 - x)/5$ and the horizontal line $y = -2$, then observe the slanted graph lies below $y = -2$ exactly when $x > 7$.



Process of Elimination

Correct: $k > 7$

Tempting wrong answer	Exactly what went wrong
$k < 7$	Divided $-k < -7$ by -1 without reversing the inequality sign, concluding $k < 7$ instead of the correct $k > 7$ — exactly backwards.
$k < -13$	Wrote the slope formula with the y-values subtracted in the wrong order relative to the x-values, using $(k + 3)/5$ instead of $(-3 - k)/5$, which flips the sign of the whole expression and leads to $k < -13$.
$k \geq 7$	Solved the inequality correctly down to the boundary $k = 7$ but reported it as $k \geq 7$ (inclusive), forgetting the original slope condition was strict less-than, which excludes the boundary $k = 7$ itself.

Problem 6

Very Hard

A factory's quality-control rule states that a machined rod fails inspection if its length L (in centimeters) differs from the target length of 18 cm by 3 cm or more: $|L - 18| \geq 3$. The factory runs three shifts per day and produces 200 rods per shift. For what values of L does a rod PASS inspection? Express your answer as a compound inequality.

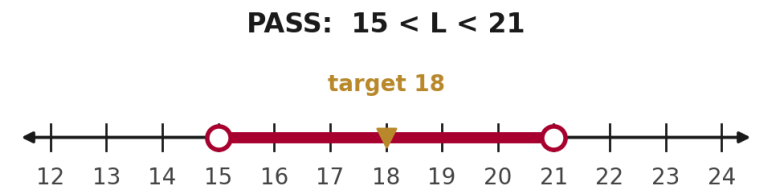
Formula / Rule

A rod passes inspection exactly when it does NOT satisfy the failing condition — negate $|L - 18| \geq 3$ to get $|L - 18| < 3$, then split the absolute value inequality into a three-part compound inequality.

Solution

1. Passing is the logical opposite of failing. Since failing means $|L - 18| \geq 3$, passing means the negation: $|L - 18| < 3$.
2. An absolute value inequality of the form $|\text{expression}| < c$ (with c positive) splits into a compound inequality: $-c < \text{expression} < c$. Here that gives $-3 < L - 18 < 3$.
3. Add 18 to all three parts of the compound inequality: $-3 + 18 < L - 18 + 18 < 3 + 18$.
4. Simplify: $15 < L < 21$.
5. Sanity check with the failing boundary: at $L = 15$, $|15 - 18| = |-3| = 3$, which satisfies the ORIGINAL failing condition (≥ 3 , since it equals 3), so $L = 15$ should indeed fail and correctly be excluded from the passing range — confirming the strict inequality ($<$ not $\leq$) is correct on both sides.
6. Note that the production details (three shifts, 200 rods per shift) are not needed to answer the question about acceptable

✓ **Answer:** $15 < L < 21$ (centimeters)





Problem 6 — Strategy & Elimination



Tip

When a problem defines a 'failing' or 'rejection' condition with $\geq$ or $>$, remember that the 'passing' condition is its exact logical complement, which flips the inequality to $<$ or $\leq$ before you even start splitting the absolute value.



Shortcut

The boundary values (15 and 21) are identical whether you solve for the failing or passing condition, so you can solve whichever direction feels more natural (here, directly splitting $|L - 18| < 3$) and just make sure the final inequality symbols match what 'passing' truly requires.



Trap

A student who solves $|L - 18| \geq 3$ directly (getting $L \leq 15$ OR $L \geq 21$, the FAILING range) and reports that as the answer to 'which rods pass' has answered the opposite question — the passing range is the complement, $15 < L < 21$.



Desmos Tip

Graph $y = \text{abs}(L - 18)$ and the horizontal line $y = 3$, then read off the L-interval where the V-shaped graph dips below the line — that's the passing range $15 < L < 21$.



Process of Elimination

Correct: $15 < L < 21$ (centimeters)

Tempting wrong answer	Exactly what went wrong
$L \leq 15$ or $L \geq 21$	Solved the given failing condition $ L - 18 \geq 3$ directly and reported that FAILING range as the answer, instead of first negating it to find the PASSING condition $ L - 18 < 3$.
$15 \leq L \leq 21$	Used $\leq$ instead of $<$ when splitting $ L - 18 < 3$ into $-3 < L - 18 < 3$, incorrectly including the boundary values 15 and 21, which actually belong to the failing range since $ 15 - 18 = 3$ satisfies ≥ 3 .
$12 < L < 24$	Doubled the tolerance value (used 6 instead of 3) when splitting the absolute value inequality, apparently confusing the ± 3 boundary with a full range width, giving $12 < L < 24$ instead of $15 < L < 21$.

Problem 7

Hard

For how many integer values of x do both inequalities $2x - 5 < 9$ and $-3x + 4 \leq 19$ hold true at the same time?

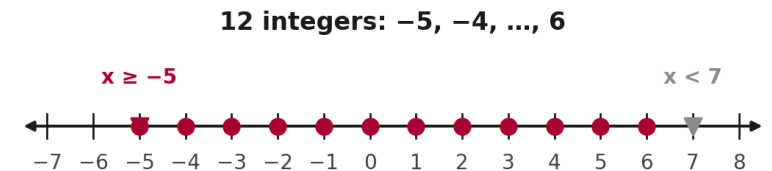
Formula / Rule

Solve each linear inequality separately, then take the intersection of the two solution sets before counting integer solutions.

Solution

1. Solve the first inequality: $2x - 5 < 9$. Add 5 to both sides: $2x < 14$. Divide by 2 (positive, no flip): $x < 7$.
2. Solve the second inequality: $-3x + 4 \leq 19$. Subtract 4 from both sides: $-3x \leq 15$. Divide by -3, which is negative, so reverse the inequality: $x \geq -5$.
3. Combine both solution sets: x must satisfy $x < 7$ AND $x \geq -5$ at the same time, giving the intersection $-5 \leq x < 7$.
4. List the integers in this range, including -5 but not including 7: -5, -4, -3, -2, -1, 0, 1, 2, 3, 4, 5, 6.
5. Count the listed integers: there are 12 of them.
6. Verify the endpoints: at $x = -5$, $-3(-5) + 4 = 19$, which satisfies ≤ 19 (boundary correctly included); at $x = 7$, $2(7) - 5 = 9$, which does NOT satisfy < 9 (boundary correctly excluded).

✓ **Answer:** 12 integer values ($x = -5$ through $x = 6$)





Problem 7 — Strategy & Elimination



Tip

When a problem asks how many integers satisfy two inequalities together, solve each one separately first, then intersect the two number lines before counting — never try to merge the original inequalities into one expression.



Shortcut

Since both inequalities solve cleanly to integer boundaries, skip listing every integer individually — just compute the upper bound minus the lower bound, $7 - (-5) = 12$, which directly gives the count since the lower bound is included and the upper bound is excluded.



Trap

A student who forgets to flip the inequality sign when dividing $-3x \leq 15$ by -3 , or who mishandles which boundary is included versus excluded, will miscount the integers in the intersection even though each inequality looks routine on its own.



Desmos Tip

Graph $y = 2x - 5$ with the line $y = 9$, and $y = -3x + 4$ with the line $y = 19$, to visually confirm the two boundary x -values -5 and 7 before counting the integers between them.



Process of Elimination

Correct: 12 integer values ($x = -5$ through $x = 6$)

Tempting wrong answer	Exactly what went wrong
13	Treated the strict inequality $2x - 5 < 9$ as if it were $\leq$, including the boundary $x = 7$, which actually fails the original strict inequality since $2(7) - 5 = 9$ is not less than 9.
11	Treated $-3x + 4 \leq 19$ as if it were strictly $<$, excluding the boundary $x = -5$, which actually satisfies the original inequality with equality since $-3(-5) + 4 = 19$.
14	Added 4 to both sides instead of subtracting it when isolating $-3x$, turning $-3x \leq 15$ into $-3x \leq 23$, which shifts the lower bound down and produces an incorrect count of 14.

Problem 8

Very Hard

For a positive constant a , the inequality $a|x - 5| \geq 12$ has solution set $x \leq -1$ or $x \geq 11$. What is the value of a ?

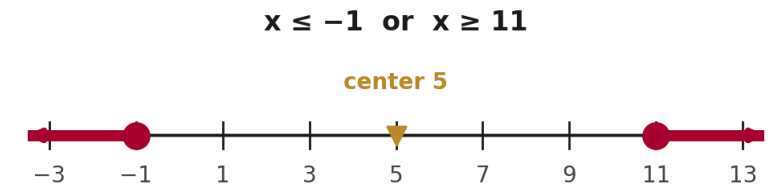
Formula / Rule

Splitting an absolute value inequality of the form $|\text{expression}| \geq c$ into two linear branches, combined with reverse-engineering an unknown coefficient from a given solution set.

Solution

1. Divide both sides by a . Since a is positive, the inequality direction stays the same: $|x - 5| \geq 12/a$.
2. Split the absolute value inequality into its two branches: $x - 5 \geq 12/a$, or $x - 5 \leq -12/a$.
3. Solve each branch for x : $x \geq 5 + 12/a$, or $x \leq 5 - 12/a$.
4. Match the first branch to the given boundary $x \geq 11$: $5 + 12/a = 11$.
5. Solve for a : $12/a = 11 - 5 = 6$, so $a = 12/6 = 2$.
6. Verify with the second branch using $a = 2$: $5 - 12/2 = 5 - 6 = -1$, which exactly matches the given boundary $x \leq -1$, confirming $a = 2$ works for both branches.
7. Since the problem specifies a is a positive constant, $a = 2$ satisfies that condition as well.

✓ **Answer:** $a = 2$





Problem 8 — Strategy & Elimination



Tip

When an unknown coefficient sits inside an absolute value inequality and you're given the final solution set, split the absolute value into its two linear branches symbolically first, then match either branch's boundary to the corresponding given boundary value to solve for the unknown.



Trap

A student who solves $5 + 12/a = 11$ by adding 5 to both sides instead of subtracting it will get $12/a = 16$ instead of $12/a = 6$, producing $a = 3/4$ instead of the correct $a = 2$.



Shortcut

Once you split into the two branches $5 + 12/a$ and $5 - 12/a$, you only need to match ONE branch to its corresponding boundary (say $5 + 12/a = 11$) to solve for a — the other branch is just a free check, not a second equation to solve.



Desmos Tip

Graph $y = 2 \cdot \text{abs}(x - 5)$ (using the solved value $a = 2$) together with the horizontal line $y = 12$, then confirm the graph crosses $y = 12$ exactly at $x = -1$ and $x = 11$, matching the given solution set.



Process of Elimination

Correct: $a = 2$

Tempting wrong answer	Exactly what went wrong
$a = 3/4$	Solved $5 + 12/a = 11$ by adding 5 to both sides instead of subtracting it, getting $12/a = 16$ instead of $12/a = 6$, which gives $a = 3/4$ instead of $a = 2$.
$a = 6$	Mishandled isolating the absolute value by subtracting a from 12 instead of dividing by a , turning $ x - 5 \geq 12/a$ into $ x - 5 \geq 12 - a$, which leads to $a = 6$ instead of $a = 2$.
$a = -2$	Dropped the negative sign on the second branch (wrote $x - 5 \leq 12/a$ instead of $x - 5 \leq -12/a$), then matched $5 + 12/a$ to the boundary -1 instead of 11 , producing $a = -2$, which also violates the problem's requirement that a be positive.

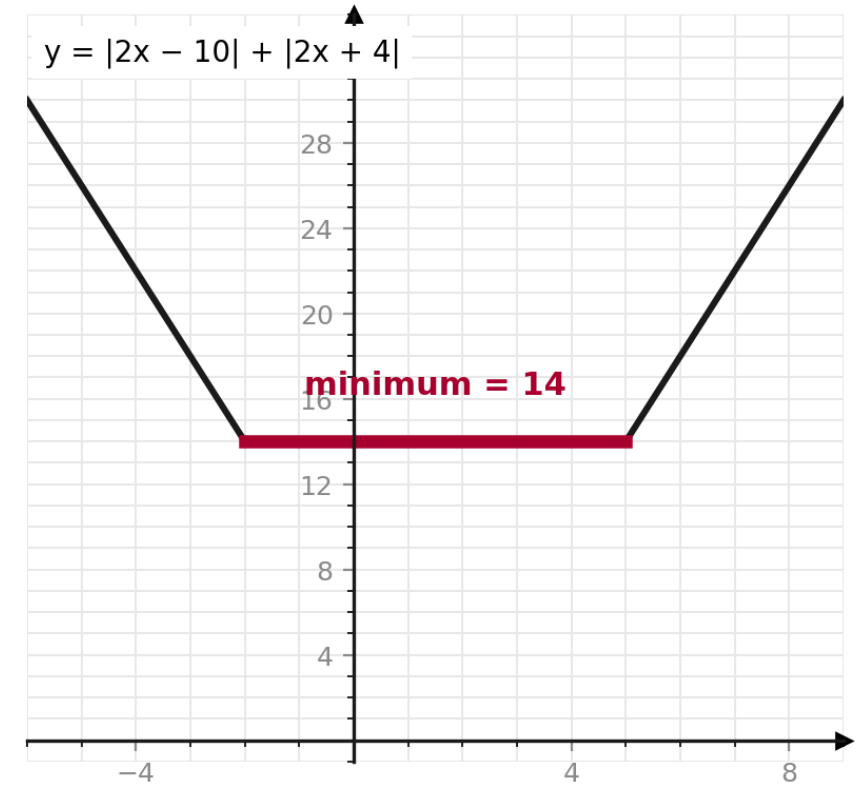
04

Absolute Value

Distance on the number line, two-case equations, and extraneous solutions.

In this section

- Key terms — English / Ελληνικά
- Explained theory with examples
- 8 worked problems (Hard / Very Hard)

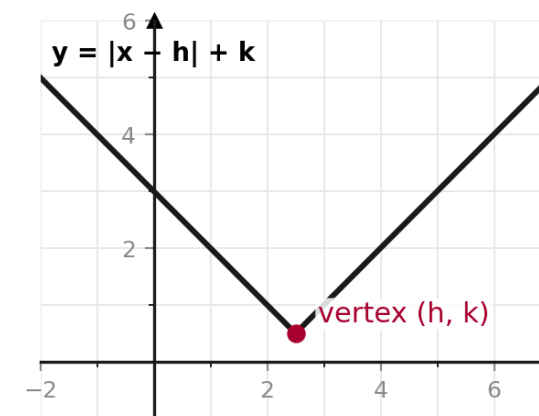


Key Terminology

Term (English)	Ελληνικά
Absolute value	Απόλυτη τιμή
Distance from zero	Απόσταση από το μηδέν
Two cases (positive/negative)	Δύο περιπτώσεις (θετική/αρνητική)
Extraneous solution	Ξένα (μη αποδεκτή) λύση

A ㄱ The big picture

- $|x|$ = distance from 0, so it is never negative.
- Every absolute-value equation splits into two cases.
- Extraneous solutions appear when a variable sits outside the bars.



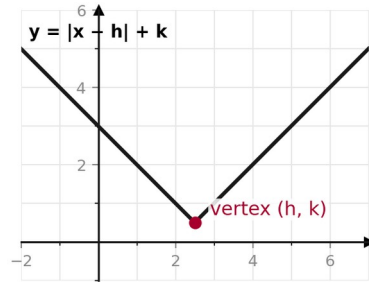
Theory Explained (1/3)

Meaning of $|x - a|$

$$|x - a| = \text{distance between } x \text{ and } a$$

WHAT IT MEANS

- $|x - 3| = 5$ means x is 5 units from 3.
- Two answers: $3 + 5$ and $3 - 5$.
- The graph of $y = |x - h| + k$ is a V with vertex (h, k) .



Example

$$|x - 3| = 5 \rightarrow x = 8 \text{ or } x = -2.$$

Solving Equations

$$|A| = B \rightarrow A = B \text{ or } A = -B \quad (B \geq 0)$$

WHAT IT MEANS

- First isolate the absolute value completely.
- If B is negative, there is NO solution.
- If B contains x , check every answer in the original.



Example

$$|2x - 5| = 3x - 4 \rightarrow x = -1 \text{ (fails) or } x = 9/5 \checkmark.$$

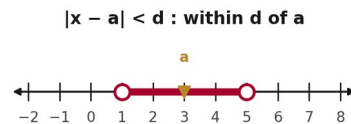
Theory Explained (2/3)

Inequalities

$$|A| < d \rightarrow -d < A < d \quad |A| > d \rightarrow A < -d \text{ or } A > d$$

WHAT IT MEANS

- 'Less than' = within a distance $\rightarrow$ one interval (AND).
- 'Greater than' = beyond a distance $\rightarrow$ two rays (OR).
- Center = the value that makes the inside zero.



Example

$$|x - 5| \geq 6 \rightarrow x \leq -1 \text{ or } x \geq 11.$$

Tolerance Problems

$$|\text{measurement} - \text{target}| \leq \text{tolerance}$$

WHAT IT MEANS

- Target = center; tolerance = allowed distance.
- Accepted window: target $\pm$ tolerance.
- 'Outside by' = distance beyond the window's edge.



Example

$$|V - 500| \leq 4.5 \rightarrow 495.5 \leq V \leq 504.5.$$

Theory Explained (3/3)

Nested & Sums

work from the outside in; $|x - a| + |x - b| \geq |a - b|$

WHAT IT MEANS

- Nested: split the outer bars first, then each inner case.
- A sum of two distances is smallest between the two points.
- Factor out shared coefficients first: $|2x - 10| = 2|x - 5|$.



Example

$||x - 4| - 7| = 3 \rightarrow$ four solutions: $-6, 0, 8, 14$.

Watch Out



Isolate before splitting

- $5 + |2x - 6| = \dots \rightarrow$ subtract 5 first.
- Never split while other terms sit outside the bars.



Check answers when x is outside the bars

- An absolute value cannot equal a negative number.
- Substitute every candidate into the ORIGINAL equation.

Problem 1

Hard

Solve for x : $|2x - 5| = 3x - 4$. Check for extraneous solutions and give only the valid solution(s).

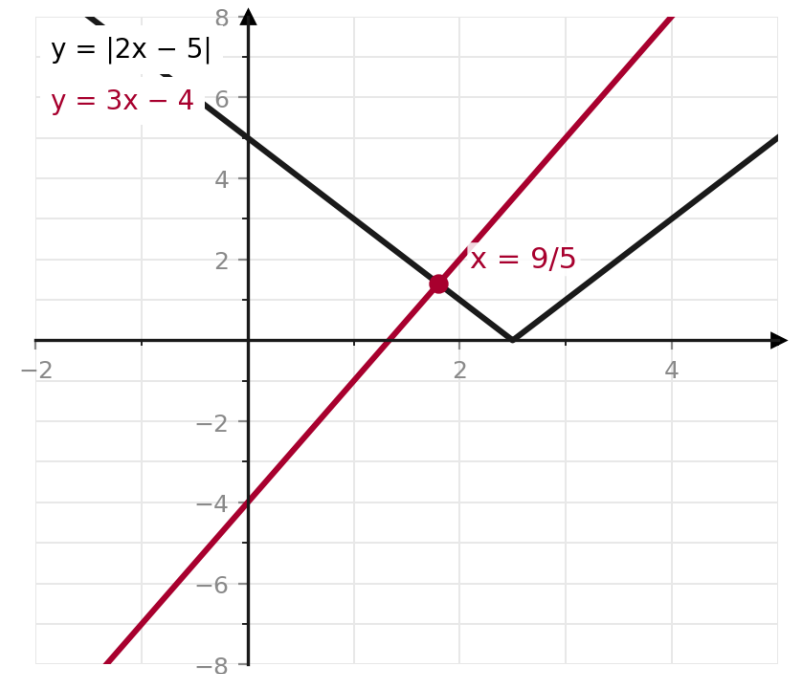
Formula / Rule

$|X| = a$ splits into $X = a$ or $X = -a$; any solution must be checked in the original equation.

Solution

1. Since the variable appears both inside and outside the absolute value bars, split into two cases.
2. Case 1: $2x - 5 = 3x - 4$. Solve: $2x - 3x = -4 + 5$, so $-x = 1$, giving $x = -1$.
3. Case 2: $2x - 5 = -(3x - 4) = -3x + 4$. Solve: $2x + 3x = 4 + 5$, so $5x = 9$, giving $x = 9/5$.
4. Check $x = -1$ in the original equation: left side = $|2(-1) - 5| = |-7| = 7$, right side = $3(-1) - 4 = -7$. Since 7 does not equal -7, $x = -1$ is extraneous and must be rejected.
5. Check $x = 9/5$: left side = $|2(9/5) - 5| = |18/5 - 25/5| = |-7/5| = 7/5$, right side = $3(9/5) - 4 = 27/5 - 20/5 = 7/5$. These match.
6. The only valid solution is $x = 9/5$.

✓ **Answer:** $x = 9/5$



only one intersection → $x = -1$ is extraneous



Problem 1 — Strategy & Elimination



Tip

Whenever the variable also sits outside the absolute value bars, treat both algebraic results as candidates only — substitute each back into the ORIGINAL equation before accepting it.



Shortcut

Before checking by substitution, you can flag a candidate as suspicious instantly by testing whether the bare side ($3x-4$) is negative at that x -value, since an absolute value can never equal a negative number — this marks $x=-1$ as doubtful before any arithmetic.



Trap

A rushed student stops after solving the two linear cases and reports $x = -1$ as a valid answer too, never noticing it forces the absolute value expression to equal a negative number (-7), which is impossible.



Desmos Tip

Graph $y = \text{abs}(2x-5)$ and $y = 3x-4$ and look at the actual intersection point — Desmos shows only one crossing, near $x = 1.8$, confirming $x = -1$ is not a real solution.



Process of Elimination

Correct: $x = 9/5$

Tempting wrong answer	Exactly what went wrong
$x = -1$	Solved Case 1 correctly to $x = -1$ but never substituted it back into the original equation, so did not catch that it makes the right side negative ($3(-1)-4 = -7$), which an absolute value can never equal.
$x = 1/5$	When expanding Case 2's negative sign, wrote $-(3x-4)$ as $-3x-4$ instead of correctly distributing to $-3x+4$, producing the wrong linear equation $2x-5 = -3x-4$ and the wrong root.
$x = -1$ and $x = 9/5$	Listed both algebraic candidates as final answers, treating the extraneous-solution check as optional instead of mandatory whenever the variable sits outside the bars.

Problem 2

Hard

Solve for all values of x : $5 + |2x - 6| = |x + 3| + 5$.

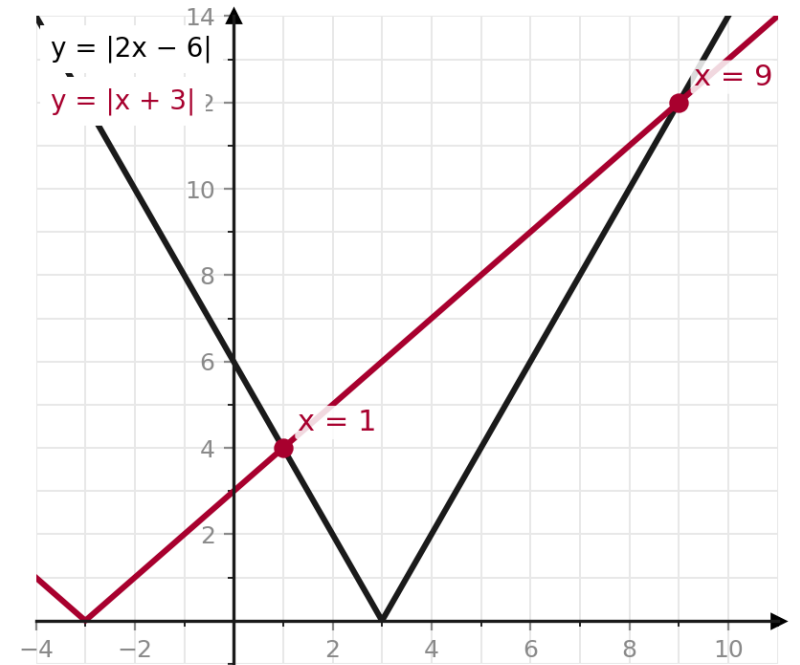
Formula / Rule

$|A| = |B|$ means $A = B$ or $A = -B$.

Solution

1. Subtract 5 from both sides to strip away the disguise: $|2x - 6| = |x + 3|$.
2. Since $|A| = |B|$ means $A = B$ or $A = -B$, set up both cases using $A = 2x - 6$ and $B = x + 3$.
3. Case 1: $2x - 6 = x + 3$. Solve: $2x - x = 3 + 6$, so $x = 9$.
4. Case 2: $2x - 6 = -(x + 3) = -x - 3$. Solve: $2x + x = -3 + 6$, so $3x = 3$, giving $x = 1$.
5. Verify $x = 9$: $|2(9) - 6| = |12| = 12$ and $|9 + 3| = |12| = 12$, so both sides of the simplified equation match.
6. Verify $x = 1$: $|2(1) - 6| = |-4| = 4$ and $|1 + 3| = |4| = 4$, so both sides match here too.
7. Both candidates check out, so the solutions are $x = 1$ and $x = 9$.

✓ **Answer:** $x = 1$ and $x = 9$





Problem 2 — Strategy & Elimination



Tip

First strip away anything added or subtracted outside the bars until the equation is purely $|A| = |B|$, then split into $A = B$ and $A = -B$.



Shortcut

Once simplified to $|2x-6| = |x+3|$, you can solve $2x - 6 = \text{plus-or-minus}(x+3)$ as one combined step instead of writing two fully separate case equations.



Trap

A rushed student splits into cases before simplifying, writing ' $2x-6+5 = x+3$ ' and ' $2x-6+5 = -(x+3)$ ' by dragging the extra +5 into the case equations, which produces the wrong x-values entirely.



Desmos Tip

Graph $y = 5 + \text{abs}(2x-6)$ and $y = \text{abs}(x+3) + 5$ and read off both intersection x-coordinates directly, which confirms $x = 1$ and $x = 9$ without any algebra.



Process of Elimination

Correct: $x = 1$ and $x = 9$

Tempting wrong answer	Exactly what went wrong
$x = 4$	Incorrectly folded the +5 into the bars, solving $2x-6+5 = x+3$ (as if $5+ 2x-6 $ meant $ 2x-6+5 $) instead of first subtracting 5 from both sides to isolate $ 2x-6 = x+3 $.
$x = -2/3$	Made the same folding-in error but from the negative-case version, solving $2x-6+5 = -(x+3)$ instead of the correct Case 2 equation $2x-6 = -(x+3)$.
$x = 9$	Solved only Case 1 ($2x-6 = x+3$) and never set up Case 2 ($2x-6 = -(x+3)$), so reported one solution and missed the second valid solution $x = 1$.

Problem 3

Very Hard

The inequality $5|x - 3| - 7 > k$ has solution set $x < -1$ or $x > 7$. What is the value of k ?

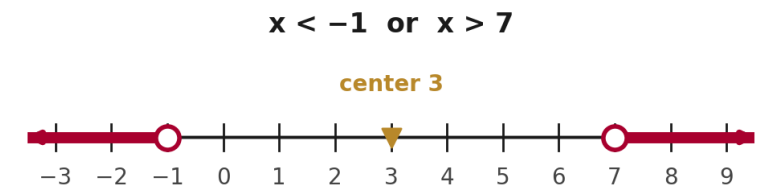
Formula / Rule

$|X| > a$ ($a > 0$) splits into $X > a$ or $X < -a$.

Solution

1. Isolate the absolute value expression first: add 7 to both sides to get $5|x - 3| > k + 7$.
2. Divide both sides by 5: $|x - 3| > (k+7)/5$.
3. An inequality of the form $|X| > a$ splits into $X > a$ or $X < -a$, so $x - 3 > (k+7)/5$ or $x - 3 < -(k+7)/5$.
4. Rewrite each branch by adding 3: $x > 3 + (k+7)/5$, or $x < 3 - (k+7)/5$.
5. Match the given upper boundary $x > 7$ to the first branch: $3 + (k+7)/5 = 7$, so $(k+7)/5 = 4$, giving $k + 7 = 20$, so $k = 13$.
6. Check against the given lower boundary $x < -1$: $3 - (k+7)/5 = -1$, so $(k+7)/5 = 4$ again, giving the same $k = 13$.
7. Both boundaries agree, so $k = 13$.

✓ **Answer:** $k = 13$





Problem 3 — Strategy & Elimination



Tip

When you're given the finished solution set of an absolute value inequality and asked for a hidden constant, rebuild the general $|x - \text{center}| > \text{distance}$ form and match its two boundary numbers to the given ones algebraically.



Shortcut

Since 3 is exactly the midpoint of the two boundary values -1 and 7, the 'radius' distance is $(7 - (-1))/2 = 4$, so $(k+7)/5 = 4$ can be written immediately without deriving both boundary equations separately.



Trap

A rushed student plugs $x = 7$ or $x = -1$ directly into the original inequality as if it were an equation, but forgets to first isolate $|x-3|$ by undoing the -7 and dividing by 5, and often reports $k = 7$ by ignoring the coefficient of 5 entirely.



Desmos Tip

Graph $y = 5 \cdot \text{abs}(x-3) - 7$ and the horizontal line $y = k$, then adjust k until the shaded region's edges land exactly at $x = -1$ and $x = 7$ to visually confirm $k = 13$.



Process of Elimination

Correct: **$k = 13$**

Tempting wrong answer	Exactly what went wrong
$k = 20$	Dropped the -7 when isolating, treating $5 x-3 -7 > k$ as $5 x-3 > k$, which gives $ x-3 > k/5$; matching the boundary $x=7$ then gives $k/5 = 4$, so $k = 20$ instead of 13.
$k = -3$	Correctly isolated the absolute value but forgot to divide by 5, using $3 + (k+7) = 7$ instead of $3 + (k+7)/5 = 7$, giving $k = -3$.
$k = -31/5$	When solving $(k+7)/5 = 4$, cross-multiplied backwards, writing $k+7 = 4/5$ instead of $k+7 = 4(5) = 20$, producing a wrong fractional value for k .

Problem 4

Hard

A quality-control station in Patras accepts a 500 mL bottle only if its actual filled volume V satisfies $|V - 500| \leq 4.5$. One bottle was rejected because its measured volume was outside the acceptable tolerance by exactly 1.8 mL on the low side. What was that bottle's measured volume?

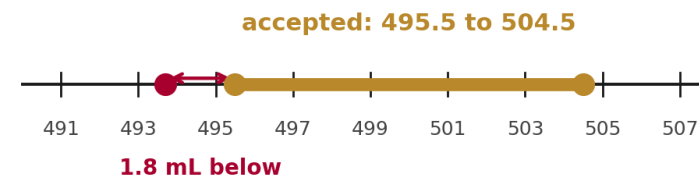
Formula / Rule

Tolerance statements translate to $|\text{actual} - \text{target}| \leq \text{tolerance}$, which unfolds into a compound inequality.

Solution

1. Translate the tolerance rule into an inequality: $|V - 500| \leq 4.5$.
2. Rewrite this without absolute value bars as a compound inequality: $-4.5 \leq V - 500 \leq 4.5$.
3. Add 500 to all three parts: $495.5 \leq V \leq 504.5$, so any volume from 495.5 mL to 504.5 mL passes inspection.
4. The bottle was rejected on the low side, meaning its measured volume fell below the minimum acceptable bound of 495.5 mL.
5. 'Outside the tolerance by exactly 1.8 mL' means the measured volume was 1.8 mL less than that lower bound of 495.5 mL, not 1.8 mL less than the target of 500 mL.
6. Compute: $V = 495.5 - 1.8 = 493.7$ mL.
7. The bottle's measured volume was 493.7 mL.

✓ **Answer:** $V = 493.7$ mL





Problem 4 — Strategy & Elimination



Tip

Translate a tolerance statement 'within c of a target T ' as $|\text{actual} - T| \leq c$, then convert it into a normal compound inequality before doing any arithmetic with the boundary values.



Shortcut

Since the failure happened only on the low side, you can skip finding the upper bound (504.5) entirely and work with just the lower bound $500 - 4.5 = 495.5$, saving a step.



Trap

A rushed student subtracts 1.8 directly from the target 500 (getting 498.2) instead of from the lower boundary 495.5, forgetting that the bottle was already outside the allowed window before being pushed 1.8 mL further away.



Desmos Tip

Type `abs(V-500)<=4.5` into Desmos as a one-variable inequality to see the accepted interval $[495.5, 504.5]$ plotted on the number line, then confirm 493.7 falls outside it on the left.



Process of Elimination

Correct: $V = 493.7$ mL

Tempting wrong answer	Exactly what went wrong
$V = 498.2$ mL	Subtracted 1.8 from the target 500 directly instead of from the lower tolerance bound 495.5, ignoring that the bottle was already at the edge of the accepted window before failing further.
$V = 506.3$ mL	Added 1.8 to the upper bound 504.5 instead of subtracting from the lower bound, misreading 'low side' as meaning the upper boundary was the one exceeded.
$V = 494.2$ mL	Mis-copied the tolerance value as 4 mL instead of 4.5 mL, computing the lower bound as $500 - 4 = 496$, then subtracting 1.8 to get 494.2 mL instead of the correct 493.7 mL.

Problem 5

Very Hard

A tutor in Athens charges 35 euros per hour and is currently reviewing the equation $|4x + 1| = x - 6$ with a student during a 50-minute session. Solve the equation for x , or determine that no real solution exists.

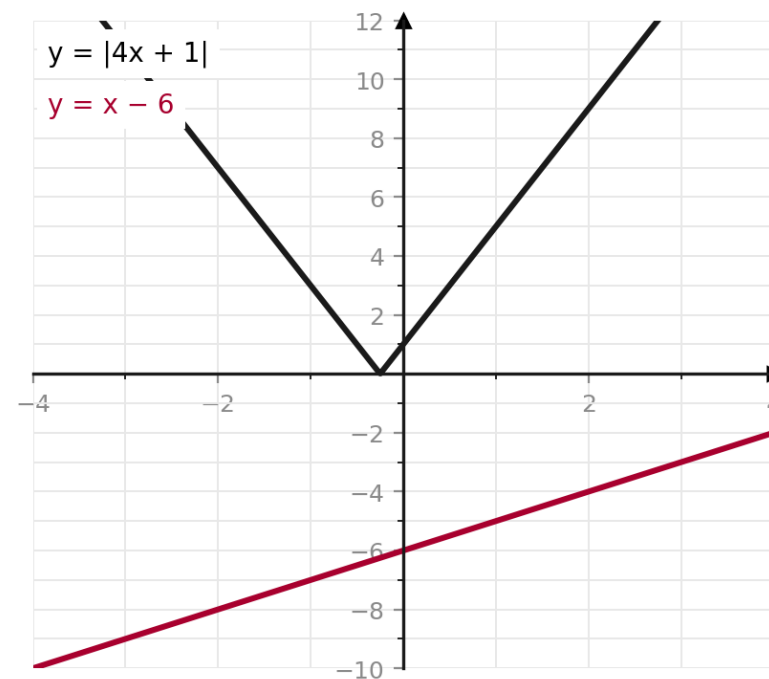
Formula / Rule

$|X|$ can never equal a negative number, so any candidate solution must make the bare side non-negative.

Solution

1. Ignore the tutoring and pricing details — they don't affect the algebra. Split the equation into two cases as usual.
2. Case 1: $4x + 1 = x - 6$. Solve: $4x - x = -6 - 1$, so $3x = -7$, giving $x = -7/3$.
3. Case 2: $4x + 1 = -(x - 6) = -x + 6$. Solve: $4x + x = 6 - 1$, so $5x = 5$, giving $x = 1$.
4. Before accepting either value, note that $|4x+1|$ is never negative, so any true solution requires the right side $x - 6$ to be greater than or equal to 0, meaning $x \geq 6$.
5. Check $x = -7/3$: this does not satisfy $x \geq 6$. Directly verify: $|4(-7/3)+1| = |-28/3+3/3| = |-25/3| = 25/3$, but $x - 6 = -7/3 - 18/3 = -25/3$. Since $25/3$ does not equal $-25/3$, this is extraneous.
6. Check $x = 1$: this also fails $x \geq 6$. Directly verify: $|4(1)+1| = 5$, but $x - 6 = -5$. Since 5 does not equal -5, this is extraneous too.
7. Both algebraic candidates fail the check, so the equation has no real solution.

✓ **Answer:** No solution



the graphs never meet → no solution



Problem 5 — Strategy & Elimination



Tip

Before trusting either algebraic case in an equation like $|\text{expr}| = (\text{linear expression})$, check whether that linear expression can even be non-negative at the candidate x -value — an absolute value can never equal a negative number.



Shortcut

You can often spot 'no solution' quickly by testing only the larger candidate against the requirement $x \geq 6$ first — since even $x = 1$ already fails, both branches are immediately suspect without doing the full substitution check on each.



Trap

A rushed student solves both linear cases, gets $x = -7/3$ and $x = 1$, and reports one or both as answers without ever substituting back into the original equation, missing that neither actually works.



Desmos Tip

Graph $y = \text{abs}(4x+1)$ and $y = x - 6$ — the two curves never intersect anywhere, visually confirming there is no solution.



Process of Elimination

Correct: **No solution**

Tempting wrong answer	Exactly what went wrong
$x = -7/3$	Solved Case 1 ($4x+1 = x-6$) correctly but never checked whether $x-6$ could be negative, submitting the algebraic root without noticing $ 4x+1 $ cannot equal the resulting negative value there.
$x = 1$	Solved Case 2 ($4x+1 = -(x-6)$) correctly but likewise never verified the result against the original equation, missing that it also forces the right side to be negative.
$x = -7/3$ and $x = 1$	Listed both algebraic case solutions as the final answer without performing the required non-negativity check on either one, treating extraneous-solution verification as unnecessary.

Problem 6

Very Hard

Let m be the larger solution to the equation $|3m - 5| = 2m + 1$. Then determine the number of real solutions of the quadratic equation $x^2 + 4x + (m - 9) = 0$ by evaluating its discriminant.

Formula / Rule

Solve $|X| = a$ in two cases first; then use the discriminant $b^2 - 4ac$ to count real roots of a quadratic.



First move

When a problem chains an absolute value equation into a second calculation, fully solve and verify the absolute value part first — including checking every candidate — before touching the second concept.

Solution

1. Solve $|3m - 5| = 2m + 1$ by splitting into two cases.
2. Case 1: $3m - 5 = 2m + 1$. Solve: $3m - 2m = 1 + 5$, so $m = 6$. Check: $|3(6) - 5| = |13| = 13$, and $2(6) + 1 = 13$. Valid.
3. Case 2: $3m - 5 = -(2m + 1) = -2m - 1$. Solve: $3m + 2m = -1 + 5$, so $5m = 4$, giving $m = 4/5$. Check: $|3(4/5) - 5| = |12/5 - 25/5| = |-13/5| = 13/5$, and $2(4/5) + 1 = 8/5 + 5/5 = 13/5$. Also valid.
4. Both $m = 6$ and $m = 4/5$ satisfy the original equation, but the problem asks for the larger one, so $m = 6$.
5. Substitute $m = 6$ into the quadratic: $x^2 + 4x + (6 - 9) = 0$, which simplifies to $x^2 + 4x - 3 = 0$.
6. Compute the discriminant: $b^2 - 4ac = 4^2 - 4(1)(-3) = 16 + 12 = 28$.
7. Since the discriminant 28 is positive, the quadratic has two distinct real solutions.



Answer: Two distinct real solutions ($m = 6$, discriminant = 28)



Problem 6 — Strategy & Elimination



Tip

When a problem chains an absolute value equation into a second calculation, fully solve and verify the absolute value part first — including checking every candidate — before touching the second concept.



Trap

A rushed student uses whichever solution of the absolute value equation they find first (often $m = 4/5$ from Case 2) instead of comparing both valid solutions and correctly selecting the larger one as the problem instructs.



Shortcut

Since both cases of $|3m-5| = 2m+1$ check out here, there's no extraneous-root elimination needed — you can go straight to comparing 6 and $4/5$ and move on to the quadratic without re-verifying anything twice.



Desmos Tip

Graph $y = \text{abs}(3m-5)$ and $y = 2m+1$ to instantly read off both intersection m -values (6 and 0.8), then graph $y = x^2+4x-3$ separately to see it crosses the x -axis twice, confirming two real solutions.



Process of Elimination

Correct: **Two distinct real solutions ($m = 6$, discriminant = 28)**

Tempting wrong answer	Exactly what went wrong
Two distinct real solutions ($m = 4/5$, discriminant = 48.8)	Used the smaller root $m = 4/5$ from Case 2 instead of correctly selecting the larger solution $m = 6$ as the problem requires, propagating the wrong m into the discriminant.
Two distinct real solutions (discriminant = 4)	Substituted $m = 6$ correctly but made a sign error simplifying the constant term, writing $9 - m = 3$ instead of $m - 9 = -3$, giving the wrong quadratic $x^2+4x+3=0$ and the wrong discriminant.
Two distinct real solutions (discriminant = 16)	Forgot to square b in the discriminant formula, using $b = 4$ directly instead of $b^2 = 16$, computing $4 - 4(1)(-3) = 16$ instead of the correct $16 - 4(1)(-3) = 28$.

Problem 7

Very Hard

Find all real solutions to the equation $||x - 4| - 7| = 3$.

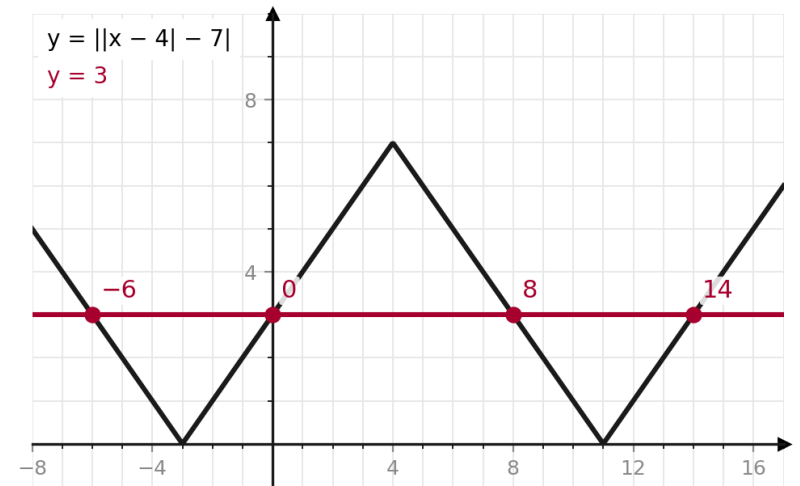
Formula / Rule

Nested absolute value: apply the $|X| = a$ rule to the outer bars first ($X = a$ or $X = -a$), then apply the same rule again to each resulting inner absolute value equation.

Solution

1. Let $X = |x - 4| - 7$, so the equation is $|X| = 3$. This splits into two outer cases: $X = 3$ or $X = -3$.
2. Outer Case A: $|x - 4| - 7 = 3$, so $|x - 4| = 10$.
3. Split Outer Case A into its own two inner cases: $x - 4 = 10$, giving $x = 14$, or $x - 4 = -10$, giving $x = -6$.
4. Outer Case B: $|x - 4| - 7 = -3$, so $|x - 4| = 4$.
5. Split Outer Case B into its own two inner cases: $x - 4 = 4$, giving $x = 8$, or $x - 4 = -4$, giving $x = 0$.
6. Since the outer equation's right side (3) is a positive constant, both outer cases are automatically valid, so no candidate needs to be discarded here.
7. Verify one candidate, $x = 0$: $|0 - 4| = 4$, then $|4 - 7| = |-3| = 3$. Matches the original equation.
8. The four real solutions are $x = -6, 0, 8, 14$.

✓ **Answer:** $x = -6, 0, 8, 14$





Problem 7 — Strategy & Elimination



Tip

When you see an absolute value nested inside another absolute value, work from the outside in — first split the outer $|X| = a$ into $X = a$ and $X = -a$ as two full equations, then split each of those inner absolute value equations again.



Shortcut

Once you find $|x-4| = 10$ and $|x-4| = 4$, you can write all four x -values immediately as 4 ± 10 and 4 ± 4 without redoing separate case equations for each.



Trap

The most common failure is treating this as a single-layer absolute value problem and solving only $|x-4| - 7 = 3$, never realizing that $|x-4| - 7 = -3$ is an equally valid second outer case, which silently discards two of the four correct solutions.



Desmos Tip

Graph $y = \text{abs}(\text{abs}(x-4)-7)$ and $y = 3$, then read off the four x -coordinates where they intersect: $-6, 0, 8$, and 14 .



Process of Elimination

Correct: $x = -6, 0, 8, 14$

Tempting wrong answer	Exactly what went wrong
$x = -6, 14$	Solved only the outer case $ x-4 -7 = 3$, forgetting to also set up $ x-4 -7 = -3$, so missed the two solutions (0 and 8) that come from the second outer case.
$x = 0, 8$	Solved only the outer case $ x-4 -7 = -3$, forgetting the other outer case $ x-4 -7 = 3$, so missed the two solutions (-6 and 14) that come from the first outer case.
$x = 8, 14$	For both inner equations $ x-4 = 10$ and $ x-4 = 4$, took only the positive branch ($x-4=10$ and $x-4=4$) and never solved the negative branch ($x-4=-10$ and $x-4=-4$), so missed $x = -6$ and $x = 0$.

Problem 8

Very Hard

For all real numbers x , what is the minimum possible value of the expression $|2x - 10| + |2x + 4|$?

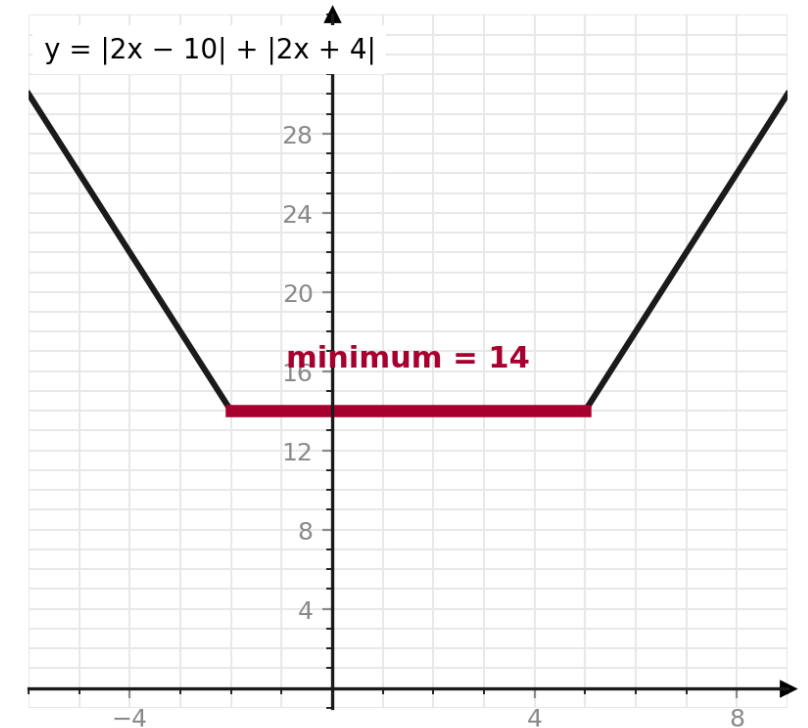
Formula / Rule

$|x-a| + |x-b|$ represents the total distance from x to two fixed points a and b on the number line; its minimum equals $|a-b|$, attained for any x between them. Factor out a shared coefficient before applying this rule.

Solution

- Factor the coefficient 2 out of each absolute value term: $|2x - 10| = |2(x-5)| = 2|x-5|$, and $|2x + 4| = |2(x+2)| = 2|x+2|$.
- The expression becomes $2|x-5| + 2|x+2| = 2(|x-5| + |x+2|)$.
- $|x-5| + |x+2|$ is the sum of the distances from x to the two fixed points 5 and -2 on the number line.
- This sum of distances is minimized whenever x lies anywhere between -2 and 5 (inclusive), and the minimum value equals the distance between the two points: $5 - (-2) = 7$.
- Multiply by the factored-out coefficient of 2: minimum value = $2 \times 7 = 14$.
- Confirm with a test point inside the interval, say $x = 0$: $|2(0)-10| + |2(0)+4| = |-10| + |4| = 10 + 4 = 14$, matching the predicted minimum.

✓ **Answer: 14**





Problem 8 — Strategy & Elimination



Tip

When a sum of two absolute values shares a common coefficient, factor it out first, then recognize $|x-a| + |x-b|$ as the total distance from x to two fixed points — a sum that is minimized between them and equals the distance between the points.



Shortcut

Once you factor out the shared coefficient, no calculus or test points are needed — the minimum of $|x-a| + |x-b|$ is always just the distance between a and b , so $2 \times 7 = 14$ can be written in one line.



Trap

A rushed student forgets to factor out the shared coefficient of 2 and reports the raw anchor-point distance $|5-(-2)| = 7$ as the final answer, never doubling it back to account for the coefficient.



Desmos Tip

Graph $y = \text{abs}(2x-10) + \text{abs}(2x+4)$ and trace the flat bottom of the V-shaped graph between $x = -2$ and $x = 5$, where y stays constant at 14, confirming the minimum.



Process of Elimination

Correct: 14

Tempting wrong answer	Exactly what went wrong
7	Forgot to factor out and reapply the shared coefficient of 2 from each term, computing only the anchor-point distance $ 5-(-2) = 7$ and never multiplying it back by 2.
18	Incorrectly factored $ 2x+4 $ as $2 x+4 $ instead of $2 x+2 $ (a slip in the factoring step), which shifted the second anchor point to -4 and gave $2 \times 5-(-4) = 2 \times 9 = 18$ instead of $2 \times 7 = 14$.
6	Added the two anchor points $(5 + (-2) = 3)$ instead of finding the distance between them $(5 - (-2) = 7)$, then multiplied by 2 to get 6 instead of 14.

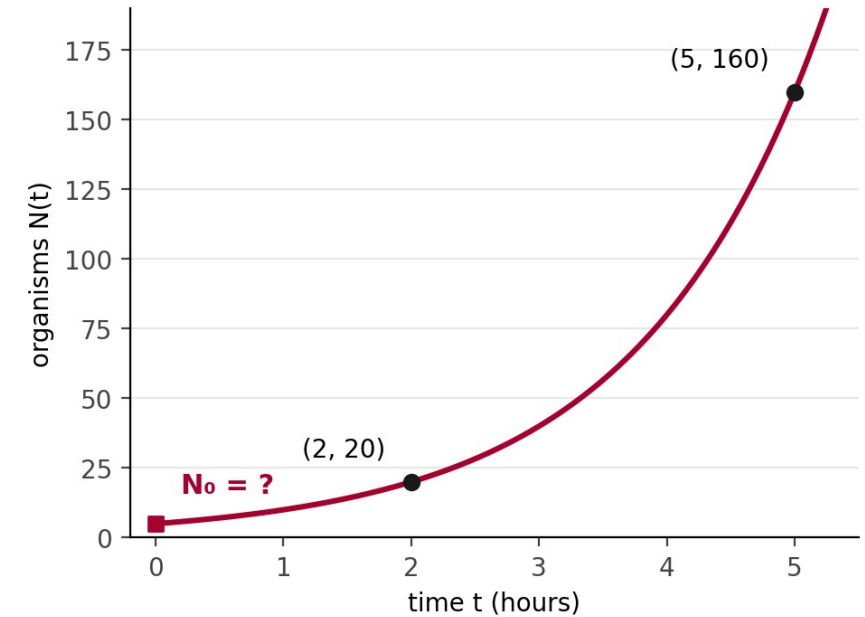
05

Exponent Rules

Simplifying powers and solving exponential equations.

In this section

- Key terms — English / Ελληνικά
- Explained theory with examples
- 8 worked problems (Hard / Very Hard)

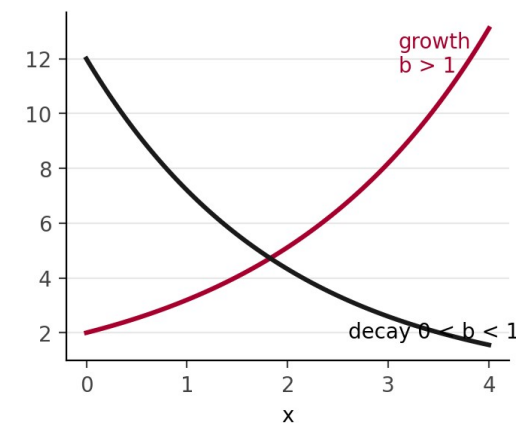


Key Terminology

Term (English)	Ελληνικά
Exponent (power)	Εκθέτης (δύναμη)
Base	Βάση
Exponent rules	Ιδιότητες δυνάμεων
Negative exponent	Αρνητικός εκθέτης
Common base	Κοινή βάση

A ㄱ The big picture

- Every rule follows from repeated multiplication.
- The SAT tests simplifying AND solving.
- Most traps involve negative exponents or common bases.



Theory Explained (1/3)

Product & Quotient

$$a^m \cdot a^n = a^{m+n} \quad a^m / a^n = a^{m-n}$$

WHAT IT MEANS

- Same base only.
- Multiply → add exponents; divide → subtract.
- Subtract top minus bottom, watching double negatives.



Example

$$x^5 / x^{-2} = x^{5-(-2)} = x^7.$$

Power Rules

$$(a^m)^n = a^{mn} \quad (ab)^n = a^n b^n \quad (a/b)^n = a^n / b^n$$

WHAT IT MEANS

- Power of a power: multiply exponents.
- The outer exponent reaches EVERY factor, including numbers.
- Never distribute over + or -: $(a + b)^2 \neq a^2 + b^2$.



Example

$$(2x^3)^4 = 2^4 x^{12} = 16x^{12}.$$

Theory Explained (2/3)

Zero & Negative

$$a^0 = 1 \quad a^{-n} = 1/a^n$$

WHAT IT MEANS

- Any nonzero base to the 0 power is 1.
- A negative exponent means reciprocal, never a negative number.
- Move a factor across the fraction bar to flip its sign.



Example

$$4^{-1} = 1/4, \text{ not } -4.$$

Fractional Exponents

$$a^{(m/n)} = (\sqrt[n]{a})^m$$

WHAT IT MEANS

- Denominator = the root; numerator = the power.
- Take the root first to keep numbers small.
- To solve $x^{(m/n)} = k$, raise both sides to n/m .



Example

$$16^{(3/2)} = (\sqrt{16})^3 = 4^3 = 64.$$

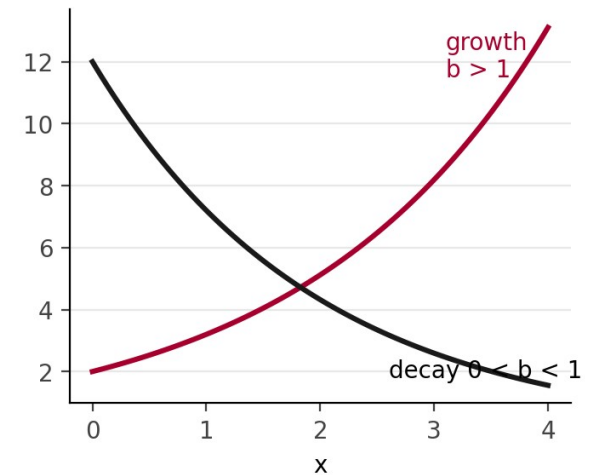
Theory Explained (3/3)

Exponential Equations & Models

$$a^m = a^n \rightarrow m = n \quad y = a \cdot b^x$$

WHAT IT MEANS

- Rewrite both sides with a common base ($9 = 3^2$, $27 = 3^3$).
- Model: a = starting value, b = growth factor.
- Divide two data points to cancel a and isolate b .



Example

$$N(5)/N(2) = 160/20 \rightarrow a^3 = 8 \rightarrow a = 2.$$

Watch Out



$(a + b)^n$ is NEVER $a^n + b^n$

- Exponents distribute over multiplication only.
- Expand sums by multiplying out.



Negative $\neq$ negative value

- $x^{-2} = 1/x^2$, which is positive.
- Rewrite with positive exponents as the final step.

Problem 1

Very Hard

Simplify to a single term with positive exponents only:

$$[(2x^{-3}y^2)^3 \cdot (4x^5y^{-1})^{-1}] / (x^2y^{-3})^0$$

Formula / Rule

Power of a product, product rule, negative exponents, and the zero-exponent rule together.

First move

When a factor in the expression is raised to the 0 power, simplify it to 1 immediately instead of expanding it — it never affects the rest of the problem.

Solution

1. Recognize that any nonzero expression raised to the power 0 equals 1, so the denominator $(x^2y^{-3})^0 = 1$ — it does not need to be expanded.
2. Apply the power rule to the first factor: $(2x^{-3}y^2)^3 = 2^3 \cdot x^{(-3 \cdot 3)} \cdot y^{(2 \cdot 3)} = 8x^{-9}y^6$.
3. Apply the power rule to the second factor: $(4x^5y^{-1})^{-1} = 4^{-1} \cdot x^{(5 \cdot -1)} \cdot y^{(-1 \cdot -1)} = (1/4)x^{-5}y$.
4. Multiply the two results using the product rule (add exponents of like bases): $8x^{-9}y^6 \cdot (1/4)x^{-5}y = ((8 \cdot 1)/4) \cdot x^{-9-5} \cdot y^{6+1} = 2x^{-14}y^7$.
5. Divide by the denominator, which is 1, so the expression is unchanged: $2x^{-14}y^7/1 = 2x^{-14}y^7$.
6. Rewrite with positive exponents only: $2x^{-14}y^7 = 2y^7/x^{14}$.

 **Answer:** $2y^7/x^{14}$



Problem 1 — Strategy & Elimination



Tip

When a factor in the expression is raised to the 0 power, simplify it to 1 immediately instead of expanding it — it never affects the rest of the problem.



Shortcut

Besides recognizing the zero-power factor as 1, there's no shortcut here — the fastest path is still applying the power rule to each bracket separately before combining, exactly as shown.



Trap

A student who doesn't recognize $(x^2y^{-3})^0 = 1$ wastes time expanding it, or worse, mistakenly distributes the 0 as if it were a real exponent to multiply through, instead of just replacing the whole denominator with 1.



Desmos Tip

Pick simple values like $x=2$, $y=1$, evaluate the original expression and the simplified answer $2y^7/x^{14}$ in Desmos's calculator, and confirm they give the same number.



Process of Elimination

Correct: $2y^7/x^{14}$

Tempting wrong answer	Exactly what went wrong
$2x^{-14}y^7$	Correctly combined all exponents but forgot the last step of rewriting negative exponents as positive — stopped before flipping x^{-14} to $1/x^{14}$.
$2y^5/x^{14}$	Mis-applied the power rule to $(4x^5y^{-1})^{-1}$, treating $y^{(-1 \cdot -1)}$ as y^{-1} instead of y^1 , which lowers the final y exponent from 7 to 5.
$-32y^7/x^{14}$	Misinterpreted the negative exponent on the coefficient, computing 4^{-1} as -4 instead of the reciprocal $1/4$, which corrupts the numeric coefficient in the final answer.

Problem 2

Hard

Solve for x : $3 \cdot 9^{(2x - 1)} = 27^{(x + 1)}$

Formula / Rule

Rewriting exponential expressions with a common base, then setting exponents equal.



First move

Convert every term to the same base first, remembering to fold any leading coefficient (like the 3-) into the exponent before comparing powers.

Solution

1. Rewrite all bases as powers of 3: $9 = 3^2$ and $27 = 3^3$.
2. Rewrite the left side: $3 \cdot 9^{(2x - 1)} = 3^1 \cdot (3^2)^{(2x - 1)} = 3^1 \cdot 3^{(4x - 2)} = 3^{(4x - 1)}$.
3. Rewrite the right side: $27^{(x + 1)} = (3^3)^{(x + 1)} = 3^{(3x + 3)}$.
4. Since both sides now have base 3, set the exponents equal: $4x - 1 = 3x + 3$.
5. Solve for x : $4x - 3x = 3 + 1$, so $x = 4$.
6. Check: left exponent = $4(4) - 1 = 15$; right exponent = $3(4) + 3 = 15$. Both sides equal 3^{15} , confirming $x = 4$.

**Answer: $x = 4$**



Problem 2 — Strategy & Elimination



Tip

Convert every term to the same base first, remembering to fold any leading coefficient (like the 3·) into the exponent before comparing powers.



Shortcut

There's no faster route than converting to a common base — but once converted, you can skip fully expanding and just compare the coefficient-adjusted exponents ($4x-1$ vs $3x+3$) in one line.



Trap

A student who ignores the leading factor of 3 and sets $2(2x-1)$ equal directly to $3(x+1)$ gets $4x-2=3x+3$, producing the wrong answer $x = 5$ instead of the correct $x = 4$.



Desmos Tip

Graph $y = 3 \cdot 9^{(2x - 1)}$ and $y = 27^{(x + 1)}$ and find their intersection point; the x-coordinate should read exactly 4.



Process of Elimination

Correct: **$x = 4$**

Tempting wrong answer	Exactly what went wrong
$x = 5$	Ignored the leading factor of 3 outside $9^{(2x - 1)}$ and set $2(2x-1)$ equal directly to $3(x+1)$, giving $4x-2=3x+3$, which solves to $x=5$.
$x = 2$	Incorrectly distributed the outer exponent on $27^{(x + 1)}$, writing it as $3^{(3x + 1)}$ instead of $3^{(3x + 3)}$ (forgetting to multiply the +1 by 3 as well), then solved $4x-1=3x+1$.
$x = 1.5$	Misremembered 27 as 3^2 instead of 3^3 , giving the right side exponent $2x+2$ instead of $3x+3$, and solved $4x-1=2x+2$.

Problem 3

Hard

A positive number k satisfies $k^{2/3} = 16$. What is the value of $k^{3/2}$?

Formula / Rule

Fractional exponents; solve by raising both sides to the reciprocal power.

First move

When you're given a fractional power of an unknown and asked for a different fractional power of it, solve for the unknown first using the reciprocal-power move, then evaluate the new expression from scratch.

Solution

1. To undo the exponent $2/3$ on k , raise both sides to the reciprocal power, $3/2$: $(k^{2/3})^{3/2} = 16^{3/2}$.
2. The left side simplifies because $(2/3)(3/2) = 1$: $k^1 = 16^{3/2}$.
3. Evaluate the right side: $16^{3/2} = (16^{1/2})^3 = 4^3 = 64$. So $k = 64$.
4. Now compute the requested value, $k^{3/2}$, using $k = 64$: $64^{3/2} = (64^{1/2})^3$.
5. Evaluate the square root first: $64^{1/2} = 8$.
6. Cube the result: $8^3 = 512$.

 **Answer:** $k^{3/2} = 512$



Problem 3 — Strategy & Elimination



Tip

When you're given a fractional power of an unknown and asked for a different fractional power of it, solve for the unknown first using the reciprocal-power move, then evaluate the new expression from scratch.



Shortcut

Since $k = 64$ works out cleanly, you can track it as $k^{(2/3)} = 16$ means $k = 4^3 = 64$, then $k^{(3/2)} = 8^3 = 512$ directly, without ever writing radical notation.



Trap

A student may notice that $3/2$ is the reciprocal of $2/3$ and wrongly assume $k^{(3/2)}$ is just $1/16$, the reciprocal of the given value — but flipping an exponent does not flip the value; k must actually be solved for first.



Desmos Tip

Graph $y = x^{(2/3)}$ and find where $y = 16$ to read off $k = 64$, then evaluate $64^{(3/2)}$ directly in the calculator to confirm 512.



Process of Elimination

Correct: $k^{(3/2)} = 512$

Tempting wrong answer	Exactly what went wrong
$k^{(3/2)} = 1/16$	Assumed that since $3/2$ is the reciprocal exponent of $2/3$, the value must also be the reciprocal of 16, without actually solving for k first.
$k^{(3/2)} = 36$	Treated the exponent $3/2$ as a multiplier rather than a power throughout, computing $16^{(3/2)}$ as $(16 \cdot 3)/2 = 24$ (instead of 64) and then repeating the same error on the final step to get $(24 \cdot 3)/2 = 36$.
$k^{(3/2)} = 8$	Correctly solved $k=64$ but only took the square root ($64^{(1/2)} = 8$) and forgot to then cube the result to finish evaluating $64^{(3/2)} = 512$.

Problem 4

Very Hard

Suppose b is a positive number such that $b^n = 9$ and $b^{(n+4)} = 729$. What is the value of $b^{(n-2)}$?

Formula / Rule

Quotient rule for exponents (dividing two equations with a common base), then the zero-exponent rule.



First move

When two exponential equations share the same base but different exponents, divide one equation by the other to cancel the base's unknown power and isolate a numeric power of b you can solve directly.

Solution

1. Divide the second equation by the first to eliminate the unknown exponent structure: $b^{n+4}/b^n = 729/9$.
2. Simplify the left side using the quotient rule: $b^{(n+4-n)} = b^4$.
3. So $b^4 = 81$. Solve for b by taking the fourth root (b is positive): $b = 81^{(1/4)} = 3$.
4. Substitute $b = 3$ into the first equation to find n : $3^n = 9 = 3^2$, so $n = 2$.
5. Evaluate the requested expression: $b^{(n-2)} = 3^{2-2} = 3^0$.
6. Apply the zero-exponent rule: $3^0 = 1$.
7. Check by a second route: $b^{n-2} = b^n/b^2 = 9/9 = 1$, confirming the answer.



Answer: $b^{(n-2)} = 1$



Problem 4 — Strategy & Elimination



Tip

When two exponential equations share the same base but different exponents, divide one equation by the other to cancel the base's unknown power and isolate a numeric power of b you can solve directly.



Trap

A student who solves $b^4 = 81$ by writing $b = 81/4 = 20.25$ (dividing instead of taking the fourth root) gets a nonsensical base, instead of correctly evaluating $b = 81^{(1/4)} = 3$.



Shortcut

You can skip solving for n entirely: $b^{n-2} = b^n/b^2$, and $b^2 = \sqrt{b^4} = \sqrt{81} = 9$, so $b^{n-2} = 9/9 = 1$ in two lines.



Process of Elimination

Correct: $b^{(n-2)} = 1$

Tempting wrong answer	Exactly what went wrong
0	Misinterpreted $b^{(n-2)}$ as b^n minus b^2 (subtracting values) rather than b^n divided by b^2 (subtracting exponents), computing $9 - 9 = 0$ instead of $9/9 = 1$.
81	Divided the two given equations in the wrong order, getting $b^{-4} = 1/81$, then misread the negative exponent as if the equation were $b^4 = 1/81$, giving $b = 1/3$ instead of 3; carrying this incorrect base through the rest of the problem produces 81 instead of the correct 1.
9	Misread the requested expression as b^n instead of $b^{(n-2)}$, plugging $n=2$ directly into $b^n = 3^2 = 9$ rather than first evaluating the exponent $n-2 = 0$.

Problem 5

Hard

Simplify to a single fraction with positive exponents only:

$$(x^{-3}y^2)/(x^2y^{-4})^{-1}$$

Formula / Rule

Negative exponents and the quotient rule — distributing a negative outer exponent across a product.



First move

When an outer exponent applies to a product of two factors, distribute it to both factors' exponents — including any factor that is already negative — before simplifying further.

Solution

1. Apply the outer exponent -1 to every factor inside the parentheses in the denominator: $(x^2y^{-4})^{-1} = x^{2 \cdot -1} \cdot y^{-4 \cdot -1} = x^{-2} \cdot y^4$.
2. Rewrite the full expression using this result: $(x^{-3}y^2)/(x^{-2}y^4)$.
3. Apply the quotient rule to the x terms: $x^{-3}/x^{-2} = x^{-3-(-2)} = x^{-1}$.
4. Apply the quotient rule to the y terms: $y^2/y^4 = y^{2-4} = y^{-2}$.
5. Combine the results: $x^{-1} \cdot y^{-2}$.
6. Rewrite with positive exponents: $x^{-1}y^{-2} = 1/(xy^2)$.



Answer: $1/(xy^2)$



Problem 5 — Strategy & Elimination



Tip

When an outer exponent applies to a product of two factors, distribute it to both factors' exponents — including any factor that is already negative — before simplifying further.



Trap

A student may distribute the outer -1 to x^2 (getting x^{-2}) but leave y^{-4} unchanged, reasoning it's 'already negative,' getting the wrong intermediate $x^{-2}y^{-4}$ instead of the correct $x^{-2}y^4$ — this flips the final sign on y and produces the wrong answer y^6/x instead of the correct $1/(xy^2)$.



Shortcut

No shortcut beats carefully distributing the outer exponent first — but once distributed, you can combine both quotient-rule subtractions ($-3-(-2)$ and $2-4$) on the same line instead of writing them separately.



Desmos Tip

Substitute a value like $x=2$, $y=3$ into both the original expression and the simplified answer $1/(xy^2)$ in Desmos's calculator to confirm they evaluate to the same number.



Process of Elimination

Correct: $1/(xy^2)$

Tempting wrong answer	Exactly what went wrong
y^6/x	Distributed the outer -1 exponent to x^2 but left y^{-4} unchanged (reasoning it was 'already negative'), producing $x^{-2}y^{-4}$ instead of $x^{-2}y^4$, which flips the sign on y 's exponent in the final step.
$1/(x^5y^2)$	Applied the quotient rule to the x terms as $x^{-3} - 2$ instead of $x^{-3} - (-2)$, mishandling the double negative and getting x^{-5} instead of x^{-1} .
y^2/x	Subtracted the y exponents in the wrong order ($4-2$ instead of $2-4$) when applying the quotient rule, getting y^2 instead of y^{-2} , which flips y into the numerator incorrectly.

Problem 6

Very Hard

A biotech lab in Rotterdam is tracking a bacterial culture that grows according to $N(t) = N_0 \cdot a^t$, where t is measured in hours and a is a constant growth factor. The technician recorded that at $t=2$ hours the population was 20 organisms, and at $t=5$ hours it was 160 organisms. The lab's petri dish can hold up to 10,000 organisms before needing to be split. Assuming the growth follows this exponential model exactly, what was the population N_0 at $t=0$, when the sample was first placed in the dish?

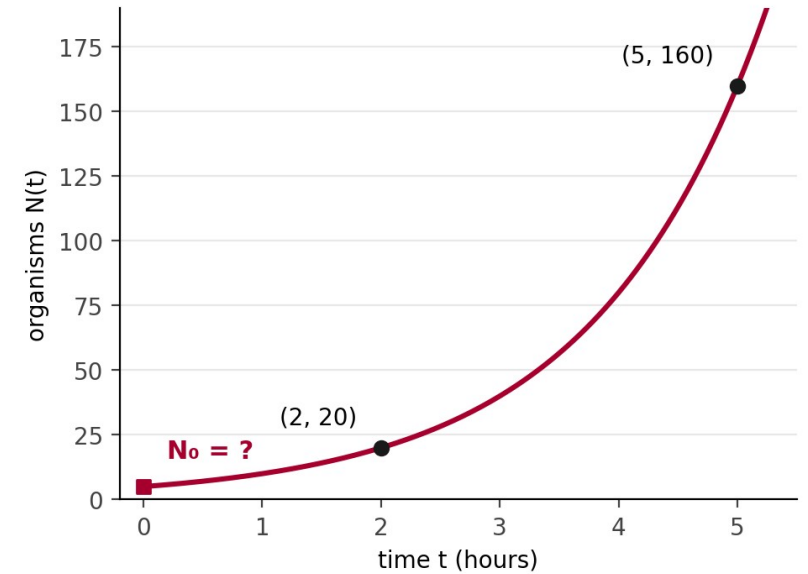
Formula / Rule

Exponential growth model $N(t) = N_0 \cdot a^t$; use the quotient rule to eliminate N_0 and solve the resulting equation.

Solution

1. Translate the two data points into equations: $N_0 \cdot a^2 = 20$ and $N_0 \cdot a^5 = 160$. (The 10,000-organism dish capacity is not needed for this calculation.)
2. Divide the second equation by the first to eliminate N_0 : $((N_0 \cdot a^5))/(N_0 \cdot a^2) = 160/20$.
3. Simplify the left side using the quotient rule: $a^{5-2} = a^3$.
4. So $a^3 = 8$, which gives $a = 2$.
5. Substitute $a = 2$ back into the first equation: $N_0 \cdot 2^2 = 20$, so $4N_0 = 20$.
6. Solve for N_0 : $N_0 = 20/4 = 5$.
7. Check: $N(5) = 5 \cdot 2^5 = 5 \cdot 32 = 160$, which matches the given data.

✓ **Answer:** $N_0 = 5$ organisms





Problem 6 — Strategy & Elimination



Tip

In a growth word problem with two data points instead of a known starting value, divide the two exponential equations to cancel N_0 first — that isolates a power of the growth factor a you can solve directly.



Trap

A rushed student divides the equations in the wrong order ($N(2)/N(5)$) instead of $N(5)/N(2)$, getting $a^{-3} = 20/160 = 1/8$, which leads to $a = 1/2$ and implies the culture is shrinking — the reverse of what the data actually shows.



Shortcut

Dividing the two given equations to eliminate N_0 immediately, as shown, is already the fastest route — solving for N_0 or a individually first (e.g., using logarithms) would take longer.



Desmos Tip

Graph $N(t) = 5 \cdot 2^t$ in Desmos and confirm the curve passes through both (2, 20) and (5, 160) to verify the fitted model before trusting $N_0 = 5$.



Process of Elimination

Correct: **$N_0 = 5$ organisms**

Tempting wrong answer	Exactly what went wrong
80 organisms	Divided the equations in the wrong order ($N(2)/N(5)$) instead of $N(5)/N(2)$, getting $a = 1/2$ instead of $a=2$, then solved $N_0 = 20/((1/2)^2) = 80$ instead of 5.
20 organisms	Misread the question and reported the population at $t=2$ (which was given directly as 20) instead of solving for N_0 , the population at $t=0$.
0.625 organisms	After correctly finding $a=2$, substituted it into the $t=5$ equation but used the $t=2$ population value (20) instead of the $t=5$ value (160), mismatching the exponent and the data point: $N_0 = 20/2^5 = 0.625$.

Problem 7

Very Hard

Simplify to a single number: $(27^{-2/3} \cdot 9^{3/2}) / 3^{-1}$

Formula / Rule

Rewriting mixed bases as powers of a common base, then combining fractional and negative exponents.



First move

When a problem mixes different bases with fractional and negative exponents, first rewrite everything as a power of the smallest common base, then combine exponents using the rules — never evaluate the messy fractional powers by estimation.

Solution

1. Rewrite 27 and 9 as powers of 3: $27 = 3^3$ and $9 = 3^2$.
2. Rewrite the first factor: $27^{-2/3} = (3^3)^{-2/3} = 3^{3 \cdot (-2/3)} = 3^{-2}$.
3. Rewrite the second factor: $9^{3/2} = (3^2)^{3/2} = 3^{2 \cdot (3/2)} = 3^3$.
4. Multiply the numerator's two factors using the product rule: $3^{-2} \cdot 3^3 = 3^{-2 + 3} = 3^1 = 3$.
5. Rewrite the denominator: $3^{-1} = 1/3$.
6. Divide the numerator by the denominator: $3 \div (1/3) = 3 \cdot 3 = 9$.
7. Check with a second route: the net exponent of 3 across the whole expression is $-2 + 3 - (-1) = 2$, so the value is $3^2 = 9$, matching.



Answer: 9



Problem 7 — Strategy & Elimination



Tip

When a problem mixes different bases with fractional and negative exponents, first rewrite everything as a power of the smallest common base, then combine exponents using the rules — never evaluate the messy fractional powers by estimation.



Shortcut

Since every base reduces to 3, you can skip writing 27 and 9 explicitly and just track the net exponent of 3 in one line: $-2 + 3 - (-1) = 2$, so the answer is $3^2 = 9$.



Trap

A student who tries to evaluate $27^{(-2/3)}$ and $9^{(3/2)}$ as decimals right away (rather than rewriting both as powers of 3 first) is likely to make an arithmetic slip on the negative fractional exponent and lose track of the final division by 3^{-1} , often forgetting that dividing by 3^{-1} means multiplying by 3.



Desmos Tip

Type the original expression directly into Desmos's calculator ($(27^{(-2/3)} \cdot 9^{(3/2)})/3^{-1}$) and confirm it evaluates to exactly 9.



Process of Elimination

Correct: 9

Tempting wrong answer	Exactly what went wrong
1	Correctly simplified the numerator to $3^1 = 3$ but then divided by 3^{-1} as if dividing by 3 (instead of multiplying by 3), computing $3/3 = 1$ instead of $3 \times 3 = 9$.
-9	Correctly reached $3^1 \div 3^{-1}$ but wrongly assumed dividing by a negative-exponent term makes the result negative, writing -9 instead of correctly computing $3^1 \cdot 3^1 = 9$.
1/81	When combining the numerator's exponents $3^{-2} \cdot 3^3$, subtracted instead of added them ($-2-3=-5$ instead of $-2+3=1$), giving $3^{-5} = 1/243$, then divided by 3^{-1} to get 1/81 instead of 9.

Problem 8

Hard

Suppose $5^m = 3$ and $5^n = 12$, where m and n are positive real numbers with $m + n < 10$ (this bound is not needed for the calculation). What is the value of $5^{(2m - n)}$?

Formula / Rule

Power and quotient rules applied directly to two given exponential equations sharing a base.

First move

When asked for an expression like $a^{(pm + qn)}$ built from two given exponential equations with the same base, rewrite it as a product or quotient of powers of the two known quantities, $(5^m)^p$ and $(5^n)^q$, rather than trying to solve for m and n individually.

Solution

1. Rewrite the target expression using exponent rules: $5^{2m-n} = 5^{2m}/5^n = (5^m)^2/5^n$. (The condition $m+n<10$ is extra information not needed here.)
2. Substitute the known value for the numerator: $5^m = 3$, so $(5^m)^2 = 3^2 = 9$.
3. Substitute the known value for the denominator: $5^n = 12$.
4. Divide: $5^{2m-n} = 9/12$.
5. Simplify the fraction: $9/12 = 3/4$.
6. Check: this used only the two given equations combined via the power and quotient rules — the bound $m+n<10$ never entered the calculation.

 **Answer:** $5^{2m-n} = 3/4$



Problem 8 — Strategy & Elimination



Tip

When asked for an expression like $a^{(pm + qn)}$ built from two given exponential equations with the same base, rewrite it as a product or quotient of powers of the two known quantities, $(5^m)^p$ and $(5^n)^q$, rather than trying to solve for m and n individually.



Trap

A student may try to solve for m and n individually using logarithms ($m = \log_5(3)$, $n = \log_5(12)$) and then compute $2m - n$ numerically, which is slower and invites rounding errors — or may get distracted trying to use the irrelevant bound $m + n < 10$ to set up a system of equations that was never needed.



Shortcut

Since $2m - n$ breaks cleanly into $2m$ minus n , you can substitute directly as $(5^m)^2/5^n = 9/12$ in one line without ever solving for m or n individually — this is already the fastest method.



Process of Elimination

Correct: $5^{2m-n} = 3/4$

Tempting wrong answer	Exactly what went wrong
1/2	Misapplied the power rule, treating $5^{(2m)}$ as $2 \cdot 5^m$ instead of $(5^m)^2$, computing $((2 \cdot 3))/12 = 1/2$ instead of $3^2/12 = 3/4$.
108	Misread the exponent as $2m+n$ instead of $2m-n$, multiplying $(5^m)^2$ by 5^n instead of dividing, getting $9 \times 12 = 108$.
4/3	Set up the quotient upside down, computing $5^n/(5^m)^2 = 12/9 = 4/3$ instead of $(5^m)^2/5^n = 9/12 = 3/4$.

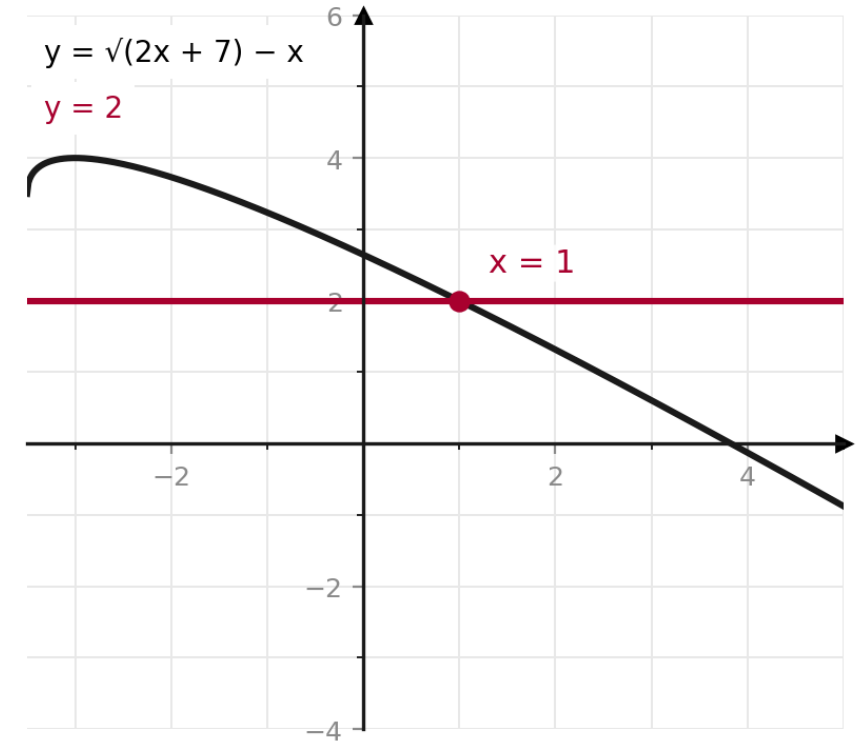
06

Radicals, Rational Exponents & Scientific Notation

Simplifying roots, radical equations, and powers of ten.

In this section

- Key terms — English / Ελληνικά
- Explained theory with examples
- 8 worked problems (Hard / Very Hard)



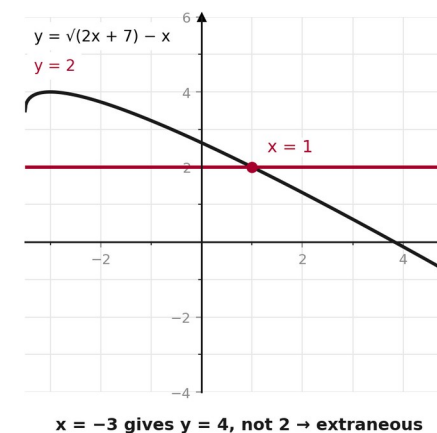
$x = -3$ gives $y = 4$, not 2 → extraneous

Key Terminology

Term (English)	Ελληνικά
Radical (root)	Ρίζα (ριζικό)
Rational exponent	Ρητός εκθέτης
Simplify a radical	Απλοποίηση ρίζας
Scientific notation	Επιστημονική σημειογραφία

A The big picture

- Radicals are fractional exponents in disguise.
- Radical equations can create false (extraneous) solutions.
- Scientific notation is exponent rules applied to 10.



Theory Explained (1/3)

Radicals ↔ Exponents

$$\sqrt[n]{a^m} = a^{(m/n)} \quad \sqrt{x} = x^{(1/2)}$$

WHAT IT MEANS

- Convert every radical to an exponent before combining.
- Then use the exponent rules normally.
- Common denominators are needed to add exponents.



Example

$$\sqrt[3]{x^2} \cdot \sqrt{x^3} = x^{(2/3 + 3/2)} = x^{(13/6)}.$$

Simplifying Radicals

$$\sqrt{ab} = \sqrt{a} \cdot \sqrt{b} \quad \sqrt{a} / \sqrt{b} = \sqrt{a/b}$$

WHAT IT MEANS

- Pull out perfect-square factors.
- Combine into one radical when dividing.
- Assume variables are positive unless told otherwise.



Example

$$\sqrt{72} = \sqrt{36} \cdot \sqrt{2} = 6\sqrt{2}.$$

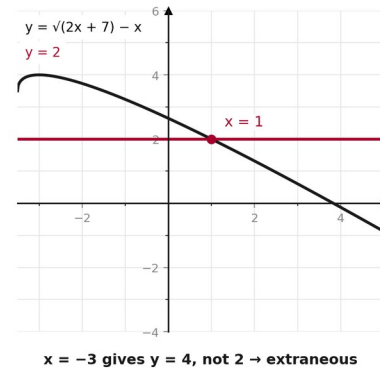
Theory Explained (2/3)

Radical Equations

isolate → square both sides → CHECK

WHAT IT MEANS

- Isolate the radical before squaring.
- Squaring can introduce extraneous roots.
- Every candidate must satisfy the ORIGINAL equation.



Example

$$\sqrt{2x + 7} - x = 2 \rightarrow x = 1 \checkmark, x = -3 \text{ X}.$$

Scientific Notation

$$a \times 10^n, 1 \leq a < 10$$

WHAT IT MEANS

- Multiply: multiply coefficients, add exponents.
- Divide: divide coefficients, subtract exponents.
- Renormalize: $0.75 \times 10^{-16} = 7.5 \times 10^{-17}$.



Example

$$(3.6 \times 10^{31}) / (4.8 \times 10^{47}) = 7.5 \times 10^{-17}.$$

Theory Explained (3/3)

Similar-Solid Scaling

$$\text{length} \times k \rightarrow \text{area} \times k^2 \rightarrow \text{volume} \times k^3$$

WHAT IT MEANS

- Area ratio $\rightarrow$ take the square root for the linear ratio.
- Then cube it for the volume ratio.
- Never apply an area ratio to volume directly.



Example

Area ratio $1/4 \rightarrow$ linear $1/2 \rightarrow$ volume $1/8$.

Watch Out



Square roots are never negative

- $\sqrt{9} = 3$, not ± 3 .
- This is why squaring creates extraneous solutions.



Coefficient between 1 and 10

- Always renormalize the final answer.
- Moving the decimal left raises the exponent.

Problem 1

Hard

Simplify completely (assume $x > 0$ and $y > 0$): $\sqrt[3]{(48x^5y^7)} / \sqrt[3]{(3xy^3)}$

Formula / Rule

Quotient rule for radicals: $\sqrt[n]{a}/\sqrt[n]{b} = \sqrt[n]{(a/b)}$; simplify using exponent subtraction.

First move

When you see a quotient of two radicals with the same index, combine them into a single radical first — it is almost always less arithmetic than simplifying each radical separately and then dividing.

Solution

1. Combine the two radicals into one using the quotient rule:
 $\sqrt[3]{48x^5y^7}/\sqrt[3]{3xy^3} = \sqrt[3]{(48x^5y^7/(3xy^3))}$.
2. Simplify the numerical coefficient inside the radical: $48/3 = 16$.
3. Simplify the x terms using exponent subtraction: $x^5/x^1 = x^{5-1} = x^4$.
4. Simplify the y terms using exponent subtraction: $y^7/y^3 = y^{7-3} = y^4$.
5. The expression under the radical is now: $\sqrt[3]{16x^4y^4}$.
6. Since 16, x^4 , and y^4 are all perfect squares, split the radical: $\sqrt[3]{16} \cdot \sqrt[3]{x^4} \cdot \sqrt[3]{y^4} = 4 \cdot x^{4/3} \cdot y^{4/3}$.
7. Final simplified expression: $4x^{4/3}y^{4/3}$.

 **Answer:** $4x^{4/3}y^{4/3}$



Problem 1 — Strategy & Elimination



Tip

When you see a quotient of two radicals with the same index, combine them into a single radical first — it is almost always less arithmetic than simplifying each radical separately and then dividing.



Trap

Simplifying $\sqrt{48x^5y^7}$ and $\sqrt{3xy^3}$ separately before dividing invites an exponent bookkeeping error — for instance writing y^7/y^3 as y^3 instead of y^4 — which lands on the wrong final answer $4x^2y^3$ instead of the correct $4x^2y^2$.



Shortcut

Combining under one radical first turns four separate simplifications into a single division, which is already the fastest route on this problem — there is no faster alternate method.



Desmos Tip

Plug in a test value like $x=2$, $y=1$ into both the original expression $\sqrt{48x^5y^7}/\sqrt{3xy^3}$ and the simplified form $4x^2y^2$ on Desmos's calculator to confirm they give the same number.



Process of Elimination

Correct: $4x^2y^2$

Tempting wrong answer	Exactly what went wrong
$4x^2y^4$	Took the square root of 16 and x^4 but left y^4 unrooted.
$4x^{(5/2)}y^2$	Forgot to divide the x-terms: kept x^5 under the root instead of x^4 .
$16x^2y^2$	Reduced inside the root to $16x^4y^4$ but never took $\sqrt{16} = 4$.

Problem 2

Hard

Solve for x : $\sqrt{2x + 7} - x = 2$. State the value(s) of x that satisfy the original equation.

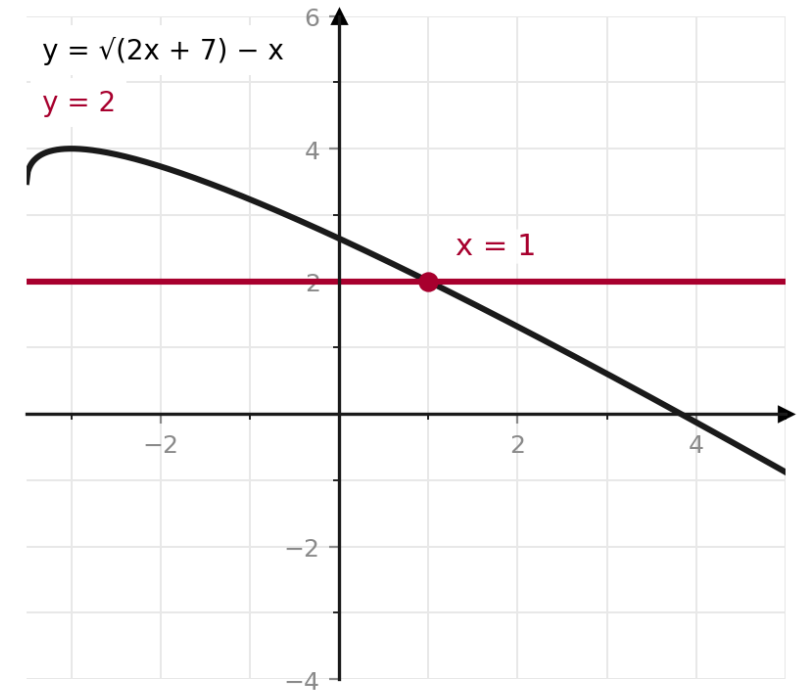
Formula / Rule

Isolate the radical, square both sides, solve the resulting equation, and check every solution in the original equation.

Solution

1. Isolate the radical by adding x to both sides: $\sqrt{2x + 7} = x + 2$.
2. Square both sides to eliminate the radical: $2x + 7 = (x + 2)^2 = x^2 + 4x + 4$.
3. Move all terms to one side: $0 = x^2 + 4x + 4 - 2x - 7 = x^2 + 2x - 3$.
4. Factor the quadratic: $x^2 + 2x - 3 = (x + 3)(x - 1)$, so $(x + 3)(x - 1) = 0$.
5. Solve for x : $x = -3$ or $x = 1$.
6. Check $x = 1$ in the ORIGINAL equation: $\sqrt{2(1) + 7} - 1 = \sqrt{9} - 1 = 3 - 1 = 2$. This works.
7. Check $x = -3$ in the ORIGINAL equation: $\sqrt{2(-3) + 7} - (-3) = \sqrt{1 + 3} = 1 + 3 = 4$, which does not equal 2. This solution is extraneous.
8. Reject $x = -3$ and keep only the valid solution.

✓ **Answer:** $x = 1$



$x = -3$ gives $y = 4$, not 2 → extraneous



Problem 2 — Strategy & Elimination



Tip

After squaring both sides of a radical equation, treat every solution as a candidate only — always substitute back into the ORIGINAL (unsquared) equation before finalizing an answer.



Shortcut

You can eliminate $x = -3$ without recomputing anything: since $\sqrt{2x + 7}$ is always non-negative, it can only equal $x + 2$ when $x + 2$ is also non-negative ($x \geq -2$); $x = -3$ fails this instantly.



Trap

A student who solves the quadratic correctly but forgets to check both roots in the original equation (not the squared one) will report both $x = 1$ and $x = -3$, missing that $x = -3$ is extraneous because the left side actually evaluates to 4, not 2.



Desmos Tip

Graph $y = \sqrt{2x + 7} - x$ and $y = 2$ and find their intersection point directly — Desmos will show only one true intersection, at $x = 1$, immediately revealing that $x = -3$ is extraneous.



Process of Elimination

Correct: **$x = 1$**

Tempting wrong answer	Exactly what went wrong
$x = 1$ and $x = -3$	Forgot to substitute both candidate roots back into the ORIGINAL (unsquared) equation, so didn't catch that $x = -3$ is extraneous, and reported both roots as valid.
$x = -3$	Assumed without checking that the negative-valued root must be the extraneous one, rejected $x = 1$ and kept $x = -3$ instead of actually verifying each root in the original equation.
$x = 3$	Mis-factored $x^2 + 2x - 3$ as $(x - 3)(x + 1)$ instead of the correct $(x + 3)(x - 1)$, producing roots $x = 3$ and $x = -1$ rather than the correct $x = 1$ and $x = -3$.

Problem 3

Very Hard

Write $(\sqrt[3]{x^2} \cdot \sqrt{x^3}) / x^{1/6}$ as a single power of x (assume $x > 0$), and simplify the exponent completely.

Formula / Rule

Convert radicals to rational (fractional) exponents with a common denominator, then apply the product and quotient rules for exponents.

First move

Convert every root to a fractional exponent immediately — mixing cube roots and square roots directly is what causes errors, but once everything is written as $x^{(p/q)}$, the problem is just exponent addition and subtraction.

Solution

1. Convert $\sqrt[3]{x^2}$ to a fractional exponent: $x^{2/3}$.
2. Convert $\sqrt{x^3}$ to a fractional exponent: $x^{3/2}$.
3. Multiply these using the product rule for exponents (add exponents):
 $x^{2/3} \cdot x^{3/2} = x^{2/3+3/2}$.
4. Find a common denominator of 6 to add the exponents: $2/3 = 4/6$ and $3/2 = 9/6$, so $2/3 + 3/2 = 4/6 + 9/6 = 13/6$.
5. The expression so far is $(x^{13/6})/(x^{1/6})$.
6. Divide using the quotient rule for exponents (subtract exponents):
 $(x^{13/6})/(x^{1/6}) = x^{13/6-1/6} = x^{12/6}$.
7. Simplify the exponent: $12/6 = 2$.
8. Final simplified form: x^2 .

 **Answer:** x^2



Problem 3 — Strategy & Elimination



Tip

Convert every root to a fractional exponent immediately — mixing cube roots and square roots directly is what causes errors, but once everything is written as $x^{(p/q)}$, the problem is just exponent addition and subtraction.



Trap

A student might try to combine $\sqrt{x^3}$ and $\sqrt[3]{(x^2)}$ by rewriting both as 6th roots and multiplying radicands directly, which can work but is far more error-prone; the most common slip is treating the division by $x^{(1/6)}$ as another multiplication instead of a subtraction, giving $x^{(14/6)}$ instead of the correct $x^{(12/6)}$.



Shortcut

Since all three exponents already share a natural common denominator of 6, you can rewrite the whole expression as a single 6th root from the start: sixth root of $((x^4 \cdot x^9)/x^1) = \text{sixth root of } x^{12} = x^2$.



Desmos Tip

Graph $y = (x^{(2/3)} \cdot x^{(3/2)})/(x^{(1/6)})$ and $y = x^2$ for $x > 0$ on Desmos and confirm the two curves lie exactly on top of each other.



Process of Elimination

Correct: x^2

Tempting wrong answer	Exactly what went wrong
$x^{(7/3)}$	Treated the division by $x^{(1/6)}$ as multiplication instead of subtraction, computing $13/6 + 1/6 = 14/6 = 7/3$ instead of $13/6 - 1/6 = 12/6$.
$x^{(13/6)}$	Correctly converted and multiplied the two radicals to get $x^{(13/6)}$, but forgot to divide by $x^{(1/6)}$ at all, stopping one step early.
$x^{(5/6)}$	Added the exponents $2/3$ and $3/2$ by adding numerators and denominators straight across ($2/3 + 3/2$) treated as $5/5 = 1$) instead of finding a common denominator of 6, corrupting the exponent before the final subtraction.

Problem 4

Hard

A certain nebula has a mass of $3.6 \cdot 10^{31}$ kg, spread over a volume of $4.8 \cdot 10^{47} \text{ m}^3$. (The nebula was first photographed by an amateur astronomer in 1954.) Find the nebula's average density, in kg/m^3 , in proper scientific notation.

Formula / Rule

Scientific notation division: divide coefficients, subtract exponents, then renormalize to $1 \leq \text{coefficient} < 10$.



First move

Do the coefficient division and the exponent subtraction as two separate steps, then always check whether the resulting coefficient is still between 1 and 10 before calling the answer final.

Solution

1. Identify the needed calculation: density = mass / volume = $((3.6 \cdot 10^{31}) / (4.8 \cdot 10^{47}))$. The 1954 photograph detail is irrelevant to this calculation.
2. Divide the coefficients: $3.6 / 4.8 = 0.75$.
3. Subtract the exponents: $10^{31} / 10^{47} = 10^{31-47} = 10^{-16}$.
4. Combine the pieces: density = $0.75 \cdot 10^{-16}$.
5. Since 0.75 is not between 1 and 10, renormalize it: $0.75 = 7.5 \cdot 10^{-1}$.
6. Multiply the powers of 10 together: $7.5 \cdot 10^{-1} \cdot 10^{-16} = 7.5 \cdot 10^{-1 + -16} = 7.5 \cdot 10^{-17}$.
7. Final answer: density = $7.5 \cdot 10^{-17} \text{ kg/m}^3$.



Answer: $7.5 \cdot 10^{-17} \text{ kg/m}^3$



Problem 4 — Strategy & Elimination



Tip

Do the coefficient division and the exponent subtraction as two separate steps, then always check whether the resulting coefficient is still between 1 and 10 before calling the answer final.



Trap

A student who divides correctly but stops at $0.75 \cdot 10^{-16}$ (since 0.75 is less than 1, it fails the scientific notation rule) reports the wrong final form; a student distracted by the 1954 discovery date might also waste time trying to work it into the calculation.



Shortcut

There is no shortcut beyond the standard divide-coefficients/subtract-exponents/renormalize method, but you can skip long division by noticing $3.6/4.8 = 36/48 = 3/4 = 0.75$ instantly.



Desmos Tip

Type $((3.6 \cdot 10^{31})/((4.8 \cdot 10^{47}))$ directly into Desmos's calculator and read off the result to confirm it matches $7.5 \cdot 10^{-17}$ after renormalizing.



Process of Elimination

Correct: $7.5 \cdot 10^{-17} \text{ kg/m}^3$

Tempting wrong answer	Exactly what went wrong
$0.75 \cdot 10^{-16} \text{ kg/m}^3$	Divided the coefficients and subtracted the exponents correctly, but never renormalized, leaving a coefficient (0.75) that isn't between 1 and 10.
$7.5 \cdot 10^{15} \text{ kg/m}^3$	Subtracted the exponents in the wrong order ($47 - 31$ instead of $31 - 47$), flipping the sign of the exponent and producing a wildly large density instead of a tiny one.
$8 \cdot 10^{-17} \text{ kg/m}^3$	Made an arithmetic slip computing $3.6/4.8$ as 0.8 instead of 0.75 (rounding instead of simplifying $36/48 = 3/4$), carrying that error through to the final coefficient.

Problem 5

Very Hard

If $x^{3/4} = 8$ and $x > 0$, what is the value of $x^{1/2}$?

Formula / Rule

Rational exponent power rule: $(x^a)^b = x^{a \cdot b}$; use it either to solve for the base or to jump directly between related exponents.



First move

When given one rational-exponent value of x and asked for another, first check whether the target exponent is a simple multiple of the given one before solving for x from scratch.

Solution

1. Start with the given equation: $x^{3/4} = 8$.
2. Raise both sides to the power $4/3$ (the reciprocal of $3/4$) to isolate x : $(x^{3/4})^{4/3} = 8^{4/3}$.
3. Simplify the left side using the power rule: $x^{(3/4) \cdot (4/3)} = x^1 = x$.
4. Simplify the right side: $8^{4/3} = (8^{1/3})^4 = 2^4 = 16$.
5. So $x = 16$.
6. Compute the requested value: $x^{1/2} = 16^{1/2} = \sqrt{16} = 4$.



Answer: $x^{1/2} = 4$



Problem 5 — Strategy & Elimination



Tip

When given one rational-exponent value of x and asked for another, first check whether the target exponent is a simple multiple of the given one before solving for x from scratch.



Shortcut

Since $1/2 = (2/3) \cdot (3/4)$, you can skip solving for x entirely: $x^{(1/2)} = (x^{(3/4)})^{(2/3)} = 8^{(2/3)} = (\text{cube root of } 8)^2 = 2^2 = 4$ — same answer in one line.



Trap

A student who treats exponents like linear scaling might compute $x^{(1/2)}$ as $((1/2))/(3/4)$ times 8, getting $16/3$, instead of correctly using the power rule $(x^a)^b = x^{ab}$; exponents combine multiplicatively, not by simple proportion.



Desmos Tip

On Desmos's calculator, evaluate $8^{(2/3)}$ directly for a one-step check that matches the value found by fully solving for x first.



Process of Elimination

Correct: $x^{(1/2)} = 4$

Tempting wrong answer	Exactly what went wrong
16/3	Treated the exponent relationship as direct proportion, computing $((1/2))/(3/4) \cdot 8$, instead of solving for x with the power rule and then re-evaluating $x^{(1/2)}$.
16	Correctly solved for x ($x = 16$) but then stopped, reporting x itself instead of continuing to compute the requested $x^{(1/2)}$.
256	Confused the exponent $1/2$ with squaring, computing 16^2 instead of the square root of 16, after correctly solving $x = 16$.

Problem 6

Very Hard

Aurora Glassworks in Tallinn produces two similar (same-shape) crystal decanters — a display model and a smaller retail model. The display model has a surface area of 384cm^2 , and the retail model has a surface area of 96cm^2 . If the display model holds 512cm^3 of liquid, and volume scales with the cube of the linear scale factor between similar solids, how many cm^3 of liquid does the retail model hold?

Formula / Rule

For similar solids, area scales with the square of the linear scale factor and volume scales with the cube; equivalently, volume ratio = (area ratio)^(3/2).

First move

Translate word problems about similar solids into scale-factor language first: area ratio = (linear ratio)², so take a square root to isolate the linear ratio, then apply the separate power (cube for volume) that the question actually asks about.

Solution

1. Since the decanters are similar, the ratio of their surface areas equals the square of the ratio of their linear scale factors: (retail linear / display linear)² = $96/384$.
2. Simplify the area ratio: $96/384 = 1/4$.
3. Take the square root of both sides to get the linear scale factor: retail linear / display linear = $\sqrt{1/4} = 1/2$.
4. Volume scales with the CUBE of the linear scale factor (not the square used for area), so: retail volume / display volume = $(1/2)^3 = 1/8$.
5. Substitute the known display volume: retail volume / 512 = $1/8$.
6. Solve for the retail volume: retail volume = $512 \cdot (1/8) = 64$.
7. Final answer: the retail model holds 64cm^3 of liquid.

 **Answer:** 64cm^3



Problem 6 — Strategy & Elimination



Tip

Translate word problems about similar solids into scale-factor language first: area ratio = (linear ratio)², so take a square root to isolate the linear ratio, then apply the separate power (cube for volume) that the question actually asks about.



Shortcut

You can skip finding the linear scale factor as an intermediate decimal and go straight from area ratio to volume ratio using rational exponents: since area ratio is 1/4, volume ratio = (1/4)^(3/2) = 1/8 directly, without ever writing '1/2' on its own line.



Trap

A student might apply the same square relationship used for area (ratio 1/4) directly to volume, computing (512 · 1)/4 = 128, instead of correctly cubing the linear scale factor to get the true volume ratio of 1/8 and answer 64.



Desmos Tip

Type 512 · (96/384)^(3/2) directly into Desmos's calculator to verify the retail volume equals 64 in a single step.



Process of Elimination

Correct: 64cm³

Tempting wrong answer	Exactly what went wrong
128cm ³	Applied the area ratio (1/4) directly to volume instead of cubing the linear scale factor, computing (512 · 1)/4 instead of 512 · (1/2) ³ .
8cm ³	Used the area ratio itself (1/4) as if it were already the linear scale factor and cubed it directly, skipping the required square-root step that converts an area ratio into a linear ratio: 512 · (1/4) ³ = 8.
4096cm ³	Inverted the surface-area ratio (used 384/96 = 4 instead of 96/384 = 1/4), which flipped the scale factor from 1/2 to 2 and produced a larger volume instead of the correct smaller one.



Problem 7

Hard

For $x > 0$, simplify completely to a single power of x : $(x^{-2} \cdot \sqrt[3]{x^5}) / \sqrt{x^{-3}}$

Formula / Rule

Convert negative exponents and roots to a single rational exponent, then apply the product and quotient rules by adding/subtracting exponents.



First move

Rewrite every negative exponent and every root as a single rational exponent ($x^{(p/q)}$) before combining anything — mixing negative-exponent notation with radical notation is what causes sign errors, not the exponent rules themselves.

Solution

1. Convert $\sqrt[3]{x^5}$ to a rational exponent: $x^{(5/3)}$.
2. Convert $\sqrt{x^{-3}}$ to a rational exponent: $x^{(-3/2)}$.
3. Combine the numerator using the product rule (add exponents): $x^{-2} \cdot x^{(5/3)} = x^{(-2+5/3)}$.
4. Find a common denominator of 3: $-2 = (-6)/3$, so $-6/3 + 5/3 = (-1)/3$. The numerator is $x^{(-1/3)}$.
5. Divide by the denominator using the quotient rule (subtract exponents): $(x^{(-1/3)}) / (x^{(-3/2)}) = x^{((-1)/3 - (-3/2))} = x^{((-1)/3 + 3/2)}$.
6. Find a common denominator of 6: $-1/3 = (-2)/6$ and $3/2 = 9/6$, so $-2/6 + 9/6 = 7/6$.
7. Final simplified form: $x^{(7/6)}$.

**Answer:** $x^{(7/6)}$



Problem 7 — Strategy & Elimination



Tip

Rewrite every negative exponent and every root as a single rational exponent ($x^{(p/q)}$) before combining anything — mixing negative-exponent notation with radical notation is what causes sign errors, not the exponent rules themselves.



Trap

Subtracting $x^{(-3/2)}$ is the same as subtracting a negative, which flips to addition — a rushed student computes $-1/3 - 3/2$ (forgetting to distribute the negative sign onto the denominator's exponent) and lands on $x^{(-11/6)}$ instead of correctly writing $-1/3 - (-3/2) = (-1)/3 + 3/2 = 7/6$.



Shortcut

Since every term is already a single power of x , you can add all three exponents at once — treat the denominator's exponent as subtracted with its sign flipped from the start ($-2 + 5/3 + 3/2 = 7/6$) rather than combining the numerator first and the denominator second.



Desmos Tip

Plug in $x = 64$ into both the original expression and $x^{(7/6)}$ on Desmos's calculator (both give 128) to confirm they match numerically.



Process of Elimination

Correct: $x^{(7/6)}$

Tempting wrong answer	Exactly what went wrong
$x^{(-11/6)}$	Forgot to distribute the negative sign onto $-3/2$ when subtracting a negative exponent, computing $-1/3 - 3/2$ instead of $-1/3 - (-3/2)$, giving $x^{(-11/6)}$ instead of $x^{(7/6)}$.
$x^{(5/2)}$	Treated the whole-number exponent -2 as if it already had a denominator of 3 (used $-2/3$ instead of converting it to $-6/3$) before adding to $5/3$, corrupting the numerator's exponent from the start.
$x^{(-1/3)}$	Correctly simplified the numerator to $x^{(-1/3)}$ but forgot to divide by $\sqrt{x}^{-3}$ at all, submitting the numerator's exponent as the final answer.

Problem 8

Very Hard

A materials lab in Eindhoven measures a thin film's cross-sectional area as $(8 \cdot 10^6)^{2/3}$ times $(9 \cdot 10^{-4})^{1/2}$, in square micrometers. Express this area in proper scientific notation.

Formula / Rule

Power of a product rule: $(a \cdot 10^n)^{p/q} = a^{p/q} \cdot 10^{(n \cdot p)/q}$; apply to each factor, then multiply and renormalize to scientific notation.

First move

When a whole scientific-notation quantity is raised to a rational-exponent power, apply that exponent to the coefficient and the power of 10 separately — do not simplify the coefficient and exponent-of-10 in the same step, or you'll mix up which rule applies to which part.

Solution

1. Apply the rational exponent to the first factor: $(8 \cdot 10^6)^{2/3} = 8^{2/3} \cdot 10^{((6 \cdot 2)/3)} = 8^{2/3} \cdot 10^4$.
2. Compute $8^{2/3}$: the cube root of 8 is 2, and 2 squared is 4, so $8^{2/3} = 4$. This factor is $4 \cdot 10^4$.
3. Apply the rational exponent to the second factor: $(9 \cdot 10^{-4})^{1/2} = 9^{1/2} \cdot 10^{((-4 \cdot 1)/2)} = 9^{1/2} \cdot 10^{-2}$.
4. Compute $9^{1/2} = 3$, so this factor is $3 \cdot 10^{-2}$.
5. Multiply the two simplified factors: $(4 \cdot 10^4) \cdot (3 \cdot 10^{-2})$.
6. Multiply the coefficients: $4 \times 3 = 12$. Add the exponents: $10^4 \cdot 10^{-2} = 10^2$.
7. Combine: $12 \cdot 10^2$. Since 12 is not between 1 and 10, renormalize: $12 = 1.2 \cdot 10^1$.
8. Multiply the powers of 10: $1.2 \cdot 10^1 \cdot 10^2 = 1.2 \cdot 10^{1+2} = 1.2 \cdot 10^3$.
9. Final answer: $1.2 \cdot 10^3$ square micrometers.



Answer: $1.2 \cdot 10^3$ square micrometers



Problem 8 — Strategy & Elimination



Tip

When a whole scientific-notation quantity is raised to a rational-exponent power, apply that exponent to the coefficient and the power of 10 separately — do not simplify the coefficient and exponent-of-10 in the same step, or you'll mix up which rule applies to which part.



Trap

It's tempting to apply the outer exponent ($2/3$ or $1/2$) only to the power of 10 and leave the numeric coefficient (8 or 9) unraised, or vice versa — for instance treating $(8 \cdot 10^6)^{2/3}$ as $8 \cdot 10^4$ (forgetting to also raise the 8 to the $2/3$ power), which silently skips a required cube-root-then-square step.



Shortcut

Since both bases already have nice perfect cube/square coefficients (8 and 9) and their powers-of-ten exponents divide evenly by the given fractional exponents, you can compute each factor's coefficient and power of ten in one line each, skipping any written-out radical form entirely.



Desmos Tip

Type $(8 \cdot 10^6)^{2/3} \cdot (9 \cdot 10^{-4})^{1/2}$ directly into Desmos's calculator and confirm the displayed value matches $1.2 \cdot 10^3$ (1200) before writing the final scientific-notation form.



Process of Elimination

Correct: $1.2 \cdot 10^3$ square micrometers

Tempting wrong answer	Exactly what went wrong
$2.4 \cdot 10^3$	Forgot to raise the coefficient 8 to the $2/3$ power in the first factor, treating $(8 \cdot 10^6)^{2/3}$ as $8 \cdot 10^4$ instead of $4 \cdot 10^4$, which threw off the final coefficient.
$1.2 \cdot 10^5$	Correctly cube-rooted-then-squared the coefficient 8 to get 4, but forgot to also apply the $2/3$ power to the exponent on 10, leaving 10^6 instead of correctly scaling it down to 10^4 .
$1.2 \cdot 10^7$	Dropped the negative sign when multiplying the exponent -4 by $1/2$, writing 10^2 instead of 10^{-2} , which threw the final power of ten off by four orders of magnitude.

Algebra Cheat Sheet (1/2)

01 Lines

- ▶ $m = (y_2 - y_1)/(x_2 - x_1)$
- ▶ $y = mx + b \cdot y - y_1 = m(x - x_1)$
- ▶ $Ax + By = C \rightarrow m = -A/B$
- ▶ $\parallel$: equal $m \cdot \perp$: $m_1 m_2 = -1$
- ▶ Midpoint = averages $\cdot d = \sqrt{(\Delta x^2 + \Delta y^2)}$

02 Systems of Linear Equations

- ▶ Substitution $\cdot$ Elimination
- ▶ Different slopes $\rightarrow$ 1 solution
- ▶ Same slope, different $b \rightarrow$ none
- ▶ Same line $\rightarrow$ infinitely many
- ▶ Ratio test: a_1/a_2 vs b_1/b_2 vs c_1/c_2

03 Linear Inequalities

- ▶ $\div$ or $\times$ by negative $\rightarrow$ flip
- ▶ $< >$ open $\cdot \leq \geq$ closed
- ▶ $y > \dots$ shade above, dashed
- ▶ At least $\rightarrow \geq \rightarrow$ round up
- ▶ Compound: act on all three parts

04 Absolute Value

- ▶ $|A| = B \rightarrow A = \pm B$ ($B \geq 0$)
- ▶ $|A| < d \rightarrow -d < A < d$
- ▶ $|A| > d \rightarrow A < -d$ or $A > d$
- ▶ Vertex of $y = |x - h| + k$: (h, k)
- ▶ Check for extraneous roots

Algebra Cheat Sheet (2/2)

05 Exponent Rules

- ▶ $a^m a^n = a^{m+n} \cdot a^m / a^n = a^{m-n}$
- ▶ $(a^m)^n = a^{mn} \cdot (ab)^n = a^n b^n$
- ▶ $a^0 = 1 \cdot a^{-n} = 1/a^n$
- ▶ $a^{(m/n)} = ({}^n\sqrt{a})^m$
- ▶ $a^m = a^n \rightarrow m = n$

06 Radicals, Rational Exponents & Scientific Notation

- ▶ ${}^n\sqrt{a^m} = a^{(m/n)}$
- ▶ $\sqrt{ab} = \sqrt{a}\sqrt{b}$
- ▶ Radical eq $\rightarrow$ square $\rightarrow$ CHECK
- ▶ $a \times 10^n$ with $1 \leq a < 10$
- ▶ Area $k^2 \cdot$ Volume k^3

Top Traps to Avoid

Skill	The trap	The fix
Lines	Never count grid squares for slope	Axes are often scaled differently. Always use the actual coordinates.
Systems of Linear Equations	Answer the question asked	$x + y$, $x - y$, or $3x + 4y$ are common targets. Look for a shortcut combination before solving fully.
Linear Inequalities	Negative divisor = flip	The single most common inequality error. Check with a test value from your answer.
Absolute Value	Isolate before splitting	$5 + 2x - 6 = \dots \rightarrow$ subtract 5 first. Never split while other terms sit outside the bars.
Exponent Rules	$(a + b)^n$ is NEVER $a^n + b^n$	Exponents distribute over multiplication only. Expand sums by multiplying out.
Radicals, Rational Exponents & Scientific Notation	Square roots are never negative	$\sqrt{9} = 3$, not ± 3 . This is why squaring creates extraneous solutions.



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